Introduction to Probability Models

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Chapter 1

Introduction To the Probability Theorey

1.1 Theorems

Theorem 1.1.

$$P\left(\bigcap_{i=1}^{n} E_i\right) = P(E_1)P(E_2|E_1)P(E_3|E_1E_2)\cdots P(E_n|E_1\cdots E_{n-1}).$$

Proof.

$$\begin{aligned}
&P(E_1)P(E_2|E_1)P(E_3|E_1E_2)\cdots P(E_n|E_1\cdots E_{n-1}) \\
&= P(E_1)\frac{P(E_1E_2)}{P(E_1)}\frac{P(E_1E_2E_3)}{P(E_1E_2)}\cdots\frac{P(E_1\cdots E_n)}{P(E_1\cdots E_{n-1})} \\
&= P(E_1\cdots E_n).
\end{aligned}$$

□

Theorem 1.2. *If the occurence of B makes A more likely, then the occurrence of A make B more likely.*

Proof.

$$\begin{aligned}
P(A|B) > P(A) &\iff P(AB) > P(A)P(B) \\
&\iff P(B|A) > P(B).
\end{aligned}$$

□

Theorem 1.3. *(a) If E and F are mutually exclusive, then*

$$P(E|F) = 0.$$

(b) If $E \subseteq F$, then

$$P(E|F) \geq P(E).$$

(c) If $F \subseteq E$, then

$$P(E|F) = 1.$$

Theorem 1.4. *Suppose all n men at a party throw their hats in the center of the room. Each man then randomly selects a hat. Show that the probability that none of the n men selects this own hat is*

$$\sum_{k=2}^{n} \frac{(-1)^k}{k!}.$$

Proof. Let

$$E_i = \text{event person } i \text{ selects own hat.}$$

$$\begin{aligned}
&P(\text{ no one selects own hat }) \\
&= 1 - P\left(\bigcup_{k=1}^{n} E_k\right) \\
&= 1 - \sum_{i_1} P(E_{i_1}) + \sum_{i_1<i_2} P(E_{i_1}E_{i_2}) - \sum_{i_1<i_2<i_3} P(E_{i_1}E_{i_2}E_{i_3}) + \cdots + (-1)^n P(E_1 \cdots E_n).
\end{aligned}$$

Let $k \in [n]$, then

$$P(E_{i_1} \cdots E_{i_k}) = \frac{(n-k)!}{n!},$$

$$\text{number of terms in summation} \sum_{i_1<\cdots<i_k} = \binom{n}{k}.$$

Then

$$\sum_{i_1<\cdots<i_k} P(E_{i_1} \cdots E_{i_k}) = \binom{n}{k} \frac{(n-k)!}{n!} = \frac{1}{k!}.$$

Note as $n \to \infty$, this converges to e^{-1}. □

Theorem 1.5. *For a fixed event B, show that the collection $P(A|B)$, defined for all events A, satisfies the three conditions for a probability. Conclude from this that*

$$P(A|B) = P(A|BC)P(C|B) + P(A|BC^c)P(C^c|B).$$

Proof.

$$\begin{aligned}
P(A|BC)P(C|B) + P(A|BC^c)P(C^c|B) &= \frac{P(ABC)}{P(BC)}\frac{P(BC)}{P(B)} + \frac{P(ABC^c)}{P(BC^c)}\frac{P(BC^c)}{P(B)} \\
&= \frac{P(AB)}{P(B)} = P(A|B).
\end{aligned}$$

□

1.2 Examples

Example 1.6. There are r players, with player i initially having n_i units, $n_i > 0$, $i = 1, \ldots, r$. At each stage, two of the players are chosen to play a game, with the winner of the game receiving 1 unit from the loser. Any player whose fortune drops to 0 is eliminated, and this continues until a single player has all $n = \sum_{i=1}^{r} n_i$ units, with that player designated as the victor. Assuming that

the results of successive games are independent, and that each game is equally likely to be won by either of its two players, find the probability that player i is the victor.

Suppose there are n players, with each player initially having 1 unit.

Consider player i. Each stage she plays will be equally likely to result in her either winning or losing 1 unit, with the results from each stage being independent.

In addition, she will continue to play stages until her fortune becomes either 0 or n. Because this is the same for all players, it follows that each player has the same chance of being the victor.

Consequently, each player has player probability $\frac{1}{n}$ of being the victor.

Now, suppose these n players are divided into r teams, with team i containing n_i players, $i = 1, \ldots, r$. That is, suppose players $1, \ldots, n_1$ constitute team 1, players $n_1 + 1, \ldots, n_1 + n_2$ constitute team 2 and so on.

Then the probability that the victor is a member of team i is $\frac{n_i}{n}$. But because team i initially has a total fortune of n_i units, $i = 1, \ldots, r$, and each game played by members of different teams results in the fortune of the winner's team increasing by 1 and that of the loser's team decreasing by 1, then the probability that the victor is from team i is exactly the desired probability. Moreover, our argument also shows that the result is true no matter how the choices of the players in each stage are made.

Example 1.7. Let E and F be mutually exclusive events in the sample space of an experiment. Suppose that the experiment is repeated until either event E or event F occurs. What does the sample space of this new super experiment look like? Show that the probability that event E occurs before event F is

$$\frac{P(E)}{P(E) + P(F)}.$$

$$\begin{aligned} P(E \text{ before } F) &= P(E \text{ before F } |\text{initial outcome is } E)P(E) \\ &\quad + P(E \text{ before F } |\text{initial outcome is } F)P(F) \\ &\quad + P(E \text{ before F } |\text{initial outcome neither } E \text{ nor } F)(1 - P(E) - P(F)) \\ &= 1 \cdot P(E) + 0 \cdot P(F) + P(E \text{ before } F)(1 - P(E) - P(F)). \end{aligned}$$

So

$$P(E \text{ before } F) = \frac{P(E)}{P(E) + P(F)}.$$

Example 1.8. The probability of winning on a single toss of the dice is p. A starts, and if he fails, he passes the dice to B, who then attempts to win on her toss. They continue tossing the dice back and forth until one of them wins. What are their respective probabilities of winning?

$$P(A \text{ wins}) = \sum_{n=0}^{\infty} P(A \text{ wins on} (2n+1)^{\text{th}} \text{ toss}) = \sum_{n=0}^{\infty} p(1-p)^{2n} = \frac{1}{2-p}.$$

Example 1.9. A and B play until one has 2 more points than the other. Assuming that each point is independently won by A with probability p, what is the probability they will play a total of $2n$ points? What is the probability that A will win? (In each trial, one point either is won by A or by B.)

Let trial 1 consist of the first two points; trial 2 the next two points, and so on.

The probability that each player wins one point in a trial is $2p(1-p)$.

Now a total of $2n$ points are played if the first $(n-1)$ trials all result in each player winning one of the points in that trial and the nth trial results in one of the players winning both points.

By independence, we obtain

$$P(2n \text{ points are needed}) = (2p(1-p))^{n-1}\left(p^2 + (1-p)^2\right),\ n \geq 1.$$

The probability that A wins on trial n is $(2p(1-p))^{n-1}p^2$ and so

$$P(A \text{ wins}) = p^2 \sum_{n=1}^{\infty} (2p(1-p))^{n-1} = \frac{p^2}{1-2p(1-p)}.$$

Example 1.10. In an election, candidate A reveives n votes and candidate B receives m votes, where $n > m$. Assume that in the count of the votes all possible orderings of the $n+m$ votes are equally likely. Let $P_{n,m}$ denote the probability that from the first vote on A is always in the head. Make a conjecture as to the value of $P_{n,m}$.

$$P_{2,1} = \frac{1}{\binom{3}{1}} = 1\frac{1}{3}.$$

$$P_{3,1} = \frac{2}{\binom{4}{1}} = 1\frac{1}{2}.$$

$$P_{3,2} = \frac{2}{\binom{5}{2}} = 1\frac{1}{5}.$$

$$P_{4,1} = \frac{3}{\binom{5}{1}} = 1\frac{3}{5}.$$

$$P_{4,2} = \frac{5}{\binom{6}{2}} = 1\frac{1}{3}.$$

$$P_{4,3} = \frac{5}{\binom{7}{3}} = 1\frac{1}{3}.$$

In all case above,

$$P_{n,m} = \frac{n-m}{n+m}.$$

Example 1.11. A deck of 52 playing cards, containing all 4 aces, is randomly divided into 4 piles of 13 cards each. Define events E_1, E_2, E_3 and E_4 as follows:

$$E_1 = \{\text{the first pile has exactly 1 ace}\},$$

$$E_2 = \{\text{the second pile has exactly 1 ace}\},$$

$$E_3 = \{\text{the third pile has exactly 1 ace}\},$$

$$E_4 = \{\text{the fourth pile has exactly 1 ace}\},$$

Find $P(E_1E_2E_3E_4)$.

$$P(E_1E_2E_3E_4) = P(E_1)P(E_2|E_1)P(E_3|E_1E_2)P(E_4|E_1E_2E_3).$$

Example 1.12. A deck of 52 playing cards, containing all 4 aces, is randomly divided into 4 piles of 13 cards each. Define events E_1, E_2, E_3 and E_4 as follows:

$$E_1 = \{\text{one of the piles contains the ace of spades}\},$$

$$E_2 = \{\text{the ace of spades and the ace of hearts are in different piles}\},$$

$$E_3 = \{\text{the ace of spades, the ace of hearts, and the ace of diamonds are in different piles}\},$$

$$E_4 = \{\text{all 4 aces are in different piles}\}.$$

Find $P(E_1E_2E_3E_4)$, the probability that each pile has an ace.

$$P(E_1) = 1,$$

$$P(E_2|E_1) = \frac{39}{51},$$

$$P(E_3|E_1E_2) = \frac{26}{50},$$

$$P(E_4|E_1E_2E_3) = \frac{13}{49}.$$

Example 1.13. A fair coin is continually flippes. What is the probability that the pattern T, H, H, H occurs before the pattern H, H, H, H?
$\frac{15}{16}$, since the only way in which the pattern H, H, H, H can appear before patten T, H, H, H is if the first four flips all land heads.

Example 1.14. Three prisoners are informed by their jailer that one of them has been chosen at random to be executed, and the other two are to be freed. Prisoner A asks the jailer to tell him privately which of his fellow prisoners will be set free, claiming that there would be no harm in divulging this information, since he already knows that at least one will go free. The jailer refuses to answer this question, pointing out that if A knew which of his fellows were to be set free, then his own probability of being executed would rise from $\frac{1}{3}$ to $\frac{1}{2}$, since he would then be one of two prisoners. What do you think of the jailer's reasoning?
Let $X \in \{B, C\}$ denote the jailer's answer to prisoners A. Now for instance,

$$\begin{aligned} P(A \text{ to be excuted}|X = B) &= \frac{P(A \text{ to be excuted},\ X = B)}{P(X = B)} \\ &= \frac{P(A \text{ to be executed})P(X = B|A \text{ to be executed})}{P(X = B)} \\ &= \frac{\frac{1}{3} \cdot \frac{1}{2}}{\frac{1}{2}} = \frac{1}{2}. \end{aligned}$$

Thus, the jailer's reasoning is invalid.

1.3 Random variables

Definition 1.15. Quantities of interest, or more formally, these real-valued functions defined on the sample space, are known as random variables. Since the value of a random variable is determined by the outcome of the experiment, we may assign probabilities to the possible values of the random variable.

Definition 1.16. The cdf

$$F(b) = P(X \leq b), \ \forall \ b \in \mathbb{R}.$$

Then

$$P(a < X \leq b) = F(b) - F(a), \ \forall \ a < b.$$

$$P(X < b) = \lim_{h \to 0^+} P(X \leq b - h) = \lim_{h \to 0^+} F(b - h).$$

Example 1.17. Consider an experiment that consists of counting the number of α-particles given off in a one-second interval by one gram of radioactive material. If we know from past experience that, on the averge, 3.2 such α-particles are given off, what is a good approximation to the probability that no more then two α-particles will appear?

If we think of the gram of radioactive material as consisting of a large number n of atoms each of which has probability $\frac{3.2}{n}$ of disintegrating and sending off an α-particle during the second considered, then to a very close approximation, the number of α-particles given off will be a Poisson random variable with parameter $\lambda = 3.2$.

Example 1.18. Suppose there are 25 different types of coupons and suppose that each time one obtain a coupon, it is equally likely to be any one of the 25 types. Compute the expected number of different types that are contained in a set of 10 coupons.

Let X denote the number of different types in the set of 10 coupons and let

$$X_i = \begin{cases} 1, & \text{if at least one type } i \text{ coupon is in the set of 10} \\ 0, & \text{otherwise} \end{cases}.$$

Then

$$X = \sum_{i=1}^{25} X_i.$$

Also,

$$E[X_i] = P(\text{at least one type } i \text{ coupon is in the set of 10}) = 1 - \left(\frac{24}{25}\right)^{10}.$$

Thus,

$$E[X] = 25\left[1 - \left(\frac{24}{25}\right)^{10}\right].$$

Example 1.19. Suppose that each coupon obtained is, independent of what has been previously obtained, equally likely to be any of m different types. Find the expected number of coupons one need to obtain in order to have at least one of each type.

Suppose the coupon collector has i different types. Let X_i denote the number of additional coupons collected until the collector has $i+1$ types. It is easy to see that the X_i are independent geometric random variables with respective paramters $\frac{n-i}{n}$, $\forall\ i = 0, \ldots, n-1$. Therefore,

$$E\left[\sum_{i=0}^{n-1} X_i\right] = \sum_{i=0}^{n-1} E[X_i] = \sum_{i=0}^{n-1} \frac{n}{n-i} = n \sum_{j=1}^{n} \frac{1}{j}.$$

Example 1.20 (Sampling from a finite population: The Hypergeometric)**.** Consider a population of N individuals, some of whom are in favor of a certain proportion p, which is unknown.
If we let

$$X_i = \begin{cases} 1, & \text{if the } i\text{th person chosen is in favor} \\ 0, & \text{otherwise} \end{cases}.$$

Then the usual estimator of p is $\overline{X}$. Since $X_i \sim$ Bernoulli(p),

$$\mathrm{Var}(X_i) = p(1-p).$$

For $i \neq j$,

$$\begin{aligned} \mathrm{Cov}(X_i, X_j) &= E[X_i X_j] - E[X_i][X_j] \\ &= P(X_i = 1, X_j = 1) - p^2 \\ &= P(X_i = 1) P(X_j = 1 | X_i = 1) - p^2 \\ &= \frac{Np}{N} \frac{Np-1}{N-1} - p^2 \end{aligned}$$

So

$$\begin{aligned} \mathrm{Var}\left(\overline{X}\right) &= \frac{1}{n^2} \sum_{i=1}^{n} \mathrm{Var}(X_i) + \frac{2}{n^2} \sum_{i=1}^{n} \sum_{i<j} \mathrm{Cov}(X_i, X_j) \\ &= \frac{p(1-p)}{n} + \frac{2}{n^2} \binom{n}{2} \left[\frac{p(Np-1)}{N-1} - p^2\right] \\ &= \frac{p(1-p)}{n} - \frac{(n-1)p(1-p)}{n(N-1)}. \end{aligned}$$

As a function of the population size N, the variance increases as N increases. As $N \to \infty$, the variance is $\frac{p(1-p)}{n}$, which is not surprising since for N large each of the X_i will be (approximately) independent r.v., and thus $\sum_{i=1}^{n} X_i$ will have an (approximately) binomial distribution.
The random variable $\sum_{i=1}^{n} X_i$ can be thought of as representing the number of white balls obtained when n balls are randomly selected from a population consisting of Np white and $N - Np$ black balls. Such a random variable is called hypergeometric and

$$P\left(\sum_{i=1}^{n} X_i = k\right) = \frac{\binom{Np}{k}\binom{N-Np}{n-k}}{\binom{N}{n}}.$$

Definition 1.21 (Order statistic)**.**

$$P(X_{(i)} \leq x) = \sum_{k=i}^{n} \binom{n}{k} F^k(x) \left(1 - F(x)\right)^{n-k}.$$

$$f_{X_{(i)}} = \frac{n!}{(n-i)!(i-1)!} f(x) F^{i-1}(x) \left(1 - F(x)\right)^{n-i}.$$

$$f_{X(1),\ldots,X_{(n)}}(x_1, \ldots, x_n) = \begin{cases} n! \prod_{k=1}^{n} f(x_k), & x_1 < \cdots < x_n \\ 0, & \text{otherwise} \end{cases}.$$

1.4 Expectation

Theorem 1.22. *For $X \geq 0$,*

$$E[X] = \int_0^\infty P(X > t) dt.$$

Corollary 1.23. If $X \in \mathbb{N}$, then

$$E[X] = \sum_{n=0}^{\infty} P(X > n).$$

Example 1.24. Let

$$X \sim \exp(1/\lambda),$$

then

$$E\left[X^{n+1}\right] = \frac{n+1}{\lambda} E[X^n].$$

Since

$$E[X] = 1/\lambda,$$

we have

$$E[X^n] = \frac{n!}{\lambda^n}.$$

Example 1.25. Let

$$X \sim \text{Binom}(n, p),$$

then for $1 \leq m \leq n$,

$$\begin{aligned}
&E\left[X(X-1)\cdots(X-m+1)\right] \\
&= \sum_{k=0}^{n} k(k-1)\cdots(k-m+1) \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k} \\
&= \sum_{k=m}^{n} \frac{n!}{(k-m)!(n-k)!} p^k (1-p)^{n-k} \\
&= n(n-1)\cdots(n-m+1) p^m \sum_{k=0}^{n-m} \frac{(n-m)!}{k!\left((n-m)-k\right)!} p^k (1-p)^{(n-m)-k} \\
&= \frac{n!}{(n-m)!} p^m.
\end{aligned}$$

1.5 MGF

The MGF of a standard normal random variable Z is

$$M_Z(t) = E[e^{tZ}] = e^{\frac{t^2}{2}}.$$

Then the MGF of $X = \sigma Z + \mu$ is

$$e^{\mu t} M_Z(\sigma t) = e^{\frac{\sigma^2 t^2}{2} + \mu t}.$$

The MGF uniquely determines the distribution.

Example 1.26. If two independent r.v.'s

$$X \sim \text{Binom}(n, p), \quad Y \sim \text{Binom}(m, p),$$

then

$$\phi_{X+Y}(t) = \phi_X(t)\phi_Y(t) = (pe^t + 1 - p)^n (pe^t + 1 - p)^m = (pe^t + 1 - p)^{m+n}.$$

So

$$X + Y \sim \text{Binom}(m + n, p).$$

Example 1.27 (The Poisson Paradigm)**.** Suppose that the trials are independent, with trial i resulting in a success with probability p_i, where all the $p_i, i \in [n]$ are small. Let

$$X_i = \begin{cases} 1, & \text{if trial } i \text{ is a success} \\ 0, & \text{otherwise} \end{cases}.$$

The total number of success

$$X = \sum_{i=1}^{n} X_i.$$

Since $X_i \sim \text{Bernoulli}(p_i)$,

$$E[e^{tX_i}] = 1 + p_i(e^t - 1) \approx e^{p_i(e^t - 1)}.$$

Then

$$E[e^{tX}] \approx \prod_{i=1}^{n} e^{p_i(e^t - 1)} = e^{\sum_{i=1}^{n} p_i(e^t - 1)},$$

which is the MGF of a Poisson random variable with mean $\sum_{i=1}^{n} p_i$.

Not only is it not necessary for the trials to have the same success probability for the number of success to approximately have a Poisson distribution, they need not even be independent, provided that their dependence is weak.

For instance, reall the matching problem, where n people randomly select hats from a set consisting of one hat from each person. We say trial i is a success if person i chooses his or her own hat, it follows that, with A_i being the event that trial i is a success,

$$P(A_i) = \frac{1}{n} \text{ and } P(A_i | A_j) = \frac{1}{n-1}, \ j \neq i.$$

Their dependence appears, for large n, to be weak. Becasue of this weak dependence, and the small trial success probabilities, it would seem that the number of matches should approximately have a Poisson distribution.

1.6 Limit theorems

Theorem 1.28 (Markov's Inequality)**.** *If $X \geq 0$, then*

$$P(X \geq a) \leq \frac{E[X]}{a}, \ \forall \ a > 0.$$

Theorem 1.29 (Chebyshev's Inequality)**.** *If $E[X] = \mu$ and $\mathrm{Var}(X) = \sigma^2$, then*

$$P\left(|X - \mu| \geq k\right) \leq \frac{\sigma^2}{k^2}, \ \forall \ k > 0.$$

Proof. Follows from Markov's inequality. □

Example 1.30. Suppose we know that the number of items produced in a factory during a week is a random variable with mean 500 and variance is 100, what can be said about the probability that this week's production will be between 400 and 600?

$$P\left(|X - 500| \geq 100\right) \leq \frac{\sigma^2}{100^2} = \frac{1}{100}.$$

Hence

$$P\left(|X - 500| < 100\right) \geq 1 - \frac{1}{100} = \frac{99}{100}.$$

Definition 1.31. If

$$P\left(X_n \to X\right) = 1,$$

then

$$X_n \xrightarrow{\text{a.s.}} X,$$

or

$$X_n \to X \text{ w.p.1..}$$

Definition 1.32. Given a collection of random variables $\{X_n\}_{n \geq 1}$, we can define for each $\omega \in \Omega$,

$$\lim \inf_{n \to \infty} X_n(\omega) = \lim_{n \to \infty} \inf_{k \geq n} X_k(\omega),$$

and

$$\lim \sup_{n \to \infty} X_n(\omega) = \lim_{n \to \infty} \sup_{k \geq n} X_k(\omega).$$

Then

$$P\left(X_n \to X\right) = 1 \Longleftrightarrow P\left(\lim \inf_{n \to \infty} X_n = \lim \sup_{n \to \infty} X_n = X\right) = 1.$$

1.7 Stochastic Processes

Definition 1.33. A *stochastic process* $\{X(t), t \in T\}$ is a collection of random variables defined on the same probability space. That is, for each $t \in T$, $X(t)$ is a random variable. The index t is often interpreted as time and, as a result, we refer to $X(t)$ as the state of the process at time t. For example, $X(t)$ might equal the total number of customers that have entered a supermarket by time t; or the number of customers in the supermarket at time t.

1.8 Examples

Let X represent the difference between the number of heads and the number of tails obtained when a coin is tossed n times, The possibles values of X are

$$-n, -n+2, -n+4, \ldots, n-2, n.$$

Example 1.34. An airline knows that 5 percent of the people making reservation on a certain flight will not show up. Consequently, their policy is to sell 52 tickets for a flight that can hold only 50 passengers. What is the probability that there will be a seat availiable for every passenger who shows up?

$$1 - 0.95^{52} - 52 \cdot 0.95^{51} \cdot 0.05.$$

Example 1.35. Suppose that two teams are playing a series of games, each of which is independently won by team A with probability p abd by team B with probability $1-p$. The winner of the series is the first team to win 3 games. Find the expected number of games that are played.

$$P(X=3) = p^3 + (1-p)^3,$$

$$\begin{aligned} P(X=4) &= P(X=4,\ \text{I has 2 wins in first 3 games}) + P(X=4,\ \text{II has 2 wins in the first 3 games}) \\ &= 3p^2(1-p)p + 3p(1-p)^2(1-p), \end{aligned}$$

and

$$\begin{aligned} P(X=5) &= P(\text{each player has 2 wins in the first 4 games}) \\ &= 6p^2(1-p)^2. \end{aligned}$$

Then

$$E[X] = \sum_{i=3}^{5} iP(X=i).$$

Example 1.36. A fair coin is independently flipped n times, k times by A and $n-k$ times by B. Show that the probability that A and B flip the same number of heads is equal to the probability that there are a total of k heads.

$$\begin{aligned} P(\text{same number of heads}) &= \sum_i P(A=i, B=i) \\ &= \sum_i \binom{k}{i} \left(\frac{1}{2}\right)^k \binom{n-k}{i} \left(\frac{1}{2}\right)^{n-k} \\ &= \sum_i \binom{k}{i} \binom{n-k}{i} \left(\frac{1}{2}\right)^n \\ &= \sum_i \binom{k}{k-i} \binom{n-k}{i} \left(\frac{1}{2}\right)^n \\ &= \binom{n}{k} \left(\frac{1}{2}\right)^n. \end{aligned}$$

Another argument is as follows: since coin is fair,

$$\begin{aligned} P(\text{number of heads of } A \ = \ \text{number of heads of B}) &= P(\text{number of tails of } A \ = \ \text{number of heads of } B) \\ &= P(k - \text{number of heads of } A \ = \ \text{number of heads of } B) \\ &= P(k \ = \ \text{total number of heads}). \end{aligned}$$

Example 1.37. Consider n independent flips of a coin having probability p of landing heads. Say a changeover occurs whenever an outcome dif and only ifers from the cone preceding it. For instance, if the results of the flips are $H\ H\ T\ H\ T\ H\ H\ T$, then there are a total of five changeovers. If $p = \frac{1}{2}$, what is the probability there are k changeovers?
Each flip after the first will, independently, result in a changeover with probability $\frac{1}{2}$. Therefore,

$$P(k \text{ changeovers}) = \binom{n-1}{k}\left(\frac{1}{2}\right)^{n-1}.$$

If p is arbitrary, compute the expected number of changeovers.
Let

$$X_i = \begin{cases} 1, & \text{if a change over results from the } i^{\text{th}} \text{ flip} \\ 0, & \text{otherwise} \end{cases}.$$

Then

$$\text{number of changeovers} = \sum_{i=2}^{n} X_i.$$

As,

$$E[X_i] = P(X_i = 1) = P(\text{flip } i-1 \neq i) = 2p(1-p).$$

Then

$$E[\text{number of changeovers}] = \sum_{i=2}^{n} E[X_i] = 2(n-1)p(1-p).$$

Example 1.38. An urn contains $n + m$ balls, of which n are red and m are black. They are withdrawn from the urn, one at a time and without replacement. Let X be the number of red balls removed before the first black ball is chosen. To obtain this quantity, number the red balls from 1 to n. Now let

$$X_i = \begin{cases} 1, & \text{if red ball } i \text{ is taken before any black ball is chosen} \\ 0, & \text{otherwise} \end{cases}.$$

Then

$$X = \sum_{i=1}^{n} X_i,$$

$$E[X_i] = P(X_i = 1) = \frac{1}{n+1},$$

since each of these $n + 1$ balls is equally likely to be the one chosen earliest.
Therefore,

$$E[X] = \frac{n}{n+1}.$$

Example 1.39. A total of r keys are to be put, one at a time, in k boxes, with each key independently being put in box i with probability p_i, $\sum_{i=1}^{k} p_i = 1$. Each time a key is put in a nonempty box, we say that a colllision occurs. Find the expected number of collisions.

Let N_i denote the number of keys in box i, $\forall\ i \in [k]$. Then, with X equal to the number of collisions we have that

$$X = \sum_{i=1}^{k} (N_i - 1)^+ = \sum_{i=1}^{k} (N_i - 1 + \mathbb{1}_{\{N_i = 0\}}).$$

Then

$$E[X] = \sum_{i=1}^{k} \left(rp_i - 1 + (1 - p_i)^r \right).$$

Example 1.40. If X is a nonnegative integer valued random variable, show that

$$E[X] = \sum_{n=1}^{\infty} P(X \geq n) = \sum_{n=0}^{\infty} P(X > n).$$

Note

$$X = \sum_{n=1}^{\infty} \mathbb{1}_{\{X \geq n\}}.$$

Then

$$E[X] = \sum_{n=1}^{\infty} P(X \geq n) = \sum_{n=0}^{\infty} P(X \geq n+1) = \sum_{n=0}^{\infty} P(X > n).$$

If X and Y are both nonnegative integer valued random variables, show that

$$E[XY] = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} P(X \geq n, Y \geq m).$$

Note

$$XY = \sum_{n=1}^{\infty} \mathbb{1}_{\{X \geq n\}} \sum_{m=1}^{\infty} \mathbb{1}_{\{Y \geq m\}} = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} \mathbb{1}_{\{X \geq n\}} \mathbb{1}_{\{Y \geq m\}} = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} \mathbb{1}_{\{X \geq n, Y \geq m\}}.$$

Then

$$E[XY] = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} E\left[\mathbb{1}_{\{X \geq n, Y \geq m\}}\right] = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} P(X \geq n, Y \geq m).$$

Example 1.41. Consider three trials, each of which is either a success or not. Let X denote the number of successes. Suppose that $E[X] = 1.8$.

(a) What is the largest possible value of $P(X = 3)$?

Let

$$X_i = \begin{cases} 1, & \text{if trial } i \text{ is a success} \\ 0, & \text{otherwise} \end{cases}.$$

If $X_1 = X_2 = X_3$, then

$$1.8 = E[X] = 3E[X_1] = 3P(X_1 = 1).$$

Then

$$P(X = 3) = P(X_1 = 1) = 0.6.$$

This is the largest value since by Markov's inequality,

$$P(X \geq 3) \leq \frac{E[X]}{3} = 0.6.$$

(b) What is the smallest possible value of $P(X = 3)$? Let

$$U \sim \text{Uni}(0, 1).$$

Define

$$X_1 = \begin{cases} 1, & \text{if } U \leq 0.6 \\ 0, & \text{otherwise} \end{cases} \qquad X_2 = \begin{cases} 1, & \text{if } U \geq 0.6 \\ 0, & \text{otherwise} \end{cases} \qquad X_3 = \begin{cases} 1, & \text{if } U \leq 0.3 \text{ or } U \geq 0.7 \\ 0, & \text{otherwise} \end{cases}$$

Then

$$P(X_1 = X_2 = X_3 = 1) = 0.$$

Example 1.42. A coin, having probability p of landing heads, is flipped until a head appears for the rth time. Let N denote the number of flips required. Calculate $E[N]$.

$$N = \sum_{i=1}^{r} X_i,$$

where X_i is the number of flips between the $(i-1)^{\text{th}}$ and i^{th} head. Hence, X_i is geometric with mean $\frac{1}{p}$. Thus,

$$E[N] = \frac{r}{p}.$$

Example 1.43. Let X and Y each take on either the value 1 or -1. Let

$$p(1,1) = P(X = 1, Y = 1),$$
$$p(1,-1) = P(X = 1, Y = -1),$$
$$p(-1,1) = P(X = -1, Y = 1),$$
$$p(-1,-1) = P(X = -1, Y = -1).$$

Suppose that $E[X] = E[Y] = 0$. Then

$$p(1,1) = p(-1,-1);$$
$$p(1,-1) = p(-1,1);$$

since

$$0 = E[X + Y] = 2p(1,1) - 2p(-1,-1),$$
$$0 = E[X - Y] = 2p(1,-1) - 2p(-1,1).$$

Also,

$$\text{Var}(X) = E[X^2] = 1 \cdot (P(X = 1) + P(X = -1)) = 1,$$

and since $p(1,1) + p(1,-1) + p(-1,1) + p(-1,-1) = 1$, we have

$$1 = 2p(1,1) + p(1,-1).$$

Example 1.44. Suppose that the joint probability mass function of X and Y is

$$P(X = i, Y = j) = \binom{j}{i} e^{-2\lambda} \lambda^j / j!, \ 0 \le i \le j.$$

Then

$$X \sim \text{Poi}(\lambda).$$

$$Y \sim \text{Poi}(2\lambda).$$

Note

$$P(X = i, Y - X = k) = P(X = i, Y = k + i) = e^{-\lambda} \frac{\lambda^i}{i!} e^{-\lambda} \frac{\lambda^k}{k!},$$

showing that X and $Y - X$ are independent Poisson random variables with mean λ. Hence,

$$P(Y - X = k) = e^{-\lambda} \frac{\lambda^k}{k!}.$$

Example 1.45. An urn contains $2n$ balls, of which r are red. The balls are randomly removed in n successive pairs. Let X denote the number of pairs in which both balls are red.
Let

$$X_i = \begin{cases} 1, & \text{if both balls of the } i^{\text{th}} \text{ withdrawn pair are red} \\ 0, & \text{otherwise.} \end{cases}.$$

Then

$$E[X_i] = P(X_i = 1) = \frac{r(r-1)}{2n(2n-1)}.$$

$$E[X_i X_j] = \frac{\binom{r}{4}}{\binom{2n}{4}}.$$

Example 1.46. Let X_1, X_2, X_3 and X_4 be independent continuous random variables with a common distribution function F and let

$$p = P(X_1 < X_2 > X_3 < X_4).$$

(a) Argue that the value of p is the same for all continuous distribution functions F.
Use the fact that $F(X_i)$ is a uniform $(0, 1)$ random variable to obtain

$$p = P\left(F(X_1) < F(X_2) > F(X_3) < F(X_4)\right) = P(U_1 < U_2 > U_3 < U_4)$$

where the $U_i, \ \forall \ i \in [4]$ are independent uniform $(0, 1)$ random variables.

(b) Find p by integrating the joint density function over the appropriate region,

$$\begin{aligned} p &= \int_0^1 \int_{x_1}^1 \int_0^{x_2} \int_{x_3}^1 dx_4 dx_3 dx_2 dx_1 \\ &= \frac{5}{24}. \end{aligned}$$

(c) Find p by using the fact that all 4! possible orderings of $X_1, \ldots, X_4$ are equally likely.
There are 5 ordering such that $X_1 < X_2 > X_3 < X_4$. They are as follows:

$$X_2 > X_4 > X_3 > X_1,$$

$$X_2 > X_4 > X_1 > X_3,$$

$$X_2 > X_1 > X_4 > X_3,$$

$$X_4 > X_2 > X_3 > X_1,$$

$$X_4 > X_2 > X_1 > X_3.$$

Example 1.47. In deciding upon the approprite premium to charge, insurance companies sometimes use the exponential principle, defined as follows. With X as the random amount we will have to pay in claims, the premium charged by the insurance company is

$$P = \frac{1}{a} \log\left(E\left[e^{aX}\right]\right),$$

where a is some specified positive constant. Find P when $X \sim \exp(1/\lambda)$ and $a = \alpha\lambda$, where $0 < \alpha < 1$.

$$E\left[e^{\alpha\lambda X}\right] = \int e^{\alpha\lambda x} \lambda e^{-\lambda x} dx = \frac{1}{1-\alpha}.$$

Therefore,

$$P = -\frac{1}{\alpha\lambda} \log(1-\alpha).$$

The inequality $\log(1-x) \leq -x,\ \forall\ 0 < x < 1$ shows that $P \geq \frac{1}{\lambda}$.

Example 1.48. Let X and Y be independent normal random variables, each having parameters μ and σ^2. Show that $X+Y$ is independent of $X-Y$.
The joint MGF

$$\begin{aligned} M_{X+Y,X-Y} &= E\left[e^{t_1(X+Y)+t_2(X-Y)}\right] = E\left[e^{(t_1+t_2)X+(t_1-t_2)Y}\right] \\ &= E\left[e^{(t_1+t_2)X}\right] E\left[e^{(t_1-t_2)Y}\right] = e^{\mu(t_1+t_2)+\frac{\sigma^2(t_1+t_2)^2}{2}} e^{\mu(t_1-t_2)+\frac{\sigma^2(t_1-t_2)^2}{2}} \\ &= e^{2\mu t_1+\sigma^2(t_1^2+t_2^2)} = e^{(2\mu)t_1+\frac{2\sigma^2 t_1^2}{2}} e^{\frac{2\sigma^2 t_2^2}{2}}. \end{aligned}$$

Remark. Since the linear combination of the independent normal random variables is also normally distributed,

$$M_{X_1,\ldots,X_n}(t_1,\ldots,t_n) = \exp\left\{\vec{\mu}^T\vec{t} + \frac{1}{2}\vec{t}^T\Sigma\vec{t}\right\} = \exp\left\{\sum_{i=1}^{n} t_i\mu_i + \frac{1}{2}\sum_{i=1}^{m}\sum_{j=1}^{m} t_i t_j \mathrm{Cov}(X_i, X_j)\right\},$$

where

$$X_i \sim N(\mu_i, \sigma^2).$$

Remark. If $M_{X_1,\ldots,X_n}(t_1,\ldots,t_n)$ is the joint MGF of $X_1,\ldots,X_n$, then

$$M_{X_i}(t) = M_{X_1,\ldots,X_n}(0,\ldots,1\cdots,0).$$

Remark. $X_1, \ldots, X_n$ are independent if and only if

$$M_{X_1,\ldots,X_n}(t_1, \ldots, t_n) = M_{X_1}(t_1) \cdots M_{X_n}(t_n), \ \forall \ t_1, \ldots, t_n.$$

One direction is obvious.
"$\Leftarrow$". Then

$$M_{X_1,\ldots,X_n}(t, \ldots, t) = M_{X_1}(t) \cdots M_{X_n}(t).$$

Then the joint MGF is that of the sum of n independent random variables the ith of which has the same distribution as X_i. As the joint MGF uniquely determines the joint distribution, the result follows.

Example 1.49. Let $X_1, X_2, \cdots$ be a sequence of independent identically distributed continuous random varaibles. We say that a record occurs at time n if $X_n > \max(X_1, \ldots, X_{n-1})$.
As the random variables are independent, identically distributed, and continuous, it follows that, with probability 1, they will all have different values. Hence the largest of $X_1, \ldots, X_n$ is equally likely to be either X_1 or $X_2, \cdots$ or X_n. Then

$$P(\text{a record occurs at } n) = \frac{1}{n}.$$

$$E[\text{number of records by time } n] = E\left[\sum_{i=1}^{n} E[\mathbb{1}_{\{\text{a record occurs at time } i\}}]\right] = \sum_{i=1}^{n} \frac{1}{i}.$$

It is easy to see that the random variables $\mathbb{1}_{\{\text{a record occurs at time } i\}}, \ \forall \ i \in [n]$ are independent. For instance, for $j < k$, knowing that X_k is the largest of $X_1, \ldots, X_j, \ldots, X_k$ clearly tells us nothing about whether or not X_j is the largest of $X_1, \ldots, X_j$. Hence,

$$\text{Var}(\text{number of records by time } n) = \sum_{i=1}^{n} \text{Var}\left(\mathbb{1}_{\{\text{a record occurs at time } i\}}\right) = \sum_{i=1}^{n} \frac{1}{i}\frac{i-1}{i}.$$

Let

$$N = \min\{n : n > 1 \text{ and a record occurs at time } n\}.$$

Then

$$P(N > n) = P(X_1 \text{ is the largest of } X_1, \ldots, X_n) = \frac{1}{n}.$$

Hence,

$$E[N] = \sum_{n=0}^{\infty} P(N > n) = 1 + \sum_{n=1}^{\infty} \frac{1}{n} = \infty.$$

Example 1.50. Let $a_1 < a_2 < \cdots < a_n$ denote a set of n numbers, and consider any permutation of these numbers. We say there is an inversion of a_i and a_j in the permutation if $i < j$ and a_j precedes a_i. Consider now a random permutation of $a_1, \ldots, a_n$−in the sense that each of the $n!$ permutations is equally likely to be chosen−and let N denote the number of inversions in this permutation. Also, let

$$N_i = \{\text{number of } k : \ k < i, a_i \text{ precedes } a_k \text{ in the permutation}\}, \ \forall \ i \in [n].$$

Then

$$N = \sum_{i=1}^{n} N_i.$$

Show $N_1, \ldots, N_n$ are independent random variables.

Knowing the values of $N_1, \ldots, N_j$ is equivalent to knowing the relative ordering of the elements $a_1, \ldots, a_j$. The independence result follows for clearly the number of $a_1, \ldots, a_j$ that follow a_{j+1} does not probabilistically depend on the relative ordering of $a_1, \ldots, a_j$.
Also,

$$P(N_i = k) = \frac{1}{i}, \ \forall \ k = 0, \ldots, i-1,$$

which follows since of the elements $a_1, \ldots, a_{j+1}$, the elements a_{j+1} is equally likely to be the first or second or $\cdots$ or $(i+1)^{\text{th}}$.

Chapter 2

Conditional Probability and Conditional Expectation

One of the most useful concepts in probability theory is that of conditional probability and conditional probability expectation. The reason is twofold. First, in practice we are often interested in calculating probabilities and expectations when some partial information is available; hence, the desired probabilities and expectations are conditional ones. Secondly, in calculation a desired probability or expectation it is often extremenly useful to first "condition" on some appropriate random variable.

2.1 The discrete case

Definition 2.1. If X and Y are discrete, the conditional probability mass function of X given $Y = y$, $\forall\ y$ such that $P(Y = y) > 0$,

$$p_{X|Y=y}(x) = P(X = x|Y = y) = \frac{P(X = x, Y = y)}{P(Y = y)} = \frac{p(x, y)}{p_Y(y)}.$$

The conditional cdf is, given $Y = y$, $\forall\ y$ such that $P(Y = y) > 0$,

$$F_{X|Y=y}(x) = P(X \leq x|Y = y) = \sum_{a \leq x} P_{X|Y}(a|y).$$

The conditional expectation, given $Y = y$, $\forall\ y$ such that $P(Y = y) > 0$,

$$E[X|Y = y] = \sum_x xP(X = x|Y = y) = \sum_x xP_{X|Y}(x|y).$$

In other words, the definition are exactly as before with the exception that everything is now conditional on the event that $Y = y$.

2.2 The continuous case

Definition 2.2. If X and Y have a joint pdf $f(x,y)$, then the conditional pdf of X, given that $Y=y,\ \forall\ y$ such that $f_Y(y)>0$,

$$f_{X|Y=y}(x)=\frac{f(x,y)}{f_Y(y)}.$$

To motivate this definition, multiply the left side by dx and the right side by $\frac{dxdy}{dy}$ to get

$$\begin{aligned}
f_{X|Y=y}(x)dx &= \frac{f(x,y)dxdy}{f_Y(y)dy}\\
&\approx \frac{P(x\le X\le x+dx, y\le Y\le y+dy)}{P(y\le Y\le y+dy)}\\
&= P(x\le X\le x+dx|y\le Y\le y+dy).
\end{aligned}$$

The conditional expectation of X, given $Y=y,\ \forall\ y$ such that $f_Y(y)>0$,

$$E[X|Y=y]=\int_{-\infty}^{\infty} x f_{X|Y=y}(x)dx.$$

Example 2.3 (The t-distribution)**.** Let

$$T=\frac{Z}{\sqrt{Y/n}},$$

where

$$Z\sim N(0,1),\ Y\sim\chi^2(n).$$

Given $Y=y$,

$$T|Y=y=\frac{Z}{\sqrt{y/n}}\sim N(0,n/y).$$

Then

$$f_{T|Y=y}(t)=\frac{1}{\sqrt{2\pi n/y}}e^{-t^2y/n}=\frac{y^{1/2}}{\sqrt{2\pi n}}e^{-t^2y/2n},\ \forall\ t\in\mathbb{R}.$$

Note

$$f_Y(y)=\frac{y^{n/2-1}e^{-y/2}}{\Gamma(n/2)2^{n/2}},\ \forall\ y\in\mathbb{R}.$$

We obtain the density function of T:

$$\begin{aligned}
f_T(t) &= \int_0^\infty f_{T,Y}(t,y)dy=\int_0^\infty f_{T|Y=y}(t)f_Y(y)dy\\
&= \frac{1}{\sqrt{\pi n}2^{(n+1)/2}\Gamma(n/2)}\int_0^\infty e^{-(t^2+n)/(2n)y}y^{(n-1)/2}dy\\
&= \frac{\Gamma\left(\frac{n+1}{2}\right)}{\sqrt{\pi n}\Gamma\left(\frac{n}{2}\right)}\left(1+\frac{t^2}{n}\right)^{-(n+1)/2},\ \forall\ t\in\mathbb{R}.
\end{aligned}$$

Example 2.4. Let X_1 and X_2 be independent exponential random variables with rates μ_1 and μ_2. Find the conditional density of X_1 given that $X_1 + X_2 = t$.
By transformation theorem,

$$f_{X_1,X_1+X_2}(x,t) = \mu_1 e^{-\mu_1 x}\mu_2 e^{-\mu_2(t-x)},\ \forall\ 0 < x < t.$$

Then

$$\begin{aligned} f_{X_1|X_1+X_2=t}(x) &= \frac{f_{X_1,X_1+X_2}(x,t)}{f_{X_1+X_2}(t)} \\ &= \frac{\mu_1 e^{-\mu_1 x}\mu_2 e^{-\mu_2(t-x)}}{f_{X_1+X_2}(t)} \\ &= C(t)e^{-(\mu_1-\mu_2)x},\ \forall\ 0 < x < t, \end{aligned}$$

where

$$C = \frac{\mu_1\mu_2 e^{-\mu_2 t}}{f_{X_1+X_2}(t)}.$$

Now, if $\mu_1 = \mu_2$, then

$$f_{X_1|X_1+X_2}(x|t) = C,\ \forall\ 0 < x < t,$$

yielding that $C = 1/t$, and that

$$X_1|X_1 + X_2 = t \sim \text{Uni}(0,t).$$

If $\mu_1 \neq \mu_2$, then we use

$$1 = \int_0^t f_{X_1|X_1+X_2}(x|t)dt = \frac{C}{\mu_1-\mu_2}\left(1 - e^{-(\mu_1-\mu_2)t}\right)$$

to obtain

$$C = \frac{\mu_1-\mu_2}{1-e^{-(\mu_1-\mu_2)t}},$$

thus yielding the result:

$$f_{X_1|X_1+X_2}(x|t) = \frac{(\mu_1-\mu_2)e^{-(\mu_1-\mu_2)x}}{1-e^{-(\mu_1-\mu_2)t}}.$$

An byproduct is

$$f_{X_1+X_2}(t) = \frac{\mu_1\mu_2 e^{-\mu_2 t}}{C} = \begin{cases} \mu^2 t e^{-\mu t}, & \text{if } \mu_1 = \mu_2 = \mu \\ \frac{\mu_1\mu_2\left(e^{-\mu_2 t} - e^{-\mu_1 t}\right)}{\mu_1-\mu_2}, & \text{if } \mu_1 \neq \mu_2 \end{cases}.$$

2.3 Computing expectation by conditioning

Let us denote $E[X|Y]$ that function of the random variable Y whose value at $Y = y$ is $E[X|Y = y]$. Note that $E[X|Y]$ is a random variable.

Theorem 2.5.

$$E[X] = E\left[E[X|Y]\right].$$

If Y is a discrete random variable, then

$$E[X] = \sum_y E[X|Y=y]P(Y=y);$$

If Y is a continuous random variable, then

$$E[X] = \int_{-\infty}^{\infty} E[X|Y=y]f_Y(y)dy.$$

Example 2.6 (The mean of a Geometric Distribution)**.** A coin, having probability p of coming heads, is to be successively flipped until the first head appears. What is the expected number of flips required?
Let N be the number of flips required, and let

$$Y_1 = \begin{cases} 1, & \text{if the first flip results in a head} \\ 0, & \text{if the first flip results in a tail} \end{cases}.$$

Now

$$\begin{aligned} E[N] &= E[N|Y_1=1]P(Y_1=1) + E[N|Y_1=0]P(Y_1=0) \\ &= p\cdot 1 + (1-p)(1+E[N]). \end{aligned}$$

So

$$E[N] = \frac{1}{p}.$$

To see why this equality holds, observe that,

$$N|Y_1 = 0 = \inf\{n \geq 2 : X_n = 1\} = 1 + (\inf\{n \geq 2 : X_n = 1\} - 1) \stackrel{d}{=} 1 + N.$$

Example 2.7 (Multinormial Covariance)**.** Consider n independent trial, each of which results in one of the outcomes $1, \ldots, r$, with respective probabilities $p_1, \ldots, p_r$, $\sum_{i=1}^r p_i = 1$. If we let N_i denote the number of trials that result in outcome i, then $(N_1, \ldots, N_r)$ is said to have a multinormial distribution. For $i \neq j$, let us compute

$$\mathrm{Cov}(N_i, N_j) = E[N_iN_j] - E[N_i]E[N_j].$$

Note

$$N_i \sim \mathrm{Binom}(n, p_i),$$

gives that $E[N_i]E[N_j] = n^2p_ip_j$.

$$\begin{aligned} E[N_iN_j] &= \sum_{k=0}^n E[N_iN_j|N_i=k]P(N_i=k) \\ &= \sum_{k=0}^n kE[N_j|N_i=k]P(N_i=k). \end{aligned}$$

Now, given that k of the n trials result in outcome i, each of the other $n-k$ trials independently results in outcome j with probability

$$P(j|\text{not } i) = \frac{p_j}{1-p_i},$$

thus showing

$$N_j|N_i = k \sim \text{Binom}\left(n-k, \frac{p_j}{1-p_i}\right).$$

Using this yields

$$\begin{aligned}
E[N_iN_j] &= \sum_{k=0}^{n} k(n-k)\frac{p_j}{1-p_i}P(N_i=k) \\
&= \frac{p_j}{1-p_i}\left(n\sum_{k=0}^{n} kP(N_i=k) - \sum_{k=0}^{n} k^2P(N_i=k)\right) \\
&= \frac{p_j}{1-p_i}\left(nE[N_i] - E[N_i^2]\right) \\
&= \frac{p_j}{1-p_i}\left[n^2p_i - np_i(1-p_i) - n^2p_i^2\right] \\
&= n(n-1)p_ip_j,
\end{aligned}$$

which yields the result

$$\text{Cov}(N_i, N_j) = -np_ip_j.$$

Example 2.8. (a) Independent trials, each of which is a success with probability p, are performed until there are k consecutive successes. What is the mean number of necessary trials?
Let M_k denote the expected number of necessary trials to obtain k consecutive success, then

$$M_k = M_{k-1} + p\cdot 1 + (1-p)(1+M_k),$$

or

$$M_k = \frac{1}{p} + \frac{M_{k-1}}{p}.$$

Clearly,

$$M_1 = \frac{1}{p}.$$

Recursively,

$$M_k = \frac{1}{p} + \cdots + \frac{1}{p^k}.$$

(b) Suppose we continue rolling a fair, six-sided die until we see the same outcome k times in a row. Let M_k be the expected total number of times we roll the die. Then

$$M_k = M_{k-1} + p\cdot 1 + (1-p)(1+M_k).$$

Clearly,

$$M_1 = 1.$$

Recursively,

$$M_k = 1 + \frac{1}{p} + \cdots + \frac{1}{p^{k-1}}.$$

Or Let N_k denote the time until the same outcome occurs k consecutive times, consider

$$E[N_k] = E\left[E[N_k|N_{k-1}]\right].$$

Now

$$E[N_k|N_{k-1}] = N_{k-1} + 1 \cdot p + E[N_k] \cdot (1-p).$$

Then

$$E[N_k] = 1 + \frac{1}{p}E[N_{k-1}] = \cdots = 1 + \frac{1}{p} + \cdots + \frac{1}{p^{k-1}}.$$

2.3.1 Computing variances by conditioning

Definition 2.9. The conditional variance of X given that $Y = y$ is defined by

$$\mathrm{Var}(X|Y=y) = E\left[(X - E[X|Y=y])^2 \middle| Y = y\right],$$

or

$$\mathrm{Var}(X|Y) = E\left[(X - E[X|Y])^2 \middle| Y\right].$$

That is, the conditional variance is defined in exactly the same manner as the ordinary variance with the exception that all probabilities are determined conditional on the event that $Y = y$. Expanding the right side of the preceding and taking expectation term by term yields

$$\mathrm{Var}(X|Y=y) = E[X^2|Y=y] - (E[X|Y=y])^2,$$

or

$$\mathrm{Var}(X|Y) = E[X^2|Y] - (E[X|Y])^2.$$

Theorem 2.10 (The conditional variance formula)**.**

$$\mathrm{Var}(X) = E\left[\mathrm{Var}(X|Y)\right] + \mathrm{Var}\left(E[X|Y]\right).$$

2.4 Computing probabilities by conditioning

Not only can we obtain expectations by first conditioning on an appropriate random varaible, but we may also use this approach to compute probabilities. To see this, let E denote an arbitrary event and define the indicator random variable X by

$$X = \begin{cases} 1, & \text{if } E \text{ occurs} \\ 0, & \text{if } E \text{ does not occurs} \end{cases}.$$

Then

$$E[X] = P(E),$$

$$E[X|Y=y] = P(E|Y=y), \ \forall \ r.v. \ Y.$$

Then if Y is discrete,

$$P(E) = \sum_y P(E|Y = y)P(Y = y);$$

if Y is continuous,

$$P(E) = \int_{-\infty}^{\infty} P(E|Y = y)f_Y(y).$$

Example 2.11. Suppose that the number of people who visit a yoga studio each day is a Poisson random variable with mean λ. Suppose further that each person who visits is, independently, female with probability p or male with probability $1 - p$. Find the joint probability that exactly n women and m men visit the academy today.
Let N_1, N_2 denote the number of women and men who visit the academy today, respectively. let $N = N_1 + N_2$ be the total number of people who visit. Then

$$\begin{aligned}
P(N_1 = n, N_2 = m) &= \sum_{i=0}^{\infty} P(N_1 = n, N_2 = m|N = i)P(N = i) \\
&= P(N_1 = n, N_2 = m|N = n + m)e^{-\lambda}\frac{\lambda^{n+m}}{(n + m)!} \\
&= \binom{n + m}{n} p^n(1 - p)^m e^{-\lambda}\frac{\lambda^{n+m}}{(n + m)!} \\
&= e^{-\lambda p}\frac{(\lambda p)^n}{n!} \cdot e^{-\lambda(1-p)}\frac{(\lambda(1 - p))^m}{m!}.
\end{aligned}$$

So N_1 and N_2 are independent, where

$$N_1 \sim \text{Poi}\,(p\lambda)\,,$$

$$N_2 \sim \text{Poi}\,((1 - p)\lambda)\,.$$

The result generalizes to the case where each of a Poisson distributed number of events, N, with mean λ is independently classified as being one of k types, with the probability that it is type i being p_i, $\forall\ i \in [k]$, $\sum_{i=1}^k p_i = 1$. If N_i is the number that are classified as type i, then $N_1, \ldots, N_k$ are independent Poisson random variables with respective means $\lambda p_1, \ldots, \lambda p_k$. This follows, since for $n = \sum_{i=1}^k n_i$,

$$\begin{aligned}
P(N_1 = n_1, \ldots, N_k = n_k) &= P(N_1 = n_1, \ldots, N_k = n_k|N = n)P(N = n) \\
&= \frac{n!}{n_1! \cdots n_k!} p_1^{n_1} \cdots, p_k^{n_k} e^{-\lambda}\lambda^n/n! \\
&= \prod_{i=1}^{n} e^{-\lambda p_i}(\lambda p_i)^{n_i}/n_i!,
\end{aligned}$$

where the second equality used that, given a total of n events, the numbers of each type has a multinormial distribution with parameters $(n, p_1, \ldots, p_k)$.

Example 2.12 (The distribution of the sum of independent Bernoulli random variables)**.** Let $X_1, \ldots, X_n$ be independent Bernoulli random variabes, with X_i having parameter p_i, $\forall\ i \in [n]$.

Find the pmf of $X_1 + \cdots X_n$.
Define

$$p_k(j) = P(X_1 + \cdots + X_k = j),\ \forall\ k \in [n], j \in [k].$$

Note

$$p_k(k) = \prod_{i=1}^{k} p_i,\ p_k(0) = \prod_{i=1}^{k}(1 - p_i).$$

For $0 < j < k$,

$$\begin{aligned} p_k(j) &= P(X_1 + \cdots X_k = j | X_k = 1)p_k + P(X_1 + \cdots X_k = j | X_k = 0)q_k \\ &= P(X_1 + \cdots X_{k-1} = j - 1 | X_k = 1)p_k \\ &\quad + P(X_1 + \cdots + X_{k-1} = j | X_k = 0)q_k \\ &= P(X_1 + \cdots X_{k-1} = j - 1)p_k + P(X_1 + \cdots + X_{k-1} = j)q_k \\ &= p_k p_{k-1}(j - 1) + q_k p_{k-1}(j). \end{aligned}$$

Starting with $p_1(1) = p_1$, $p_1(0) = 1 - p_1$, the preceding equations can be recursively solved to obtain $p_2(j), p_3(j)$, up to $p_n(j)$.

Example 2.13 (The best prize problem)**.** Suppose that we are to be presented with n distinct prizes in sequence. After being presented with a prize we must immediately decide whether to accept it or reject it and consider the next prize. The only information we are given when deciding whether to accept a prize is the relative rank of that prize compared to ones already seen. That is, for instance, when the fifth prize is presented we learn how it compares with the first four prizes already seen. Suppose that once a prize is rejected it is lost, and that our objective is to maximize the probability of obtaining the best prize. Assuming that all $n!$ orderings of the prizes are equally likely, how well can we do?
To see this, fix a value $k, 0 \le k < n$, and consider the stategy that rejects the first k prizes and then accepts the first one that is better than all of those first k. Let $P_k(\text{best})$ denote the probability that the best prize is selected when this stategy is employed. To compute this probability, condition on X, the position of the best prize. This gives

$$P_k(\text{best}) = \sum_{i=1}^{n} P_k(\text{best}|X = i)P(X = i) = \frac{1}{n}\sum_{i=1}^{n} P_k(\text{best}|X = i).$$

Note

$$P_k(\text{best}|X = i) = 0,\ \forall\ i \le k.$$

$$P_k(\text{best}|X = i) = P(\text{best of first } i - 1 \text{ is among the first } k) = \frac{k}{i - 1},\ \forall\ i > k.$$

Then

$$\begin{aligned} P_k(\text{best}) &= \frac{k}{n}\sum_{i=k+1}^{n}\frac{1}{i - 1} \approx \frac{k}{n}\int_{k}^{n-1}\frac{1}{x}dx \\ &= \frac{k}{n}\log\left(\frac{n - 1}{k}\right) \approx \frac{k}{n}\log\left(\frac{n}{k}\right). \end{aligned}$$

Take derivative, we have

$$x = \frac{n}{e}.$$

Example 2.14. At a party n men take off their hats. The hats are then mixed up and each man randomly selects one. We say that a match occurs if a man selects his own hat. What is the probability of no mathes? What is the probability of exactly k matches?
Let E denote the event that no matches occur, and to make explicit the dependence on n, write $P_n = P(E)$. Let

$$M = \mathbb{1}_{\{\text{the first man selects his own hat}\}}.$$

Then

$$P_n = P(E) = P(E|M)P(M) + P(E|M^c)P(M^c) = P(E|M^c)\frac{n-1}{n}.$$

Now $P(E|M^c)$ is the probability of no matches when $n-1$ men select from a set of $n-1$ hats that does not contain the hat of one of these men. This can happen in either of two mutually exclusive ways. Either there are no matches and the extra man does not select extra hat with probability P_{n-1}, which is seen by rearding the extra hat as "belonging" to the extra man. or there are no matches and the extra man does select the extra hat with probability $\frac{1}{n-1}P_{n-2}$.
So

$$P(E|M^c) = P_{n-1} + \frac{1}{n-1}P_{n-2}.$$

Thus,

$$P_n = \frac{n-1}{n}P_{n-1} + \frac{1}{n}P_{n-2},$$

or, equivalently,

$$P_n - P_{n-1} = -\frac{1}{n}(P_{n-1} - P_{n-2}).$$

Clearly,

$$P_1 = 0, \quad P_2 = \frac{1}{2}.$$

Revursively, we have

$$P_n = \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \cdots + \frac{(-1)^n}{n!}.$$

To obtain the probability of exactly k matches, we consider any fixed group of k men. The probability that they, and only they, select their own hat is

$$\frac{1}{n}\frac{1}{n-1}\cdots\frac{1}{n-(k-1))}P_{n-k} = \frac{(n-k)!}{n!}P_{n-k}.$$

Because there are $\binom{n}{k}$ choices of a set of k men, the desired probability of exactly k matches is

$$\frac{P_{n-k}}{k!} = \frac{\frac{1}{2!} - \frac{1}{3!} + \cdots + \frac{(-1)^{n-k}}{(n-k)!}}{k!},$$

which for n large, is approximately equal to $\frac{e^{-1}}{k!}$.
The recursive equation, could also have beed obtained by using the concept of a cycle, where we say the sequence of distinct individuals $i_1, \ldots, i_k$ constitutes a cycle if i_1 chooses i_2's hat, $\cdots$, i_{k-1} chooses i_k's hat, and i_k chooses i_1's hat. Note that every individual is part of a cycle. Let E be the event that no matches occur, then

$$P_n = P(E) = \sum_{k=1}^{n} P(E|C=k)P(C=k),$$

where C is the size of the cycle that contains person 1. Now call person 1 the first person, then

$$P(C=k)=\frac{n-1}{n}\frac{n-2}{n-1}\cdots\frac{n-k+1}{n-k+2}\cdot\frac{n-k+1}{1}=\frac{1}{n}.$$

Also,

$$P(E|C=1)=0.$$

On the other hand, if $C=k$, the set of hats chosen by the k individuals in this cycle is exactly the set of hats of these individuals. Then

$$P(E|C=k)=P_{n-k}.$$

Thus,

$$P_n=\frac{1}{n}\sum_{k=2}^{n}P_{n-k}.$$

Example 2.15 (The Ballot Problem)**.** In an election, candidate A receives n votes, and candidate B receives m votes where $n>m$. Assuming that all orderings are equally likely, show that the probability that A is always ahead in the count of votes is $\frac{n-m}{n+m}$.
Let $P_{n,m}$ denote the desired probability. Then

$$\begin{aligned}P_{n,m}&=P(A\text{ always ahead}|A\text{ receives last vote})\frac{n}{n+m}\\&\quad+P(A\text{ always ahead}|B\text{ receives last vote})\frac{m}{n+m}\\&=\frac{n}{n+m}P_{n-1,m}+\frac{m}{n+m}P_{n,m-1},\end{aligned}$$

which proves the result by induction.
The ballot problem has some interesting applications. For example, consider successive flips of a coin that always land on "heads" with probability p, let us determine the probability distribution of the first time, after beginning, that the total number of heads is equal to the total number of tails. By conditioning on whomever receives the last vote,

$$\begin{aligned}&P(\text{first time equal }2n)\\&=P(\text{first time equal }2n|n\text{ heads in first }2n)\binom{2n}{n}p^n(1-p)^n.\\&=\left(\frac{n}{n-1+n}+\frac{n-1}{n-1+n}\right)P_{n,n-1}\binom{2n}{n}p^n(1-p)^n\\&=\frac{\binom{2n}{n}p^n(1-p)^n}{2n-1}.\end{aligned}$$

2.5 Further properties of conditional expectation

$$E[X|Y]=E\left[E[X|Y,W]|Y\right].$$

In the case where Y and W are both discrete,

$$E[X|Y=y]=\sum_{w\in S_w}E[X|Y=y,W=w]P(W=w|Y=y),$$

and in the case where Y and W are jointly continuous,

$$E[X|Y=y] = \int_{-\infty}^{\infty} E[X|Y=y, W=w] f_{W|Y=y}(w)dw.$$

An even more general version can also be state in the following proposition.

Proposition 2.16.

$$E[X|Y_1, \ldots, Y_m] = E\left[E[X|Y_1, \ldots, Y_m, W_1, \ldots, W_n]|Y_1, \ldots, Y_m\right].$$

Example 2.17. An automobile insurance comany classifies each of its policyholders as being of one of the types $i = 1, \ldots, k$. It supposes that the numbers of accidents that a type i policyholder has in successive years are independent Poisson random variables with mean λ_i, $\forall\ i \in [k]$. The probability that a newly insured policyholder is type i is p_i, $\sum_{i=1}^{k} p_i = 1$. Given that a policyholder had n accidents in her first year, what is the expected number that she has in her second year? What is the conditional probability that she has m accidents in her second year?
Let N_i denote the number of accidents the policyholder has in year i, $i = 1, 2$. Let T be her risk type.
Then

$$\begin{aligned}
E[N_2|N_1 = n] &= \sum_{j=1}^{k} E[N_2|N_1 = n, T = j]P(T = j|N_1 = n) \\
&= \sum_{j=1}^{k} E[N_2|T = j]P(T = j|N_1 = n) \\
&= \sum_{j=1}^{k} \lambda_j P(T = j|N_1 = n),
\end{aligned}$$

where $P(T = j|N_1 = n)$ can be found by the Bayesian formula.
Similarly,

$$\begin{aligned}
P(N_2 = m|N_1 = n) &= \sum_{j=1}^{k} P(N_2 = m|T = j, N_1 = n)P(T = j|N_1 = n) \\
&= \sum_{j=1}^{k} P(N_2 = m|T = j)P(T = j|N_1 = n).
\end{aligned}$$

2.6 Examples

Example 2.18. Suppose $p(x, y, z)$, the joint probability mass function of the random variables X, Y and Z is given by

$$p(1,1,1) = \frac{1}{8}, \quad p(2,1,1) = \frac{1}{4},$$
$$p(1,1,2) = \frac{1}{8}, \quad p(2,1,2) = \frac{3}{16},$$

$$p(1,2,1) = \frac{1}{16}, \quad p(2,2,1) = 0,$$
$$p(1,2,2) = 0, \quad p(2,2,2) = \frac{1}{4},$$

What is $E[X|Y=2]$? What is $E[X|Y=2, Z=1]$?
Note

$$p(Y=2) = \frac{5}{16}.$$

Then given $Y=2$, the conditional distribution of X and Z is

$$P(1,1|Y=2) = \frac{1}{5},$$
$$P(1,2|Y=2) = 0,$$
$$P(2,1|Y=2) = 0,$$
$$P(2,2|Y=2) = \frac{4}{5}.$$

Then

$$P(X=1|Y=2) = \frac{1}{5},$$
$$P(X=2|Y=2) = \frac{4}{5}.$$

So

$$E[X|Y=2] = \frac{1}{5} + \frac{8}{5} = \frac{9}{5}.$$

Clearly,

$$p(X=1|Y=2, Z=1) = \frac{1/16}{1/16} = 1.$$

Thus,

$$E[X|Y=2, Z=1] = 1.$$

Example 2.19. An unbiased die is successively rolled. Let X and Y denote, respectively, the number of rolls necessary to obtain a six and a five. Find $E[X]$, $E[X|Y=1]$, $E[X|Y=5]$.

$$E[X] = \frac{1}{6} + (1 + E[X])\frac{5}{6}.$$

So

$$E[X] = 6.$$

Or: geometric distribution.

$$E[X|Y=1] = 1 + E[X] = 7.$$

Next,

$$E[X|Y=5] = 1 \cdot \frac{1}{5} + 2 \cdot \frac{4}{5}\frac{1}{5} + 3 \cdot \left(\frac{4}{5}\right)^2 \frac{1}{5} + 4 \cdot \left(\frac{4}{5}\right)^3 \frac{1}{5} + 6 \cdot \left(\frac{4}{5}\right)^4 \frac{1}{6}$$
$$+ 7 \cdot \left(\frac{4}{5}\right)^4 \frac{5}{6}\frac{1}{6} + 8 \cdot \left(\frac{4}{5}\right)^4 \left(\frac{5}{6}\right)^2 \frac{1}{6}.$$

Example 2.20. The joint density of X and Y is

$$f(x,y) = \frac{(y^2 - x^2)}{8} e^{-y},\ 0 < y < \infty, -y \le x \le y.$$

Show that $E[X|Y = y] = 0$.

$$f_Y(y) = \int_{-y}^{y} \frac{y^2 - x^2}{8} e^{-y} dx = e^{-y} \frac{y^2 x - \frac{x^3}{3}}{8} \Bigg|_{-y}^{y} = e^{-y} \frac{y^3}{6}.$$

Then

$$E[X|Y = y] = \int_{-y}^{y} x \cdot \frac{3}{4} \frac{y^2 - x^2}{y^3} dx = \frac{3}{4y^3} \int_{-y}^{y} (y^2 - x^2) dx = 0.$$

Or

$$E[X|Y = y] = C(y) \int_{-y}^{y} x(y^2 - x^2) dx = 0.$$

Example 2.21. If X is a discrete r.v. and Y is a continuous r.v., we have

$$f_{Y|X=i}(y) = \frac{P(X = i|Y = y) f_Y(y)}{P(X = i)}.$$

Example 2.22. An individual whose level of exposure to a certain pathogen is x will contract the disease caused by this pathogen with robability $P(x)$. If the exposure level of a randomly chosen member of the population has pdf f, determine the conditional probability density of the exposure level of that member given that he or she (1) has the disease; (2) does not have the disease.

$$f(x|\text{disease}) = \frac{P(\text{disease}|x) f(x)}{\int P(\text{disease}|x) f(x) dx} = \frac{P(x) f(x)}{\int P(x) f(x) dx}.$$

$$f(x|\text{no disease}) = \frac{(1 - P(x))\, f(x)}{\int (1 - P(x))\, f(x) dx}.$$

Example 2.23. A miner is trapped in a mine containing three doors. The first door leads to a tunnel that takes him to safety after two hours of travel. The second door leads to a tunnel that returns him to the mine after three hours of travel. The third door leads to a tunnel that returns him to his mine after five hours. Assuming that the miner is at all times equally likely to choose any one of the doors.
Let N denote the total number of doors selected before the miner reeaches safety. Also, let T_i denote the traval time corresponding to the ith choice, $i \ge 1$. Again let X denote the time when the miner reaches safty.

(a) Given an identity that relates X to N and the T_i.

$$X = \sum_{i=1}^{N} T_i.$$

(b) Since $N \sim \text{Geo}(1/3)$,

$$E[N] = 3.$$

(c)

$$E[T_N] = 3.$$

(d)

$$\begin{aligned} E\left[\sum_{i=1}^{N} T_i \middle| N = n\right] &= E\left[\sum_{i=1}^{n} T_i \middle| N = n\right] \\ &= E\left[\sum_{i=1}^{n-1} T_i \middle| N = n\right] + E[T_n | N = n] \\ &= 4(n-1) + 2, \end{aligned}$$

where 4 is from the travel times $T_i, i \in [n-1]$ are each equally likely to be either 3 or 5.

Example 2.24. A coin having probability p of coming up heads is successively flipped until two of the the most recent three flips are heads. Let N denote the number of flips. (Note that if the first two flips are heads, then $N = 2$.) Find $E[N]$.
Let X denote the first time a head appears. Then

$$\begin{aligned} E[N|X] &= E[N|X,h,h]p^2 + E[N|X,h,t]pq \\ &\quad + E[N|X,t,h]pq + E[N|X,t,t]q^2 \\ &= (X+1)p^2 + (X+1)pq + (X+2)pq + ((X+2) + E[N])\, q^2. \end{aligned}$$

Taking expecation, we have

$$E[N] = p + p^2 + q + pq + q + 2pq + q^2/p + 2q^2 + q^2 E[N].$$

So

$$E[N] = \frac{2 + 2q + q^2/p}{1 - q^2}.$$

Example 2.25. A coin, having probability p of landing heads, is continually flipped until at least one head and one tail have been flipped.

(a) Find the expected number of flips need.

$$\begin{aligned} E[X] &= E[X|h]p + E[X|t](1-p) \\ &= \left(1 + \frac{1}{1-p}\right)p + \left(1 + \frac{1}{p}\right)(1-p) \\ &= 1 + \frac{p}{1-p} + \frac{1-p}{p}. \end{aligned}$$

(b) Find the expected number of flips that land on heads.

$$\begin{aligned} E[X] &= (1 + E[\text{number of heads before first tail}])\, p + 1(1-p) \\ &= 1 \cdot \left(1 + \frac{1}{1-p} - 1\right)p + 1 - p = 1 - p + \frac{p}{1-p}, \end{aligned}$$

where

$$\text{number of heads before first tail} \sim \text{Geo}_2(1-p),$$

so

$$E[\text{number of heads before first tail}] = \frac{1}{1-p} - 1.$$

(c) Find the expected number of flips that land on tails.

$$p + \frac{1-p}{p}.$$

(d) Repeat part (a) in the case where flipping is continued until a total of at least two heads and one tail have been flipped.

Conditioning on the first flip,

$$\begin{aligned} E[X] &= E[X|h]p + E[X|t](1-p) \\ &= \left(1 + \frac{p}{1-p} + \frac{1-p}{p}\right)^2 p + \left(1 + \frac{2}{p}\right)(1-p). \end{aligned}$$

Example 2.26. Independent trials, resulting in one of the outcomes $1, 2, 3$ with respectively probabilities p_1, p_2, p_3, $\sum_{i=1}^{3} p_i = 1$, are performed.

(a) Let N denote the number of trials needed until the initial outcome has occurred exactly 3 times. For instance, if the trial results are $3, 2, 1, 2, 3, 2, 3$, then $N = 7$.

$$E[N] = \sum_{i=1}^{3} E[N|i]p_i = \sum_{i=1}^{3} \left(1 + \frac{2}{p_i}\right) p_i = 1 + 6 = 7.$$

(b) Find the expected number of trials needed until both outcome 1 and outcome 2 have occurred. Let $N_{1,2}$ be the number of trials until both outcome 1 and outcome 2 have occurred. Then

$$\begin{aligned} E[N_{1,2}] &= E[N_{1,2}|1]p_1 + E[N_{1,2}|2]p_2 + E[N_{1,2}|3]p_3 \\ &= \left(1 + \frac{1}{p_2}\right) p_1 + \left(1 + \frac{1}{p_1}\right) p_2 + (1 + E[N_{1,2}])\, p_3. \end{aligned}$$

Hence,

$$E[N_{1,2}] = \frac{1 + \frac{p_1}{p_2} + \frac{p_2}{p_1}}{p_1 + p_2}.$$

Example 2.27. You have two opponents with whom you alternate play. Whenever you play A you win with probability p_A; whenever you play B, you win with probability p_B, where $p_B > p_A$. If your objective is to minimize the expected number of games you need to play to win two in a row (two consective success), should you start with A or B?
Let N_A and N_B denote the number of games needed given that you start with A and given that you start with B.

$$E[N_A] = E[N_A|w]p_A + E[N_A|l](1-p_A),$$

where

$$\begin{aligned} E[N_A|w] &= E[N_A|ww]p_B + E[N_A|wl](1-p_B) \\ &= 2p_B + (2+E[N_A])(1-p_B) \\ &= 2 + (1-p_B)E[N_A]. \end{aligned}$$

Since $E[N_A|l] = 1 + E[N_B]$, we obtain

$$E[N_A] = 1 + p_A + p_A(1-p_B)E[N_A] + (1-P_A)E[N_B].$$

Similarly,

$$E[N_B] = 1 + p_B + p_B(1-p_A)E[N_B] + (1-P_B)E[N_A].$$

Substracting gives

$$[1 + (1-p_A)(1-p_B)]\,(E[N_A] - E[N_B]) = p_A - p_B < 0,$$

showing that playing A first is better.

Example 2.28. A coin that comes up heads with probability p is continually flipped until the pattern T, T, H appears. Let X denote the number of flips made, and find $E[X]$.

$$E[X] = E[X|H]p + E[X|T](1-p) = (1+E[X])p + E[X|T](1-p).$$

$$E[X|T] = E[X|TH]p + E[X|TT](1-p) = (2+E[X])\,p + \left(2 + \frac{1}{p}\right)(1-p),$$

where the final equality follows since given that the first two flips are tails the number of additional flips is just the number of flips needed to obtain a head. Thus,

$$E[X] = \frac{1}{p(1-p)^2}.$$

Example 2.29. Polya's urn model supposes that an urn initially contains r red and b blue balls, At each stage a ball is randomly selected from the urn is then returned along with m other balls of the same color. Let X_k be the number of red balls drawn in the first k selections. Find $E[X_k]$. Let

$$Y_i = \begin{cases} 1, & \text{selction } i \text{ is red} \\ 0, & \text{otherwise} \end{cases}.$$

Then

$$E[X_k] = \sum_{i=1}^{k} E[Y_i].$$

$$E[Y_1] = \frac{r}{r+b} = E[X_1].$$

$$\begin{aligned} E[Y_2] &= E\left[E[Y_2|X_1]\right] = E\left[\frac{r+mX_1}{r+b+m}\right] \\ &= \frac{r + m\frac{r}{r+b}}{r+b+m} = \frac{r}{r+b}. \end{aligned}$$

Then

$$E[X_2] = 2\frac{r}{r+b}.$$

Claim

$$E[Y_k] = \frac{r}{r+b}.$$

By induction,

$$\begin{aligned} E[Y_k] = E[Y_k|X_{k-1}] &= E\left[\frac{r+mX_{k-1}}{r+b+(k-1)m}\right] \\ &= \frac{r+mE\left[\sum_{i=1}^{k-1} Y_i\right]}{r+b+(k-1)m} = \frac{r+m(k-1)\frac{r}{r+b}}{r+b+(k-1)m} \\ &= \frac{r}{r+b}. \end{aligned}$$

Thus,

$$E[X_k] = \frac{kr}{r+b}.$$

Example 2.30. Two players take turns shooting at a target, with each shot by player i hitting the target with probability $p_i, i = 1, 2$. Shooting ends when two consecutive shots hit the target. Let μ_i denote the mean number of shots taken when player i shoots first, $i = 1, 2$. Let h_i denote the mean number of times that the target is hit when player i shoots first, $i = 1, 2$. Find h_1 and h_2.

$$\begin{aligned} \mu_1 &= E[N|h]p_1 + E[N|m]q_1 \\ &= p_1\left(E[N|h,h]p_2 + E[N|h,m]q_2\right) + (1+\mu_2)q_1 \\ &= 2p_1p_2 + (2+\mu_1)p_1q_2 + (1+\mu_2)q_1. \end{aligned}$$

Then

$$\mu_1(1-p_1q_2) = 1 + p_1 + \mu_2 q_1.$$

Similarly,

$$\mu_2(1-p_2q_1) = 1 + p_2 + \mu_1 q_2.$$

Next,

$$\begin{aligned} h_1 &= E[H|h]p_1 + E[H|m]q_1 \\ &= p_1\left(E[H,h,h]p_2 + E[H|h,m]q_2\right) + h_2q_1 \\ &= 2p_1p_2 + (1+h_1)p_1q_2 + h_2q_1. \end{aligned}$$

Similarly,

$$h_2 = 2p_1p_2 + (1+h_2)p_2q_1 + h_1q_2.$$

Example 2.31. Let $X_i, i \geq 0$ be independent and identically distributed random varibales with probability mass function

$$p(j) = P(X_i = j),\ \forall\ j \in [m],\ \sum_{j=1}^{m} p(j) = 1.$$

Find $E[N]$, where $N = \min\{n > 0 : X_n = X_0\}$.

$$E[N] = \sum_{j=1}^{m} E[N|X_0 = j]p(j) = \sum_{j=1}^{m} \frac{1}{p(j)} p(j) = m.$$

Example 2.32. Each element in a sequence of binary data is either 1 with probability p or 0 with probability $1-p$. A maximal subsequence of consecutive values having identical outcomes is called a run. For instance, if the outcome sequence is $1, 1, 0, 1, 1, 1, 0$, the first run is of length 2, the second is of length 1, and the third is of length 3.

(a) Find the expected length of the first run.

$$\begin{aligned} E[L_1] &= E[L_1|1]p + E[L_1|0](1-p) \\ &= \frac{1}{1-p}p + \frac{1}{p}(1-p). \end{aligned}$$

(b) Find the expected length of the second run.

$$\begin{aligned} E[L_2] &= E[L_2|1]p + E[L_2|0](1-p) \\ &= \frac{1}{p}p + \frac{1}{1-p}(1-p) \\ &= 2, \end{aligned}$$

where we still are conditioning on the very initial value. !!!

Example 2.33. Independent trials, each resulting in success with probability p, are performed.

(a) Find the expected number of trials need for there to have been both at least n successes and at least m failures. Let T be the number of trials needed for both at least n successes and m failures. Let N be the number of successes in the first $n+m$ trials. Then

$$E[T] = \sum_{i=0}^{n+m} E[T|N = i] \binom{n+m}{i} p^i (1-p)^{n+m-i}.$$

Now use

$$E[T|N = i] = \begin{cases} n + m + \frac{n-i}{p}, & i \le n \\ n + m + \frac{i-n}{1-p}, & i > n. \end{cases}.$$

(b) Find the expected number of trials needed for there to have been either at least n successes or at least m failures. Let S be the number of trials needed for n successes, and let F be the number needed for m failures. Then $T = \max(S, F)$. Taking expectations of the identity

$$\min(S, F) + \max(S, F) = S + F,$$

yields the result

$$E[\min(S, F)] = \frac{n}{p} + \frac{m}{1-p} - E[T].$$

Example 2.34. If R_i denotes the random amount that is earned in period i, then $\sum_{i=1}^{\infty} \beta^{i-1} R_i$, where $0 < \beta < 1$ is a specified constant, is called the total discounted reward with discount factor β. Let T be a geometric random variable with parameter $1-\beta$ that is independent of the R_i. Show that the expected total discounted reward is equal to the expected total (undiscounted) reward earned by time T. That is, show that

$$E\left[\sum_{i=1}^{\infty} \beta^{i-1} R_i\right] = E\left[\sum_{i=1}^{T} R_i\right].$$

Note

$$\begin{aligned} E\left[\sum_{i=1}^{T} R_i\right] &= E\left[\sum_{i=1}^{\infty} \mathbb{1}_{\{T \geq i\}} R_i\right] = \sum_{i=1}^{\infty} E\left[\mathbb{1}_{\{T \geq i\}} R_i\right] \\ &= \sum_{i=1}^{\infty} E\left[\mathbb{1}_{\{T \geq i\}}\right] E[R_i] = \sum_{i=1}^{\infty} P(T \geq i) E[R_i] \\ &= \sum_{i=1}^{\infty} \beta^{i-1} E[R_i] = E\left[\sum_{i=1}^{\infty} \beta^{i-1} R_i\right] \end{aligned}$$

Example 2.35. A set of n dices is throw. All those that land on six are put aside, and the others are again thrown. This is repeated until all the dice have landed on six. Let N denote the number of throws needed. Let $m_n = E[N]$.

(a) Derive a recursive formular for m_n and use it to calculate $m_i, i = 2, 3, 4$ and to show that $m_5 \approx 13.024$.

$$\begin{aligned} m_n &= \sum_{i=0}^{n} E[N|X = i] \binom{n}{i} (1/6)^i (5/6)^{n-i} \\ &= \sum_{i=0}^{n} (1 + m_{n-i}) \binom{n}{i} (1/6)^i (5/6)^{n-i} \\ &= 1 + m_n (5/6)^n + \sum_{i=1}^{n-1} m_{n-i} \binom{n}{i} (1/6)^i (5/6)^{n-i}, \end{aligned}$$

implying that

$$m_n = \frac{1 + \sum_{i=1}^{n-1} m_{n-i} \binom{n}{i} (1/6)^i (5/6)^{n-i}}{1 - (5/6)^n}.$$

(b) Let X_i denote the number of dice rolled on the ith throw. Find $E\left[\sum_{i=1}^{N} X_i\right]$.
Since each die rolled will land on six with probability 1/6, the total number of dice rolled will equal the number of times one must roll **a** die until six appears n times. Therefore,

$$E\left[\sum_{i=1}^{N} X_i\right] = 6n.$$

Example 2.36. Consider n multinomial trials, where each trial independently results in outcome i with probability p_i, $\sum_{i=1}^k p_i = 1$. With X_i equal to the number of trials that result in outcome i, find $E[X_1|X_2 > 0]$.

$$\begin{aligned} np_1 = E[X_1] &= E[X_1|X_2 = 0](1-p_2)^n + E[X_1|X_2 > 0]\left[1-(1-p_2)^n\right] \\ &= n\frac{p_1}{1-p_2}(1-p_2)^n + E[X_1|X_2 > 0]\left[1-(1-p_2)^n\right], \end{aligned}$$

yielding the result

$$E[X_1|X_2 > 0] = \frac{np_1\left(1-(1-p_2)^{n-1}\right)}{1-(1-p_2)^n}.$$

Example 2.37. Let $p_0 = P(X = 0)$ and suppose that $p_0 \in (0, 1)$. Let $\mu = E[X]$ and $\sigma^2 = \mathrm{Var}(X)$. Find $\mathrm{Var}(X|X \neq 0)$.

$$E[X] = E[X|X \neq 0](1-p_0) + E[X|X = 0]p_0,$$

yielding

$$E[X|X \neq 0] = \frac{E[X]}{1-p_0}.$$

Similarly,

$$E[X^2|X \neq 0] = \frac{E[X^2]}{1-p_0}.$$

Hence,

$$\mathrm{Var}(X|X \neq 0) = \frac{E[X^2]}{1-p_0} - \frac{E^2[X]}{(1-p_0)^2}.$$

Example 2.38. A deck of n cards, numbered 1 through n, is randomly shuffled so that all $n!$ possible permutations are equally likely. The cards are then turned over one at a time until card number 1 appears. The upturned cards constitute the first cycle. We now determine (by looking at the upturned cards) the lowest numbered card that has not yet appeared, and we continue to turn the cards face up until that card appears, and we continue to turn the cards face up until that card appears. This new set of cards represents the second cycle. We again determine the lowest numbered of the remaining cards and turn the cards until it appears, and so on until all cards have been turned over. Let m_n denote the mean number of cycles.

Let N denote the number of cycles, and let X be the position of card 1.

$$m_n = \frac{1}{n}\sum_{i=1}^n E[N|X = i] = \frac{1}{n}\sum_{i=1}^n (1 + m_{i-1}) = 1 + \frac{1}{n}\sum_{j=1}^{n-1} m_j.$$

By induction,

$$m_n = 1 + 1/2 + 1/3 + \cdots + 1/n.$$

Example 2.39. Suppose that we want to predict the value of a random variable X by using one of the predictors $Y_1, \ldots, Y_n$, each of which satisfies $E[Y_i|X] = X$. Show that the predictor Y_i that

minimizes $E[(Y_i - X)^2]$ is the one whose variance is smallest.

$$\begin{aligned}
\mathrm{Var}(Y_i) &= E\left[\mathrm{Var}(Y_i|X)\right] + \mathrm{Var}\left(E[Y_i|X]\right) \\
&= E\left[\mathrm{Var}(Y_i|X)\right] + \mathrm{Var}(X) \\
&= E\left[E\left[(Y_i - E[Y_i|X])^2|X\right]\right] + \mathrm{Var}(X) \\
&= E\left[E\left[(Y_i - X)^2|X\right]\right] + \mathrm{Var}(X) \\
&= E\left[(Y_i - X)^2\right] + \mathrm{Var}(X).
\end{aligned}$$

Example 2.40. A and B play a series of games with A winning each game with probability p. The overall winner is the first player to have won two more games than the other.

(a) Find the probability that A is the overall winner.

Let A be the event that A is the overall winner, and let Y equal the number of wins for A in the first two games.

$$\begin{aligned}
P(A) &= P(A|Y=0)P(Y=0) + P(A|Y=1)P(Y=1) + P(A|Y=2)P(Y=2) \\
&= 0 + P(A)2p(1-p) + p^2.
\end{aligned}$$

Thus,

$$P(A) = \frac{p^2}{1 - 2p(1-p)}.$$

(b) Find the expected number of games played.

Let X be the number of games played.

$$\begin{aligned}
E[X] &= E[X|Y=0]P(Y=0) + E[X|Y=1]P(Y=1) + E[X|Y=2]P(Y=2) \\
&= 2(1-p^2) + (2 + E[X])2p(1-p) + 2p^2.
\end{aligned}$$

Thus,

$$E[X] = \frac{2}{1 - 2p(1-p)}.$$

Example 2.41. Suppose each new coupon collected is, independent of the past, a type i coupon with probability p_i. A total of n coupons is to be collected. Let A_i be the event that there is at least on type i in this set. For $i \neq j$, compute $P(A_iA_j)$ by

(a) Conditioning on N_i, the number of type i coupons in the set of n coupons;

$$\begin{aligned}
P(A_iA_j) &= \sum_{k=0}^{n} P(A_iA_j|N_i = k)\binom{n}{k}p_i^k(1-p_i)^{n-k} \\
&= \sum_{k=1}^{n} P(A_j|N_i = k)\binom{n}{k}p_i^k(1-p_i)^{n-k} \\
&= \sum_{k=1}^{n-1}\left[1 - \left(1 - \frac{p_j}{1-p_i}\right)^{n-k}\right]\binom{n}{k}p_i^k(1-p_i)^{n-k}.
\end{aligned}$$

(b) Conditioning on F_i, the first time a type i coupon is collected;

$$\begin{aligned}
P(A_iA_j) &= \sum_{k=1}^{n} P(A_iA_j|F_i = k)p_i(1-p_i)^{k-1} + P(A_iA_j|F_i > n)(1-p_i)^n \\
&= \sum_{k=1}^{n} P(A_j|F_i = k)p_i(1-p_i)^{k-1} \\
&= \sum_{k=1}^{n} \left[1 - \left(1 - \frac{p_j}{1-p_i}\right)^{k-1} (1-p_j)^{n-k}\right] p_i(1-p_i)^{k-1}.
\end{aligned}$$

(c) using the identity $P(A_i \cup A_j) = P(A_i) + A(A_j) - P(A_iA_j)$.

$$\begin{aligned}
P(A_iA_j) &= P(A_i) + P(A_j) - P(A_i \cup A_j) \\
&= 1 - (1-p_i)^n + 1 - (1-p_j)^n - [1 - (1-p_i-p_j)^n] \\
&= 1 + (1-p_i-p_j)^n - (1-p_i)^n - (1-p_j)^n.
\end{aligned}$$

Example 2.42. A, B and C are evenly matched tennis players. Initially A and B play a set, and the winner then plays C. This continues, with the winner always playing the waiting player, until one of the player has won two sets **in a row**. That player is then declared the overall winner. Find the probability that A is the overall winner.

Let W and L stands for the events that player A wins a game and loses a game, respectively. Let $P(A)$ be the probability that A wins and let $P(C)$ be the probability that C wins after just winnning one game.

$$\begin{aligned}
P(A) &= \frac{1}{2}P(A|W) + \frac{1}{2}P(A|L) \\
&= \frac{1}{2}\left[\frac{1}{2} + \frac{1}{2}P(A|WL)\right] + \frac{1}{2}\frac{1}{2}P(C) \\
&= \frac{1}{4} + \frac{1}{4}\frac{1}{2}P(C) + \frac{1}{4}P(C) \\
&= \frac{1}{4} + \frac{3}{8}P(C).
\end{aligned}$$

Also,

$$P(C) = \frac{1}{2}P(A|W) = \frac{1}{4} + \frac{1}{8}P(C).$$

Example 2.43. Suppose there are n types of coupons, and that the type of each new coupon obtained is independent of past selections and is equally likely to be any of the n types. Suppose one continues collecting until a complete set of at least one of each type is obtained.

(a) Find the probability that there is exactly one type i coupon in the final collection.
Let T be the number of types that are collected before the first type i appears and S_i be the event

there is only one type i in the final set.

$$\begin{aligned} P(S_i) &= \sum_{j=0}^{n-1} P(S_i|T=j)P(T=j) \\ &= \frac{1}{n}\sum_{j=0}^{n-1} P(S_i|T=j) \\ &= \frac{1}{n}\sum_{j=0}^{n-1} \frac{1}{n-j}? \end{aligned}$$

(b) Find the expected number of types that appear exactly once in the final collection.

$$E\left[\sum_{i=1}^{n} S_i\right] = nE[S_i] = \sum_{k=1}^{n} \frac{1}{k}.$$

Example 2.44. The opponents of soccer team A are of two types: either they are a class 1 or a class 2 team. The number of goals team A scores agaist a class i opponent is a Poisson random variable with mean λ_i, where $\lambda_1 = 2, \lambda_2 = 3$. This weekend the team has two games agaist teams they arer not familiar with. Assuming that the fist team play is a class 1 team with probability 0.6 and the second is, independently of the class of the first team, a class 1 team with probability 0.3, determine

(a) the expected number of goals team A will scrore this weekend.

$$E[G_1 + G_2] = E[G_1] + E[G_2] = (0.6)2 + (0.4)3 + (0.3)2 + (0.7)3 = 5.1.$$

(b) the probability that team A will score a total of five goals.

Conditioning on the types and using that the sum of independent Poissons is Poisson gives the solution

$$P(5) = 0.18e^{-4}\frac{4^5}{5!} + 0.54e^{-5}e^{-5}5^5/5! + 0.28e^{-6}6^5/5!.$$

Example 2.45. A coin having probability p of coming up heads is continually flipped. Let $P_j(n)$ denote the probability that a run of j successive heads occurs within the first n flips. Argue that

$$P_j(n) = P_j(n-1) + p^j(1-p)\left[1 - P_j(n-j-1)\right].$$

A run of j successive heads can occur in the following mutually exclusive ways: (i) either there is a run of j in the first $n-1$ flips, or (ii) there is no j-run in the first $n-j-1$ flips, flip $n-j$ is a tail, and the next j flips are all heads. Consequently, it follows.

Example 2.46. In the match problem, say that $(i,j), i < j$, is a pair if i chooses j's hat and j chooses i's hat.

(a) Find the expected number of pairs. Let

$$I_{(i,j)} = \begin{cases} 1, & i,j \text{ are a pair} \\ 0, & \text{otherwise} \end{cases}.$$

Then

$$E\sum_{i<j} I_{(i,j)} = \binom{n}{2}\frac{1}{n}\frac{1}{n-1} = \frac{1}{2}.$$

(b) Let Q_n denote the probability that there are no pairs, and derive a recursive formula for Q_n in terms of $Q_j, j < n$.

Let X be the size of the cycle containing person 1. Then

$$Q_n = \sum_{i=1}^{n} P(\text{no pairs}|X = i)\frac{1}{n} = \frac{1}{n}\sum_{i\neq 2} Q_{n-i},$$

where $P(X = i)$ is from previous reuslt.

Example 2.47. Suppose that we are continually roll a die until the sum of all throws exceeds 100. What is the most likely value of this total when you stop?

Condition on the value of the sum prior to going over 100. In all cases the most likely value is 101. For instance, if the sum is 98, then the final sum is equally likely to be either 101, 101,103 or 104.

Example 2.48. Prove

$$E[XY] = E\left[YE[X|Y]\right].$$

Proof.

$$E[XY] = E\left[E[XY|Y]\right] = E\left[YE[X|Y]\right].$$

Example 2.49. Let $X_i, i \geq 1$ be independent Uniform$(0, 1)$ random variables. Then

$$\begin{aligned} P(x < X_1 < \cdots < X_n) &= P(X_1 < \cdots < X_n|X_i > x, i = 1, \ldots, n)P(X_i > x, i = 1, \ldots, n) \\ &= \frac{1}{n!}(1-x)^n. \end{aligned}$$

Example 2.50. Let $X_1, X_2, \cdots$ be independent continuous random variables with a common distribution function F and density $f = F'$, and for $k \geq 1$ let

$$N_k = \min\{n \geq k : X_n = k^{\text{th}} \text{ largest of } X_1, \ldots, X_n\}.$$

(a) Show that $P(N_k = n) = \frac{k-1}{n(n-1)}, n \geq k$.
Let A_i denote the event that X_i is the kth largest of $X_1, \ldots, X_i$, $\forall\ k \leq i \leq n$. Then $A_k, \ldots, A_n$ are independent events with $P(A_i) = \frac{1}{i}$, $\forall\ k \leq i \leq n$.

$$\begin{aligned} P(N_k = n) &= P(A_k^c A_{k+1}^c \cdots A_{n-1}^c A_n) = (1 - P(A_k)) \cdots P(A_n) \\ &= \frac{k-1}{k}\frac{k}{k+1}\cdots\frac{n-2}{n-1}\frac{1}{n} = \frac{k-1}{n(n-1)}. \end{aligned}$$

(b) Argue that

$$f_{X_{N_k}}(x) = f(x)\left(\overline{F}(x)\right)^{k-1}\sum_{i=0}^{\infty}\binom{i+k-2}{i}\left(F(x)\right)^i.$$

Proof. Since knowledge of the set of values $\{X_1, \ldots, X_n\}$ gives us no information about the order of these random variables it follows that given $N_k = n$, the conditional distribution of X_{N_k} is the same as the distribution of kth **largest** of n random variables having distribution F (!!!).

$$f_{X_{N_k}}(x) = \sum_{n=k}^{\infty} \frac{k-1}{n(n-1)} \frac{n!}{(n-k)!(k-1)!} \left(F(x)\right)^{n-k} \left(\overline{F}(x)\right)^{k-1} f(x).$$

Then make the change of variable $i = n - k$.

2.7 Challenging problems

Example 2.51. Draw two subsets from the integers $\{1, \ldots, n\}$ at random (uniformly at random and allow empty set), and call these subsets A and B. Then

$$\begin{aligned} P(B \subseteq A) &= \sum_{k=0}^{n} P\left(B \subseteq A \ \mid |B| = k\right) P\left(|B| = k\right) \\ &= \sum_{k=0}^{n} \left(\frac{1}{2}\right)^{n-k} \frac{\binom{n}{k}}{2^n} = \frac{1}{4^n} \sum_{k=0}^{n} \binom{n}{k} 2^k 1^{n-k} = \left(\frac{3}{4}\right)^n. \end{aligned}$$

Alternative:

$$\begin{aligned} P(B \subseteq A) &= \sum_{k=0}^{n} P\left(B \subseteq A \ \mid |A| = k\right) P\left(|A| = k\right) \\ &= \sum_{k=0}^{n} \frac{2^k}{2^n} \frac{\binom{n}{k}}{2^n} = \frac{1}{4^n} \sum_{k=0}^{n} \binom{n}{k} 2^k 1^{n-k} = \left(\frac{3}{4}\right)^n. \end{aligned}$$

Chapter 3

Bernoulli Processes

Lemma 3.1. For any $\omega \in \Omega, k = 1, 2$ and $n \geq k$, we have

(a) $T_k(\omega) \leq n$ if and only if $N_n(\omega) \geq k$,

(b) $T_k(\omega) = n$ if and only if $N_{n-1}(\omega) = k - 1$, $X_n(\omega) = 1$.

Theorem 3.2. *For $n \in \mathbb{Z}^+$,*

$$P(T_k \leq n) = P(N_n \geq k) = \sum_{j=k}^{n} \binom{n}{j} p^j q^{n-j}, \ n = k, k+1, \cdots;$$

$$P(T_k = n) = P(N_{n-1} = k-1)P(X_n = 1) = \binom{n-1}{k-1} p^k q^{n-k}.$$

Chapter 4

Markov Chains

4.1 Introduction

Definition 4.1. Let $\{X_n, n = 0, 1, 2, \ldots, \}$ be a stochastic process that takes on a finite or countable number of possible values. If $X_n = i$, then the process is said to be in state i at time n. We suppose that whenever the process is in state i, there is a fixed probability P_{ij} that it will next be in state j. That is, we suppose that

$$P\left(X_{n+1} = j | X_n = i, X_{n-1} = i_{n-1}, \ldots, X_0 = i_0\right) = P\left(X_{n+1} = j | X_n = i\right),$$

for all states $i_0, \ldots, i_{n-1}, i, j$ and all $n \in \mathbb{N}$. Such a stochastic process is known as a *Markov chain*.

Definition 4.2. A DTMC $\{X_n\}_{n\geq 0}$ is said to be *time-homogeneous* if for each $n \in \mathbb{N}$, we have for each $i, j \in E$ that

$$P\left(X_{n+1} = j | X_n = i\right) = P\left(X_1 = j | X_0 = i\right).$$

Hence, we can associate with a time-homogeneous DTMC both an initial distribution $\alpha = \{\alpha_i\}_{i\in E}$, which we often treat as a possibly infinitely long row vector, and a transition probability matrix $P = [p_{i,j}]_{i,j\in E}$, where for each $i, j \in E$,

$$p_{i,j} = P(X_1 = j | X_0 = i).$$

Proposition 4.3. For each $n \in \mathbb{N}$ and each sequence of states $i_0, \ldots, i_n \in E$, we have

$$P\left(X_0 = i_0, \ldots, X_n = i_n\right) = \alpha_{i_0} \prod_{k=1}^{n} p_{i_{k-1}, i_k}.$$

Theorem 4.4. *Suppose $\{Z_k\}_{k\geq 1}$ is a sequence of i.i.d. discrete random variables taking values in a countable set F, and let X_0 be a random variable, independent of $\{X_k\}_{k\geq 1}$ and taking values in a countable space E. Given a function $g : E \times F \to E$, we define for each $n \in \mathbb{N}$,*

$$X_{n+1} = g(X_n, Z_{n+1}).$$

Then $\{X_n\}_{n\geq 0}$ is a DTMC.

Example 4.5. Suppose $\{Z_n\}_{n\geq 0}$ is a sequence of i.i.d. integer-valued random variables, and

$$S_n = \sum_{k=0}^{n} Z_k, \ \forall\ n \in \mathbb{N}.$$

Since

$$S_{n+1} = \sum_{k=0}^{n+1} Z_k = S_n + Z_{n+1} = g(S_n, Z_{n+1}),$$

where

$$\begin{array}{rl} g : \mathbb{Z} \times \mathbb{Z} & \to \mathbb{Z} \\ (i,j) & \mapsto i+j \end{array},$$

we have $\{S_n\}_{n\geq 0}$ is a DTMC.

Example 4.6. Suppose $\{Z_n\}_{n\geq 0}$ is a sequence of i.i.d. integer-valued random variables, and for each $n \in \mathbb{N}$, define

$$M_n = \max_{0\leq k\leq n} Z_k.$$

Since

$$M_{n+1} = \max(M_n, Z_n + 1),$$

we have $\{M_n\}_{n\geq 0}$ is a DTMC.

Example 4.7. In most of Europe and Asia annual automobile insurance premiums are determined by use of a Bonus Malus (Latin for Good-Bad) system. Each policyholder is given a positive integer valued state and the annual premium is a function of this state (along, of course, with the type of car being insured and the level of insurance). A policyholder's state changes from year to year in response to the number of claims made by that policyholder. Because lower numbered states correspond to lower annual premiums, a policyholder's state will usually decrease if he or she had no claims in the preceding year, and will generally increase if he or she had at least one claim. (Thus, no claims is good and typically results in a decreased premium, while claims are bad and typically result in a higher premium.)

For a given Bonus Malus system, let $s_i(k)$ denote the next state of a policyholder who was in state i in the previous year and who made a total of k claims in that year. If we suppose that the number of yearly claims made by a particular policyholder is a Poisson random variable with parameter λ, then the successive states of this policyholder will constitute a Markov chain with transition probabilities

$$P_{i,j} = \sum_{k:s_i(k)=j} e^{-\lambda}\frac{\lambda^k}{k!}, \ j \geq 0.$$

4.2 Chapman-Kolmogorov Equations

Definition 4.8.

$$p_{i,j}^{(n)} := P(X_n = j | X_0 = i),$$

$$P^{(n)} = \left[p_{i,j}^{(n)}\right]_{i,j\in E}.$$

Proposition 4.9. Let $P^0 = I$, then

$$P^{(n)} = P^n, \ \forall\, n \in \mathbb{N}.$$

Proof. By induction,

$$\begin{aligned}
p_{ij}^{(n+m)} &= P(X_{n+m} = j | X_0 = i) \\
&= \sum_{k\in E} P(X_{n+m} = j, X_n = k | X_0 = i) \\
&= \sum_{k\in E} P(X_{n+m} = j | X_n = k, X_0 = i) P(X_n = k | X_0 = i) \\
&= \sum_{k\in E} p_{kj}^{(m)} p_{ik}^{(n)} \\
&= \sum_{k\in E} P_{ik}^n P_{kj}^m \text{ (by induction)} \\
&= (P^n P^m)_{ij} \\
&= P_{ij}^{n+m}.
\end{aligned}$$

□

Theorem 4.10 (Chapman-Kolmogorov equations)**.**

$$P_{ij}^{n+m} = \sum_{k\in E} P_{ik}^n P_{kj}^m, \ \forall\, n, m \in \mathbb{N}, \ \forall\, i, j \in E.$$

Remark. Let

$$N_j(n) := \sum_{k=0}^{n} \mathbb{1}_{\{X_k = j\}}.$$

Then

$$E\left[N_j(n) | X_0 = i\right] = E\left[\sum_{k=0}^{n} \mathbb{1}_{\{X_k=j\}} \middle| X_0 = i\right] = \sum_{k=0}^{n} P(X_k = j | X_0 = i) = \sum_{k=0}^{n} P_{i,j}^k,$$

which is simply the $(i,j)^{\text{th}}$ element of $\sum_{k=0}^{n} P^k$.

Example 4.11. For each collection of integers $0 \le m_0 < m_1 < \cdots < m_n < m_{n+1}$ and each collection of states $i_0, i_1, \ldots, i_n, i_{n+1}$, we have

$$P\left(X_{m_0} = i_0, X_{m_1} = i_1, \ldots, X_{m_n} = i_n, X_{m_{n+1}} = i_{n+1}\right) = \sum_{l\in E} \alpha_l P_{l,i_0}^{m_0} P_{i_0,i_1}^{m_1 - m_0} \cdots P_{i_n,i_{n+1}}^{m_{n+1}-m_n}.$$

Example 4.12 (Number of success in Bernoulli process)**.** Let N_n denote the number of success in n Bernoulli trials, where the probability of a success in any on trial is p. Then $\{N_n, n \ge 0\}$ is Markov chain. Here the state space is $\{0, 1, 2, \cdots\}$, the initial distribution is $\alpha(0) = 1, \alpha(j) = 0, j \ge 1$. Then the transition matrix is

$$P = \begin{pmatrix} q & p & & & \\ & q & p & & \\ & & q & p & \\ & & & & \\ & & & & \vdots \end{pmatrix}.$$

Then

$$\begin{aligned} P^n(i,j) &= P\left(N_{k+n} = j | N_k = i\right) = P\left(N_{k+n} - N_k = j - i\right) \\ &= P(N_n = j - i) = \binom{n}{j-i} p^{j-i} q^{n-j+i}, \ j = i, \ldots, n+i. \end{aligned}$$

Example 4.13. Let T_n be the time of the nth success in a Bernoulli process. Then $\{T_n; n \geq 0\}$ is a MC. The state space is $\{0, 1, 2, \cdots\}$; $T_0 = 0$, so that the initial distribution is $\alpha(0) = 1, \alpha(1) = \alpha(2) = \cdots = 0$. Then the transition matrix is

$$P = \begin{pmatrix} 0 & p & pq & pq^2 & pq^3 & \cdots \\ 0 & 0 & p & pq & pq^2 & \cdots \\ 0 & 0 & 0 & p & pq & \cdots \\ & & & \vdots & & \end{pmatrix}.$$

Then

$$\begin{aligned} P^n(i,j) &= P\left(T_{k+n} = j | T_k = i\right) = P\left(T_{k+n} - T_k = j - i\right) \\ &= P(T_n = j - i) = \binom{j-i-1}{n-1} p^n q^{j-i-n}, \ j \geq i + n. \end{aligned}$$

Example 4.14 (Independent trials process)**.** Let $X_0, X_1, \cdots$ be independent **discrete** random variables with the common distribution. Then $\{X_n, n \geq 0\}$ is a MC. Then all the rows of P are identical and $P^n = P, \ \forall \ n \geq 1$.

Example 4.15. An urn always contains 2 balls. Ball colors are red and blue. At each stage a ball is randomly chosen and then replaced by a new ball, which with probability 0.8 is the same color, and with probability 0.2 is the opposite color, as the ball it replaces. If initially both balls are red, find the probability that the fifth ball selected is red.

Let X_n to be the number of red balls in the urn after the nth selection and subsequent replacement. Then $\{X_n\}_{n \geq 0}$ is a Markov chain with state $0, 1, 2$ and with transition probability matrix P give by

$$\begin{pmatrix} 0.8 & 0.2 & 0 \\ 0.1 & 0.8 & 0.1 \\ 0 & 0.2 & 0.8 \end{pmatrix}.$$

Then

$$\begin{aligned} P(\text{fifth selection is red} | X_0 = 2) &= \sum_{i=0}^{2} P(\text{fifth selection is red} | X_4 = i) P(X_4 = i | X_0 = 2) \\ &= 0 + 0.5 \cdot P^4_{2,1} + 1 \cdot P^4_{22}. \end{aligned}$$

Remark.

$$P^m S = S \Lambda^m.$$

4.3 First-Step Analysis

Many types of probabilities and expectations associated with a DTMC can be computed via a technique known as first-step analysis. This technique often consists of two parts:

(a) conditioning on the next state visited, given the current state, and applying the Markov property to generate a linear system of equations,

(b) solving the linear system.

Example 4.16. Consider a DTMC $\{X_n\}_{n\geq 0}$ with state space $E = \{0, 1, 2\}$ and transition probability matrix P, where

$$P = \begin{pmatrix} 0.4 & 0.6 & 0 \\ 0 & 0.9 & 0.1 \\ 0 & 0 & 1 \end{pmatrix}.$$

Compute the expected amount of time it takes the chain to travel from state 0 to state 2, i.e., compute

$$E[T|X_0 = 0],$$

where

$$T = \min\{n \geq 0 : X_n = 2\}.$$

$$\begin{aligned} E[T|X_0 = 0] &= \sum_{j=0}^{2} E[T|X_1 = j, X_0 = 0]P(X_1 = j|X_0 = 0) \\ &= (1 + E[T|X_0 = 0])\, P_{00} + (1 + E[T|X_0 = 1])\, P_{01}, \end{aligned}$$

where

$$\begin{aligned} E[T|X_0 = 1] &= \sum_{j=0}^{2} E[T|X_1 = j, X_0 = 1]P(X_1 = j|X_0 = 1) \\ &= (1 + E[T|X_0 = 1])\, P_{11} + P_{12}. \end{aligned}$$

Example 4.17 (The Gambler's Ruin problem)**.** Consider a gambler who at each play of the game has probability p of winning one unit and probability $q = 1 - p$ of losing one unit. Assuming that successive plays of the game are independent, what is the probability that, starting with i units, the gambler's fortune will reach N before reaching 0?

If we let X_n denote the player's fortune at time n, then the process $\{X_n, n \in \mathbb{N}\}$ is a MC with transition probabilities

$$P_{00} = P_{NN} = 1,$$

$$P_{i,i+1} = p = 1 - P_{i,i-1},\ i = 1, \ldots, N - 1.$$

This MC has three classes, namely, $\{0\}, \{1, \ldots, N - 1\}$ and $\{N\}$; the first and third class being recurrent and the second transient. Since each transient state is visited only finitely often, it follows that, after some finite amount of time, the gambler will either attain his goal of N or go broke. Then it is equivalent to compute

$$P_i := P\left(X_T = N|X_0 = i\right),$$

where

$$T = \min\{n \geq 0 : X_n \in \{0, N\}\}.$$

Clearly,

$$P_0 = 0,$$
$$P_N = 1.$$

By the first step analysis, we obtain

$$P_i = pP_{i+1} + qP_{i-1}, \ \forall \ i \in [N-1].$$

Then we have

$$P_i = \begin{cases} \frac{1-(q/p)^i}{1-(q/p)} P_1, & q/p \neq 1 \\ iP_1, & q/p = 1 \end{cases}.$$

Now, using the fact that $P_N = 1$, we obtain the value of P_1. As a result,

$$P_i = \begin{cases} \frac{1-(q/p)^i}{1-(q/p)^N}, & p \neq 1/2 \\ i/N, & p = 1/2 \end{cases}.$$

Next,

$$m_i := E[T|X_0 = i] = 1 + (1-p)m_{i-1} + pm_{i+1}.$$

Then

$$m_0 = m_N = 0,$$

and

$$p\left(m_{i+1} - m_i\right) = (1-p)\left(m_i - m_{i-1}\right) - 1.$$

Letting $x_i = m_i - m_{i-1}$, we have

$$x_{i+1} = \frac{1-p}{p} x_i - \frac{1}{p}.$$

Then find the general formula by observing the first terms and note

$$\sum_{i=1}^{N} x_i = m_N - m_0 = 0.$$

Remark. Consider a MC with transition probabilities P_{ij}. Let $\mathcal{A}$ be a set of states, and suppose we are interested in the probability that the MC ever enters any of the states in A by time m. That is, for a given state $i \notin A$, we are interested in determining

$$\beta = P(X_k \in \mathcal{A} \text{ for some } k = 1, \ldots, m | X_0 = i).$$

Define a MC $\{W_n, n \geq 0\}$, whose states are the states that are not in A plus an additional state, which we will call A in our general discussion. Once the MC $\{W_n\}$ enters state A, it remains there forever.

Thee new MC is defined as follows. Letting X_n denote the state at time n of the MC with transition probabilities $P_{i,j}$, define

$$N = \min\{n : X_n \in \mathcal{A}\},$$

and let $N = \infty$ if $X_n \notin \mathcal{A}$, $\forall\, n$. In words, N is the first time the MC enters the set of states $\mathcal{A}$. Now, define

$$W_n = \begin{cases} X_n, & n < N \\ A, & n \geq N \end{cases}.$$

So the state of the $\{W_n\}$ process is equal to the state of the original MC up to the point when the original MC enters a state in $\mathcal{A}$. At that time the new process goes to state A and remains there forever. The transition probabilities $Q_{i,j}$, given by

$$Q_{i,j} = P_{i,j},\ i \notin \mathcal{A}, j \notin \mathcal{A},$$

$$Q_{i,A} = \sum_{j \in \mathcal{A}} P_{i,j},\ i \notin \mathcal{A},$$

$$Q_{A,A} = 1.$$

Because the original MC will have entered a state in $\mathcal{A}$ by time m if and only if the state at time m of the new MC is A, we see that

$$P(X_k \in \mathcal{A} \text{ for some } k = 1, \ldots, m | X_0 = i) = P(W_m = A | X_0 = i) = P(W_m = A | W_0 = i) = Q^m_{i,A}.$$

Example 4.18. In a sequence of independent flips of a fair coin, let N denote the number of flips until there is a run of three consecutive heads. Find

(a) $P(N \leq 8)$. Define a MC with states $0, 1, 2, 3$ where for i ($0 \leq i < 3$) means that we currently are on a run of i consecutive heads, and where state 3 means that a run of three consecutive heads has already occurred. (!!! 3 is an absorbing state)Then

$$P = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ \frac{1}{2} & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Because there would be a run of three consecutive heads within the first eight flips if and only if $X_8 = 3$, the desired probability is $P^8_{0,3}$.

(b) $P(N = 8)$.

$$P(N = 8) = P(N \leq 8) - P(N \leq 7) = P^8_{0,3} - P^7_{0,3}.$$

Another way to determine $P(N = 8)$ is to consider a MC with states $0, 1, 2, 3, 4$ where as before, for $i < 3$ state i means that we currently are on a run of i consecutive heads, state 3 means that the first run of size 3 has just occurred, and state 4 that a run of size 3 occurred in the past. (3 and 4 are both absorbing state) That is,

$$Q = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 \\ \frac{1}{2} & 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

Then N will 8 if starting at state 0 the preceding MC is in state 3 after 8 transitions.

$$P(N = 8) = Q^8_{0,3}.$$

Example 4.19. Consider a MC with states $1, 2, 3, 4, 5$ and suppose that we want to compute

$$P(X_4 = 2, X_3 \leq 2, X_2 \leq 2, X_1 \leq 2 | X_0 = 1).$$

That is, we want the probability that, starting in state 1, the chain is in state 2 at time 4 and has never entered any of the states in the set $\mathcal{A} = \{3, 4, 5\}$. Suppose that

$$P_{11} = 0.3 \quad P_{12} = 0.3$$

$$P_{21} = 0.1 \quad P_{22} = 0.2$$

Then we consider the MC having states 1,2,3 (we are giving state $\mathcal{A}$ the name 3), having the transition probability matrix

$$A = \begin{pmatrix} 0.3 & 0.3 & 0.4 \\ 0.1 & 0.2 & 0.7 \\ 0 & 0 & 1 \end{pmatrix}.$$

As a result, the desired probability is $\alpha = Q_{12}^4 = 0.0285$.

When $i \notin \mathcal{A}$, but $j \in \mathcal{A}$, we can determine the probability

$$\begin{aligned}
\alpha &= P(X_m = j, X_k \notin \mathcal{A}, k = 1, \ldots, m-1 | X_0 = i) \\
&= \sum_{i \notin \mathcal{A}} P(X_m = j, X_{m-1} = r, X_k \notin \mathcal{A}, k = 1, \ldots, m-2 | X_0 = i) \\
&= \sum_{i \notin \mathcal{A}} P(X_m = j | X_{m-1} = r, X_k \notin \mathcal{A}, k = 1, \ldots, m, X_0 = i) \\
&\quad \cdot P(X_{m-1} = r, X_k \notin \mathcal{A}, k = 1, \ldots, m-2 | X_0 = i) \\
&= \sum_{r \notin \mathcal{A}} P_{r,j} P(X_{m-1} = r, X_k \notin \mathcal{A}, k = 1, \ldots, m-2 | X_0 = i) \\
&= \sum_{r \notin \mathcal{A}} P_{r,j} Q_{i,r}^{m-1}.
\end{aligned}$$

When $i \in \mathcal{A}$,

$$\begin{aligned}
\alpha &= P(X_m = j, X_k \notin \mathcal{A}, k = 1, \ldots, m-1 | X_0 = i) \\
&= \sum_{r \notin \mathcal{A}} P(X_m = j, X_k \notin \mathcal{A}, k = 1, \ldots, m-1 | X_0 = i, X_1 = r) P(X_1 = r | X_0 = i) \\
&= \sum_{r \notin \mathcal{A}} P(X_{m-1} = j, X_k \notin \mathcal{A}, k = 1, \ldots, m-2 | X_0 = r) P_{i,r}.
\end{aligned}$$

For instance, if $i \in \mathcal{A}, j \notin \mathcal{A}$, then the preceding equation yields

$$P(X_m = j, X_k \notin \mathcal{A}, k = 1, \ldots, m-1 | X_0 = i) = \sum_{r \notin \mathcal{A}} Q_{r,j}^{m-1} P_{i,r}.$$

Similarly, $\forall\ i, j \notin \mathcal{A}$,

$$\begin{aligned}
P(X_n = j | X_0 = i, X_k \notin \mathcal{A}, k = 1, \ldots, n) &= \frac{P(X_n = j, X_k \notin \mathcal{A}, k = 1, \ldots, n | X_0 = i)}{P(X_k \notin \mathcal{A}, k = 1, \ldots, n | X_0 = i)} \\
&= \frac{Q_{i,j}^n}{\sum_{r \notin \mathcal{A}} Q_{i,r}^n}.
\end{aligned}$$

Remark. So far, all of the probabilities we have considered are conditional probabilities. If the unconditional distribution of the state at time n is desired, it is necessary to specify the probability distribution of the initial state. Let us denote this by

$$\alpha_i = P(X_0 = i),\ i \geq 0,\ \sum_{i=0}^{\infty} \alpha_i = 1.$$

All unconditional probabilities may be computed by conditioning on the intial state. That is,

$$P(X_n = j) = \sum_{i=0}^{\infty} P(X_n = j | X_0 = i) p(X_0 = i) = \sum_{i=0}^{\infty} P_{ij}^n \alpha_i.$$

4.4 Classification of States

Definition 4.20. State j is said to be *accessible* from state i if $P_{ij}^n > 0$ for some $n \geq 0$.

Definition 4.21. Two states i and j that are *accessible to each other* are said to communicate, and we write $i \leftrightarrow j$.

Remark. Note that any state communicates with itself since

$$P_{ii}^0 = P(X_0 = i | X_0 = i) = 1.$$

Remark. Communication is an equivalent relation. Two states that communicate are said to be in the same class. Then any two classes of states are either identical or disjoint. In other words, the concept of communication divides the state space up into a number of seperate class, i.e., letting

$$C_i = \{l \in E : i \leftrightarrow l\},$$

then $\forall\ i, j \in E$, either $C_i = C_j$ or $C_i \cap C_j = \emptyset$, so the set of communicating classes forms a partition of E.

Definition 4.22. We say that a communicating class C is *closed*, if for each state $i \in C$,

$$P(X_n \in C | X_0 = i) = 1,\ \forall\ n \in \mathbb{N}.$$

Definition 4.23. The MC is said to be *irreducible* if there is only one class, that is, if all states communicate with each other.

4.4.1 Transience and Recurrence

Definition 4.24 (Strong Markov property)**.** The independence of past and future given the present state, hold even if the present state is determined by a stopping time.

Definition 4.25.

$$\eta_j = \min\{n \geq 1 : X_n = j\}.$$

$$f_i = P(\eta_j < \infty | X_0 = j).$$

Definition 4.26. State j is called *recurrent* if

$$f_i = 1.$$

State j is called *tansient* if

$$f_i < 1.$$

A recurrent state j is called *positive* if

$$E[\eta_j | X_0 = j] < \infty.$$

Corollary 4.27. Let

$$N_i = \sum_{n=0}^{\infty} \mathbb{1}_{\{X_n = i\}}.$$

The state i is recurrent if

$$\sum_{n=0}^{\infty} P_{ii}^n = E[N_i | X_0 = i] = \infty.$$

Proof. Note

$$P(N_i = 1 | X_0 = i) = P(\eta_i = \infty | X_0 = i) = 1 - f_i.$$

Since η_i is a stopping time, by the strong Markov property,

$$P(N_i = 2 | X_0 = i) = P(\eta_i < \infty | X_0 = i) P(\eta_i = \infty | X_0 = i) = f_i(1 - f_i).$$

In general,

$$P(N_i = k | X_0 = i) = f_i^{k-1}(1 - f_i), \ \forall \ k \in \mathbb{Z}^+.$$

Then

$$P(N_i \geq k | X_0 = i) = f_i^{k-1}, \ \forall \ k \in \mathbb{Z}^+.$$

Thus,

$$E[N_i | X_0 = i] = \sum_{k=1}^{\infty} P(N_i \geq k | X_0 = i) = \sum_{k=1}^{\infty} f_i^{k-1} = \sum_{k=0}^{\infty} f_i^k. \qquad \square$$

Remark. Suppose that the process starts in state i and i is recurrent. Hence, with probability 1, the process will eventually reenter state i. However, by the definition of a Markov chain, it follows that the process will be starting over again when it reenters state i and, therefore, state i will eventually be visited again. Continual repetition of this argument leads to the conclusion that if state i is recurrent then starting in state i, the process will reenter state i again and again-in fact, infinitely often.

On the other hand, suppose that state i is transient. Hence, each time the process enters state i, there will be a positive probability, namely, $1 - f_i$, that it will never again enter that state. Therefore, starting in state i, the probability that the process will be in state i for exactly n time periods equals $f_i^{n-1}(1 - f_i)$, which is $\text{Geo}(1 - f_i)$. So the mean time periods that the process will be in state i is $\frac{1}{1-f_i}$.

Corollary 4.28. In a finite-state MC, not all states can be transient.

Corollary 4.29. If state i is recurrent, and state i communicates with state j, then state j is recurrent. Namely, recurrence is a class property.

Proof. Since $i \leftrightarrow j$, $\exists\ k, m \in \mathbb{N}$ such that $P_{ij}^k > 0$ and $P_{ji}^m > 0$. Now, $\forall\ n \in \mathbb{N}$,

$$P_{jj}^{m+n+k} \geq P_{ji}^m P_{ii}^n P_{ij}^k.$$

Then since the state i is recurrent,

$$\sum_{n=1}^{\infty} P_{jj}^{m+n+k} \geq P_{ji}^m P_{ij}^k \sum_{n=1}^{\infty} P_{ii}^n = \infty.$$

□

Corollary 4.30. Transisience is a class property.

Proof. By contradiction. □

Theorem 4.31. *If C is a finite, closed communicating class, then all of the states within it must be recurrent.*

Example 4.32. Consider the DTMC on $\{0, 1, 2, 3, 4\}$, where

$$P = \begin{pmatrix} 0.2 & 0 & 0 & 0.8 & 0 \\ 0 & 0.7 & 0.3 & 0 & 0 \\ 0 & 0.9 & 0.1 & 0 & 0 \\ 0.5 & 0 & 0 & 0.5 & 0 \\ 0.1 & 0.1 & 0.2 & 0.2 & 0.4 \end{pmatrix}.$$

Then its communicating classes are

$$\{0, 3\},\ \{1, 2\},\ \{4\}.$$

By the submatrix, we have $\{0, 3\}$ is a closed, so 0 and 3 are recurrent. Similarly, 1 and 2 are also recurrent and 4 is transient.

Theorem 4.33. *Suppose i is a recurrent state. Then C_i, the communicating class containing state i, must be closed.*

4.4.2 The simple Random Walk

Theorem 4.34. *For $0 < p < 1$, the chain of random walk is recurrent if and only if $p = \frac{1}{2}$.*

Proof. Clearly, this DTMC is irreducible, let us consider the state 0,

$$\sum_{n=1}^{\infty} P_{00}^n = \sum_{n=1}^{\infty} P_{00}^{2n} = \sum_{n=1}^{\infty} \binom{2n}{n} p^n (1-p)^n = \sum_{n=1}^{\infty} \frac{(2n)!}{n!} n! \left(p(1-p)\right)^n,\ n = 1, 2, 3, \cdots.$$

By using an approximation, due to Stirling, which asserts that

$$n! \sim n^{n+1/2} e^{-n} \sqrt{2\pi}.$$

Then

$$\frac{(2n)!}{n!} n! \left(p(1-p)\right)^n \sim \frac{\left(4p(1-p)\right)^n}{\sqrt{\pi n}}.$$

Now it is easy to verify, for positive a_n, b_n, that if $a_n \sim b_n$, then $\sum_n a_n < \infty$ if and only if $\sum_n b_n < \infty$. Hence $\sum_{n=1}^{\infty} P_{00}^n$ will converge if and only if

$$\sum_{n=1}^{\infty} \frac{\left(4p(1-p)\right)^n}{\sqrt{\pi n}}$$

does. However, $4p(1-p) \leq 1$ with equality holding if and only if $p = \frac{1}{2}$. Hence, $\sum_{n=1}^{\infty} P_{00}^n = \infty$ if and only if $p = \frac{1}{2}$. □

Remark. We can also look at symmeric random walks in more than one dimension. For instance, in the two-dimensional symmetric random walk the process would, at each transition, either take one step to the left, right, up or down, each having probability $\frac{1}{4}$. That is, the state is the pair of integers (i, j) and the transition probabilities are given by

$$P_{(i,j),(i+1,j)} = P_{(i,j),(i-1,j)} = P_{(i,j),(i,j+1)} = P_{(i,j)(i,j-1)} = \frac{1}{4}.$$

Similarly, we consider $P_{(0,0),(0,0)}^{2n}$. Now after $2n$ steps, the chain will be back in its original location if for some $i, 0 \leq i \leq n$, the $2n$ steps consist of i steps to the left, i to the right, $n-i$ up and $n-i$ down. Since each step will be either of these four types with probability $\frac{1}{4}$, it follows that the desired probability is a multinormial probability. That is,

$$\begin{aligned}
P_{00}^{2n} &= \sum_{i=0}^{n} \frac{(2n)!}{i!i!(n-i)!(n-i)!} \left(\frac{1}{4}\right)^{2n} \\
&= \sum_{i=0}^{n} \frac{(2n)!}{n!n!} \frac{n!}{(n-i)!i!} \frac{n!}{(n-i)!i!} \left(\frac{1}{4}\right)^{2n} \\
&= \left(\frac{1}{4}\right)^{2n} \binom{2n}{n} \sum_{i=0}^{n} \binom{n}{i} \binom{n}{n-i} \\
&= \frac{1}{4} \binom{2n}{n} \binom{2n}{n}.
\end{aligned}$$

Similarly,

$$P_{00}^{2n} \sim \frac{1}{\pi n}.$$

Thus, all states are recurrent.

Theorem 4.35. *In the random walk of* $0 < p < 1$*,*

$$P(\textit{ever return 0}) = 2 \min\left(p, 1-p\right).$$

Proof. Let X_1 be the initial transition. Then

$$P(\text{ever return } 0) = P(\text{evet reurn } 0 | X_1 = 1)p + P(\text{ever return } 0 | X_1 = -1)(1-p).$$

Let

$$\begin{aligned}
\alpha &= P(\text{ever reurn } i-1|X_1 = i) \\
&= P(\text{ever reurn } 0|X_1 = 1) \\
&= P(\text{ever reurn } 0|X_1 = 1, X_2 = 0)(1-p) + P(\text{ever return } 0|X_1 = 1, X_2 = 2)p \\
&= 1 - p + P(\text{ever return } 0|X_1 = 1, X_2 = 2)p \\
&= 1 - p + p\alpha^2.
\end{aligned}$$

The two roots are $\alpha = 1$ and $\alpha = \frac{1-p}{p}$.
If $p = \frac{1}{2}$, then $\alpha = 1$ and then

$$P(\text{ever return } 0) = 1.$$

Suppose now that $p > \frac{1}{2}$, then

$$P(\text{ever return to } 0|X_1 = -1) = 1.$$

The equation reduces to

$$\beta = \alpha p + 1 - p.$$

Because the random walk is transient in this case we know that $\beta < 1$, showing that $\alpha \neq 1$. Therefore, $\alpha = \frac{1-p}{p}$, yielding that

$$\beta = 2(1-p),\ p > \frac{1}{2}.$$

Similarly, when $p < \frac{1}{2}$, we can show that $\beta = 2p$. Thus, in general

$$P(\text{ever return } 0) = 2\min(p, 1-p). \qquad \square$$

Remark. For a (slightly less than rigorous) probabilistic proof of Stirling's approximation, let $X_1, X_2, \cdots$ be independent Poisson random variables each having mean 1. Let $S_n = \sum_{i=1}^n X_i$, and note that both mean and variance of S_n are equal to n. Now,

$$\begin{aligned}
\frac{e^{-n}n^n}{n!} &= P(S_n = n) \\
&= P(n-1 < S_n \leq n) \\
&= P\left(-1/\sqrt{n} < (S_n - n)\sqrt{n} \leq 0\right) \\
&\approx \int_{-1/\sqrt{n}}^{0} (2\pi)^{-1/2} e^{-x^2/2} dx \\
&\approx (2\pi)^{-1/2}(1/\sqrt{n}) \\
&= (2\pi n)^{-1/2}.
\end{aligned}$$

Or, equivalently,

$$n! \approx n^{n+1/2} e^{-n} \sqrt{2\pi}.$$

4.4.3 Computations

In this last subsection, we show how to compute various probabilities and expected values associated with both transient and recurrent states. Suppose $\{X_n\}_{n\geq 0}$ si a DTMC on the space

$$E = \{0, \ldots, m, m+1, \ldots, M\},$$

where the states $T = \{0, \ldots, m\}$ are all transient states, and the states $R = \{m+1, \ldots, M\}$ are all recurrent states. The transition matrix P for this DTMC is then of the form

$$P = \begin{pmatrix} Q & U \\ 0 & S \end{pmatrix}.$$

Let $0 \leq i, j \leq m$, and let

$$N_j = \sum_{n=0}^{\infty} \mathbb{1}_{\{X_n = j\}}.$$

By MCT,

$$E_i[N_j] = \sum_{n=0}^{\infty} P_i(X_n = j) = \sum_{n=0}^{\infty} P_{i,j}^n < \infty,$$

since state j is a transient state.

Note $P_{i,j}^n$ is the sum of all probabilities associated with each path of length n that starts at state i at time zero, and ends in state j at time n, and since i and j are both transient, all such feasible paths consist of only transient states. Thus,

$$P_{ij}^n = Q_{i,j}^n.$$

Then $E_i[N_j]$ is the $(i,j)^{\text{th}}$ element of

$$\sum_{n=0}^{\infty} Q^n.$$

Since each expected value is finite, it follows that

$$\sum_{n=0}^{\infty} Q^n = (I - Q)^{-1}.$$

Our nex goal is to compute, $\forall\ i \in \{0, \ldots, m\}$ and $\forall\ j \in \{m+1, \ldots, M\}$, the probability $P_i(\eta = j)$, where

$$\eta = \min\{n \geq 1 : X_n \in R\}$$

is the first time the Markov chain visits a recurrent state. In other words, given we are currently at a transient state i, what is the probability the first recurrent state we visit is state j?

Note

$$\begin{aligned} P_i(X_\eta = j) &= \sum_{k \in T} P_i(X_\eta = j | X_1 = k) P_{ik} + \sum_{k \in R} P_i(X_\eta = j | X_1 = k) P_{ik} \\ &= \sum_{k \in T} Q_{ik} P_k(X_\eta = j) + u_{ij}, \end{aligned}$$

or equivalently, if we let $H = [P_k(X_\eta = j)]_{k\in T, j\in R}$ be a matrix containing all such probabilities, we see instead that

$$H = QH + U.$$

Then

$$H = (I - Q)^{-1}U.$$

Our final example consist of deriving the probability mass function of η when assuming

$$\alpha_i = P(X_0 = i),\ \forall\ i \in T,$$

where

$$\sum_{i\in T} \alpha_i = 1.$$

It suffices to compute $P(\eta > n),\ \forall\ n \in \mathbb{Z}^+$. Here

$$\begin{aligned} P(\eta > n) &= \sum_{j\in T} P(X_n = j) = \sum_{j\in T}\sum_{i\in T} P(X_n = j|X_0 = i)\alpha_i \\ &= \sum_{i\in T} \alpha_i \sum_{j\in T} P^n_{ij} = \alpha Q^n \mathbb{1}_{m+1}. \end{aligned}$$

Here η is sometimes referred to as a discrete phase-type random variable.

4.5 Visits to a Fixed state

Let $X = \{X_n, n \in \mathbb{N}\}$ be a MC with state space E and transition matrix P.

Let $T_1(\omega), \ldots, T_m(\omega)$ be the successive indices $n \geq 1$ for which $X_n(\omega) = j$ as long long there are such n. If j appears only finite number m of times, then we put $T_{m+1}(\omega) - T_m(\omega) = T_{m+2}(\omega) - T_{m+1}(\omega) = \cdots = \infty$. Then $T_1, T_2, \cdots$ are the times of successive vistis to state j.

For any $n \in \mathbb{N}$, $T_m(\omega) \leq n$ if and only if j appears in $\{X_1(\omega), \ldots, X_n(\omega)\}$ at least m times. Thus, every T_m is a stopping for the chain X.

4.6 Stationary and limiting behavior

In this section, we will be interested in existence of limits of the form

$$\lim_{n\to\infty} \frac{1}{n} \sum_{k=1}^{n} \mathbb{1}_{\{X_k = j\}},$$

and

$$\lim_{n\to\infty} P(X_n = j|X_0 = i),\ \forall\ i, j \in E.$$

Definition 4.36. Suppose $\nu := \{\nu_j\}_{j\in E}$ is a vector, where $\nu_j \geq 0\ \forall\ j \in E$. Then ν is called a *measure* on E.

Definition 4.37. A measure $\nu = \{\nu_i\}_{i\in E}$ is said to be an *invariant measure* with respect to a DTMC $\{X_n\}_{n\geq 0}$ if

$$\nu = \nu P.$$

We refer to $\nu = \nu P$ as the balance equations of the DTMC.

Definition 4.38. We say that an invariant measure $\pi = \{\pi_j\}_{j\in E}$ is said to be a *stationary distribution* if

$$\sum_{j\in E} \pi_j = 1.$$

In most of our applications, we will be interested in deriving stationary distributions, since they reveal a lot of useful information about Markov chains.

Remark. To see where the name "stationary distribution" comes from, suppose that π is a stationary distribution for $\{X_n\}_{n\in\mathbb{N}}$ and suppose further that

$$P(X_0 = j) = \pi_j, \ \forall\ j \in E,$$

which means the initial distribution is the stationary distribution.

Under this assumption, it is easy to show that the pmf association with each X_n is also π since by induction,

$$\begin{aligned} P(X_n = j) &= \sum_i P(X_n = j | X_{n-1} = i) P(X_{n-1} = i) \\ &= \sum_i P_{ij}\pi_i \\ &= \pi_j. \end{aligned}$$

Moreover, entire finite-dimensional distribution can also be shifted $\forall\ n \in \mathbb{N}, k \in \mathbb{Z}^+, i_0, \ldots, i_n \in E$, we have

$$\begin{aligned} P\left(X_k = i_0, \ldots, X_{k+n} = i_n\right) &= P\left(X_{k+1} = i_1, \ldots, X_{k+n} = i_n | X_k = i_0\right) P\left(X_k = i_0\right) \\ &= P\left(X_1 = i_1, \ldots, X_n = i_n | X_0 = i_0\right) P\left(X_0 = i_0\right) \\ &= P(X_0 = i_0, \ldots, X_n = i_n). \end{aligned}$$

Theorem 4.39. *By setting the distribution of X_0 to be stationary distribution π, the distribution of X_n is also π for any $n \in \mathbb{N}$, then this is enough to say that π_j can be interpreted as the long run proportion of time the Markov chain spends in state j.*

4.6.1 Periodicity and limiting Distributions

We are now ready to discuss when limits of the form $\lim_{n\to\infty} P_{ij}^n$ exist. Establishing that such limits exist will require having an understanding of what is known as periodicity.

Definition 4.40. The *period* $d(i)$ of a state $i \in E$ is defined as

$$d(i) = \gcd\{n \geq 1 : P_{ii}^n > 0\}.$$

we say state i is *aperiodic* if $d(i) = 1$.

Proposition 4.41. Suppose i and j are two states that communicate, Then $d(i) = d(j)$.

Theorem 4.42. *Aperiodicity is needed to ensure that when $\{X_n\}$ is both irreducible and positive recurrent, its unique stationary distribution also serves as its limiting distribution, i.e.,*

$$\pi_j = \lim_{n\to\infty} P_{ij}^n, \ \forall\ i, j \in E.$$

Here we refer to π as the unique limiting distribution of this MC.

Or equivalently, if $\{X_n\}$ is ergodic, its unique stationary distribution also serves as its unique limiting distribution.

Example 4.43. Suppose $\{X_n\}_{n\in\mathbb{N}}$ is a DTMC, whose state space $E = \{0, 1\}$ and

$$P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Since it is irreducible and all states are finite, this chain has a unique stationary distribution.
It is easy to show

$$\pi_0 = \pi_1 = \frac{1}{2}.$$

At the same time, the period of each state is 2 since $P_{00}^{2k+1} = 0$, but $P_{00}^{2k} = 1$, $\forall\ k \in \mathbb{Z}^+$. Furthermore,

$$\lim_{n\to\infty} \sup P_{00}^n = 1 > 0 = \lim_{n\to\infty} \inf P_{00}^n,$$

which proves this chain does not have a limiting distribution.

Definition 4.44.

$$f_k(i, j) = P(\eta_1 = k | X_0 = i),$$

which is the probabiity that first time to visit j is at k, starting in state i.
Define

$$f(i, j) = P(X_n = j \text{ for some } n > 0 | X_0 = i) = P(T_1 < \infty | X_0 = i) = \sum_{k=1}^{\infty} f_k(i, j),$$

which is the probability that the Markov chain, starting in state i, will ever visit state j.

Remark.

$$f_j = f(j, j).$$

Proposition 4.45. Let N_j be the total number of visits to j in $\{X_0, X_1, \cdots\}$. Then

$$P(N_j = m | X_0 = j) = f_j^{m-1}\left(1 - f_j\right),\ m = 1, 2, \cdots.$$

For $i \neq j$,

$$P(N_j = m | X_0 = i) = \begin{cases} 1 - f(i, j). & m = 0 \\ f(i, j) f_j^{m-1}(1 - f_i), & m = 1, 2, \cdots \end{cases}.$$

Proposition 4.46. If i is recurrent and i communicates with j, then $f_{i,j} = 1$.

Proof. Because i and j communicate there is a value n such that $P^n_{i,j} > 0$.

Let $X_0 = i$ and say that the first opportunity is a success if $X_n = j$, and note that the first opportunity is a success with probability $P^n_{i,j} > 0$. If the first opportunity is not a success then consider the next time (**after** time n) that the chain enters state i. Say that the second opportunity is a success if n time periods later the MC is in state j. Continuing in this manner, we can define an unlimited number of opportunities, each of which is a success with the same positive probability P^n_{ij}. Because the number of opportunities until the first success occurs is geometric with parameter $P^n_{i,j}$, it follows that with probability 1 a success will eventually occur. □

Definition 4.47.

$$m_j = E[\eta_j | X_0 = j] = \sum_{k=1}^{\infty} k f_k(j,j),$$

which is the expected number of transitions that it takes the MC when starting in state j to return to j.

Definition 4.48. The recurrent state j is *positive recurrent* if $m_j < \infty$ and it is *null recurrent* if $m_j = \infty$.

Theorem 4.49. *Let π_j denote the long-run proportion of time that the* MC *is in state j. If the* MC *is irreducible and* ***recurrent****, then for any initial state*

$$\pi_j = \frac{1}{m_j},$$

which is also the long-run rate of returning to state j.

Proof. Suppose that the MC starts in state i, and let T_1 denote the number of transitions until the chain enters state j; let T_2 denote the additional number of transitions from time T_1 until the MC next enters state j; let T_3 denote the additional number of transitions from time $T_1 + T_2$ until the MC next enters state j, and so on. We have shown $T_1 < \infty$. Because T_n is the number of transitions between the $(n-1)^{\text{th}}$ and the nth transition into state j, it follows from the Markovian property that $T_2, T_2, \cdots$ are independent and identically distributed with mean m_j. Because the nth transition into state j occurs at time $T_1 + \cdots + T_n$ we obtain that

$$\begin{aligned}
\pi_j &= \lim_{n\to\infty} \frac{n}{\sum_{i=1}^n T_i} \\
&= \lim_{n\to\infty} \frac{1}{\frac{1}{n}\sum_{i=1}^n T_i} \\
&= \lim_{n\to\infty} \frac{1}{\frac{T_1}{n} + \frac{T_2+\cdots+T_n}{n}} \\
&= \frac{1}{m_j},
\end{aligned}$$

where the last equality follows from the SLLN,

$$\lim_{n\to\infty} \frac{T_2 + \cdots + T_n}{n} = \lim_{n\to\infty} \frac{T_2 + \cdots T_n}{n-1} \frac{n-1}{n} = m_j.$$ □

Remark. Because $m_j < \infty$ is equivalent to $\frac{1}{m_j} > 0$, it follows from the preceding that state j is positive recurrent if and only if $\pi_j > 0$.

4.6.2 Time average

Each element of a stationary distribution π happens to have a nice interpretation. Namely, when π is the stationary distribution of an irreducible, positive recurrent DTMC $\{X_n\}_{n\in\mathbb{N}}$, one can show that each element $\pi_j, j \in E$, can be interpreted as the long-run proportion of time the chain spends on state j. More precisely,

$$\lim_{N\to\infty} \frac{1}{N+1} \sum_{n=0}^{N} \mathbb{1}_{\{X_n=j\}} = \pi_j, \text{ w.p.1..}$$

Our main result

Proposition 4.50. Suppose $f : E \to [0, \infty)$ is a nonnegative function. Moreover, suppose our MC is irreducible and positive recurrent, then for any initial state

$$\lim_{N\to\infty} \frac{1}{N} \sum_{n=0}^{N} f(X_n) = \sum_{j\in E} f(j)\pi_j, \text{ w.p.1..}$$

4.6.3 Constructing invariant measures

Our first result establishes conditions for an invariant measure to exist. Our constructions will require working with hitting-time random variables of the form

$$\eta_i = \min\{n \geq 1 : X_n = i\}.$$

Proposition 4.51. Suppose state i is **recurrent**. Then an invariant measure is given by

$$\nu_j^{(i)} = E_i \left[\sum_{n=0}^{\eta_i - 1} \mathbb{1}_{\{X_n=j\}} \right].$$

4.6.4 To what extent are invariant measures unique?

Proposition 4.52. Suppose $\{X_n\}$ is irreducible and recurrent. Then for each state $i \in E$, **all** invariant measures of $\{X_n\}$ are scalar multilples of $\nu^{(i)}$.

Moreover, if $\sum_{j\in E} \nu_j^{(i)} = E_i[\eta_i] < \infty$, i.e, i is positive recurrent, a **unique stationary** distribution π exists, and

$$\pi_j = \frac{\nu_j^{(i)}}{\sum_{j\in E} \nu_j^{(i)}} = \frac{E_i\left[\sum_{n=0}^{\eta_i-1} \mathbb{1}_{\{X_n=j\}}\right]}{E_i[\eta_i]}.$$

Proof. By the renewal reward theorem. □

Remark. Suppose $\{X_n\}_{n\in\mathbb{N}}$ is an irreducible DTMC. Assume there is a measure π on E satisfying

$$\pi = \pi P,$$

$$\sum_{j\in E} \pi_j = 1.$$

Then

(a) the MC is positive recurrent,

(b) $\pi_i > 0,\ \forall\ i \in E$,

(c) the stationary distribution is unqiue.

Theorem 4.53. *If i is positive recurrent and $i \leftrightarrow j$, then j is positive recurrent. Namely, positive recurrent is a class property.*

Proof. Let n be such that $P_{ij}^n > 0$.

$$\begin{aligned} 0 < \pi_i P_{ij}^n &= \text{long run proportion of time the chain is in } i \text{ and will be in } j \text{ after } n \text{ transitions} \\ &= \text{long-run proportion of time the chain is in } j \text{ and was in } i \ n \text{ transitions ago} \\ &\leq \text{long-run proportion of time the chain is in } j \\ &= \pi_j. \end{aligned}$$

□

Corollary 4.54. Null recurrent is a class property.

Theorem 4.55. *An irreducible finite state* MC *must be positive recurrent.*

Proof. For we know that such a chain must be recurrent; hence, all its states are either positive recurrent or null recurrent. If that were null recurrent, then all the long run proportions would equal to 0, which is impossible when there only finite number of states. □

To determine the long-run proportions $\{\pi_j, j \geq 1\}$, note, because π_i is the long-run proportion of transitions that come **from** state i, that

$$\pi_i P_{ij} = \text{long-run proportion of transitions that go from state } i \text{ to state } j.$$

Then

$$\pi_j = \sum_i \pi_i P_{i,j}.$$

Also,

$$\pi_i = \pi_i \sum_j P_{ij}.$$

Theorem 4.56. *Consider an irreducible MC. If the chain is positive recurrent, then the long-run proportions are the unique solution of the equations*

$$\pi_j = \sum_i \pi_i P_{ij}$$

$$\sum_j \pi_j = 1.$$

Moreover, if there is no solution of the preceding linear equations, then the Markov chain is either transient or null recurrent and all $\pi_j = 0$.

Example 4.57. Suppose the number of families that check into a hotel on successive days are independent Poisson random variables with mean λ. Also suppose that the number of days that a family stays in the hotel is a geometric random variable with parameter $p, 0 < p < 1$ (**Thus, a family who spent the previous night in the hotel will, independent of how long they have already spent in the hotel, check out the next day with probability** p). Also suppose that all families act independently of each other. If X_n denotes the number of families that are checked in the hotel at the beginning of day n, then $\{X_n, n \geq 0\}$ is a Markov chain.

(a) The transition probabilities of this Markov chain;

To find $P_{i,j}$, suppose there are i families checked into the hotel at the beginning of a day. Because each of these i families will stay for another day with probability $q = 1 - p$. It follows that R_i, the number of these families that remain another day, is a binoniam(i, q) random variable. Let N be the number of new families that check in that day. Then

$$\begin{aligned}
P_{i,j} &= P(R_i + N = j) \\
&= \sum_{k=0}^{i} P(R_i + N = j | R_i = k) \binom{i}{k} q^k p^{i-k} \\
&= \sum_{k=0}^{i} P(N = j - k | R_i = k) \binom{i}{k} q^k p^{i-k} \\
&= \sum_{k=0}^{\min(i,j)} P(N = j - k) \binom{i}{k} q^k p^{i-k} \\
&= \sum_{k=0}^{\min(i,j)} e^{-\lambda} \frac{\lambda^{j-k}}{(j-k)!} \binom{i}{k} q^k p^{i-k}
\end{aligned}$$

(b) $E[X_n | X_0 = i]$;

$$E[X_n | X_{n-1} = i] = E[R_i + N] = iq + \lambda.$$

Then

$$E[X_n | X_{n-1}] = X_{n-1} q + \lambda.$$

Taking expectations of both sides yields

$$\begin{aligned}
E[X_n] &= \lambda + qE[X_{n-1}] \\
&= \lambda + q(\lambda + qE[X_{n-2}]) \\
&= \lambda + q\lambda + q^2 E[X_{n-2}] \\
&= \vdots \\
&= \lambda(1 + q + \cdots q^{n-1}) + q^n E[X_0],
\end{aligned}$$

yielding the result

$$E[X_n | X_0 = i] = \frac{\lambda(1 - q^n)}{p} + q^n i.$$

(c) the stationary probabilities of this Markov chain.

Rather we will make use of the fact that the stationary probability distribution is the only distribution on the initial state that results in the next state having the same distribution. Now, suppose that the initial state X_0 has a Poisson distribution with mean α. That is, assume that the number of these families initially in the hotel is Poisson with mean α. Let R denote the number of families that remain in the hotel at the beginning of the next day. Note if each of a Poisson distributed (with mean α) number of events occurs with probability q, then the total number of these events that occur is Poisson distributed with mean αq, it follows that R is a Poisson random variable with mean αq.

In addition, the number of new families that check in during the day, call it N, is Poisson with mean λ, and is independent of R. Hence,

$$R + N \sim \text{Poi}(\lambda + \alpha q).$$

Consequently, if we choose α so that

$$\alpha = \lambda + \alpha q,$$

then

$$\alpha = \frac{\lambda}{p}.$$

Thus, the stationary probabilities are

$$\pi_i = e^{-\lambda/p}(\lambda/p)^i/i!, \ i \geq 0.$$

Remark. The preceding model has an important generalization. Namely, consider an organization whose workers are of r distinct types. For instance, the organization could be a law firm and it lawyers could either be juniors, associates, or partners. Suppose that a worker who is current type i will in the next period become type j with probability $q_{i,j}$ for $j = 1, \ldots, r$ or will leave the organization with probability $1 - \sum_{j=1}^{r} q_{ij}$. In addition, suppose that new workers are hired each period, and that the number of types $1, \ldots, r$ workers hired are independent Poisson random variables with means $\lambda_1, \ldots, \lambda_r$. If we let $\vec{X}_n = (X_n(1), \ldots, X_n(r))$, where $X_n(i)$ is the number of type i workers in the organization at the beginning of period n, then $\{\vec{X}_n, n \geq 0\}$ is a MC.

To compute its stationary probability distribution, suppose that the initial state is chosen so that the number of workers of different types are independent Poisson random variables, with α_i being the mean number of type i workers. That is, suppose that $X_0(1), \ldots, X_0(r)$ are independent Poisson random variables with respective means $\alpha_1, \ldots, \alpha_i$. Also, let $N_j, j = 1, \ldots, r$ be the number of new type j workers hired during the initial period. Now fix i, and for $j = 1, \ldots, r$, let $M_i(j)$ be the number of the $X_0(i)$ type i workers who become type j in the next period. Then because each of the Poisson number $X_0(i)$ of type i workers will independently become type j with probability $q_{i,j}, j = 1, \ldots, r$, we have $M_i(1), \ldots, M_i(r)$ are independent Poisson random variables with $M_i(j)$ having mean $\alpha_i q_{i,j}$. Because $X_0(1), \ldots, X_0(r)$ are, by assumption, independent, we also conclude that the random variables $M_i(j), i, j = 1, \ldots, r$ are all independent. Then

$$X_1(j) = N_j + \sum_{i=1}^{r} M_i(j), \ \forall \ j = 1, \ldots, r$$

are independent Poisson variables with means

$$E[X_1(j)] = \lambda_j + \sum_{i=1}^{r} \alpha_i q_{ij}.$$

Hence, if $\alpha_1, \ldots, \alpha_r$ satisfied

$$\alpha_j = \lambda_j + \sum_{i=1}^{r} \alpha_i q_{i,j}, \ j = 1, \ldots, r,$$

then $\vec{X}_1$ would have the same distribution as $\vec{X}_0$.

Theorem 4.58. *(a) If j is transient or recurrent null, then for any $i \in E$,*

$$\lim_{n\to\infty} P_{ij}^n = 0.$$

(b) If j is recurrent non-null aperiodic, then

$$\pi_j = \lim_{n\to\infty} P_{jj}^n > 0,$$

and

$$\lim_{n\to\infty} P_{ij}^n = f_{ij}\pi_j.$$

Example 4.59. Suppose $\{X_n\}_{n\geq 0}$ is a homogeneous birth-death process having state space $E = \{0, 1, 2, 3 \cdots\}$, meaning its transition matrix is as follows :

$$P_{0,0} = 1 - p, \ P_{0,1} = p,$$

$$P_{i,i-1} = q, \ P_{i,i} = r, \ P_{i,i+1} = p, \ \forall \ i \in \mathbb{Z}^+.$$

Assume further that $p, q > 0$: under this assumption, $\{X_n\}_{n\in bbN}$ is clearly an irreducible Markov chain.

We can obtain if $p < q$,

$$\pi_i = \left(1 - \frac{p}{q}\right)\left(\frac{q}{q}\right)^i.$$

4.7 Examples

Consider a process $\{X_n, n = 0, 1, \cdots\}$, which takes on the values $0, 1$ or 2. Suppose

$$P(X_{n+1} = j | X_n = i, X_{n-1} = i_{n-1}, \ldots, X_0 = i_0) = \begin{cases} P_{ij}^I, & \text{when } n \text{ is even} \\ P_{ij}^{II}, & \text{when } n \text{ is odd} \end{cases},$$

where $\sum_{j=0}^{2} P_{ij}^{\mathrm{I}} = \sum_{j=0}^{2} P_{ij}^{\mathrm{II}} = 1, \ i = 0, 1, 2$. Is $\{X_n, n \geq 0\}$ a Markov chain? If not, then show how, by enlarging the state space, we may transform it into a Markov chain.

No. Let the state space be $S = \{0, 1, 2, \bar{0}, \bar{1}, \bar{2}\}$, where state $i(\bar{i})$ signifies that the present value is i, and the present day is even (odd).

Example 4.60. Suppose that coin 1 has probability 0.7 of coming up heads, and coin 2 has probability 0.6 of coming up heads. If the coin flipped today comes up heads, then we select coin 1 to flip tomorrow, and if it comes up tails, then we select coin 2 to flip tomorrow. If the coin initially flipped is equally likely to be coin 1 or coin 2, then what is the probability that the coin flipped on the third day after the initial flip is coin 1? Suppose that the coin flipped on Monday comes up heads. What is the probability that the coin flipped on Friday of the same week also comes up heads?

Let the state on any day be the number of the coin that is flipped on that day. Then

$$P = \begin{pmatrix} 0.7 & 0.3 \\ 0.6 & 0.4 \end{pmatrix}.$$

Then

$$P^3 = \begin{pmatrix} 0.667 & 0.333 \\ 0.666 & 0.334 \end{pmatrix}.$$

Then the probability that the coin flipped on the third day is coin 1 is

$$\frac{1}{2}\left[P^3_{11} + P^3_{21}\right] = 0.6665.$$

If we let the state be 0 when the most recent flip lands heads and let it equal 1 when it lands tails, then the sequence of states is a MC with

$$P = \begin{pmatrix} 0.7 & 0.3 \\ 0.6 & 0.4 \end{pmatrix}.$$

The desired probability is

$$P^4_{0,0} = 0.6667.$$

Example 4.61. Let $E = \{0, 1, 2\}$ and

$$P = \begin{pmatrix} 0.5 & 0.4 & 0.1 \\ 0.3 & 0.4 & 0.3 \\ 0.2 & 0.3 & 0.5 \end{pmatrix}.$$

Find the probability that it is not in state 2 on any the following 3 steps. The answer is $1 - Q^3_{0,2}$ with

$$Q = \begin{pmatrix} 0.5 & 0.4 & 0.1 \\ 0.3 & 0.4 & 0.3 \\ 0 & 0 & 1 \end{pmatrix}.$$

It was in state 3 four steps ago. Given that it hasn't in state 0, what is the probability it is in state 2 in this step?

Since 2 is a absording state given the Q, if it is in state 2 now, it never enter the state 2 in these four steps. So the answer is $\frac{Q^4_{22}}{1-Q^4_{2,0}}$ with

$$Q = \begin{pmatrix} 1 & 0 & 0 \\ 0.3 & 0.4 & 0.3 \\ 0.2 & 0.3 & 0.5 \end{pmatrix}.$$

Example 4.62.

$$P = \begin{pmatrix} \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{2} & 0 & \frac{1}{2} \end{pmatrix}.$$

Then $\{0, 2\}$ is recurrent and $\{1\}$ is transient.

Example 4.63.

$$P = \begin{pmatrix} \frac{1}{4} & \frac{3}{4} & 0 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \frac{1}{3} & \frac{2}{3} & 0 \\ 1 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

Then $\{0, 1\}$ is recurrent, $\{2\}$ is recurrent, $\{3\}$ is transient, $\{4\}$ is transient.

Example 4.64. Show that if state i is recurrent and state i does not communicate with state j, then $P_{ij} = 0$. This implies that once a process enters a recurrent class of states it can never leave that class. For this reason, a recurrent class is often referred to as a closed class.
Proof. Assume P_{ij} were positive. Note P^n_{ji} would be 0 for all n, otherwise, i and j would communicate. But then the process, starting in i, has a positive probability of at least P_{ij} of never returning to i. This contradicts the recurrence of i. Hence $P_{ij} = 0$.

Example 4.65. For a random walk with $0 < p < 1$, use the SLLN to give another proof that the MC is transient when $p \neq \frac{1}{2}$.
Proof. Note the state at time n can be written as $\sum_{i=1}^n Y_i$, where Y_i's are iid with $P(Y_i = 1) = p = 1 - P(Y_i = -1)$. By the SLLN,

$$\sum_{i=1}^n \frac{Y_i}{n} \overset{a.s.}{\to} E[Y] = 2p - 1.$$

Hence if $p > 1/2$, then $E[Y] > 0$, and so the average of the Y_i's converges in this case to a positive number, which implies $\sum_{i=1}^n Y_i \to \infty$. Hence, state 0 can be visited only a finite number of times and so must be transient. Similarly, for $p < 1/2$.

Example 4.66. Coin 1 comes up heads with probability 0.6 and coin 2 with probability 0.5. A coin is continually flipped until it comes up tails, at which time that coin is put aside and we start flipping the other one.

(a) What proportion of flips use coin 1?

Let the state at time n is the coin i $(i = 1, 2)$ to be flipped , then

$$P = \begin{pmatrix} 0.6 & 0.4 \\ 0.5 & 0.5 \end{pmatrix}.$$

Then we have

$$\pi_1 = \frac{5}{9}, \quad \pi_2 = \frac{4}{9}.$$

So the proportion of flips that use coin 1 is $\frac{5}{9}$.

(b) If we start the process with coin 1, what is the probability that coin 2 is used on the fifth flip.

$$P_{12}^4 = 0.4444.$$

Example 4.67. Let Y_n be the sum of n independent rolls of a fair die. Find

$$\lim_{n\to\infty} P(Y_n \text{ is a multiple of } 13).$$

Let X_n denote the value of Y_n modulo 13. Now X_n is a Markov chain with states $0, 1, \ldots, 12$. It is easy to verify that $\sum_i P_{ij} = 1,\ \forall\ j$. For instance, for $j = 3$,

$$\sum_i P_{ij} = P_{2,3} + P_{1,3} + P_{0,3} + P_{12,3} + P_{11,3} + P_{10,3} = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = 1.$$

So $\{Y_n\}_{n\in\mathbb{N}}$ is **finitely** doubly stochastic. Hence, $r_i = \frac{1}{13}$.

Example 4.68. Each morning an individual leaves his house and goes for a run. He is equally likely to leave either from his front or back door. Upon leaving the house, he chooses a pair of running shoes (or goes running barefoot if there are no shoes at the door from which he departed). On his return he is equally likely to enter, and leave his running shoes, either by the front or back door. If he owns a total of k pairs of running shoes, what proportion of the time does he run barefooted?

Alternatively, let X_n denote the number of pairs of shoes at the front door right before the nth run, and let D_n equal either F if he leaves from the front door when he goes for the nth run, or B if he leaves from the front door when he goes for the nth run. One can see that $\{(X_n, D_n)\}_{n\geq 0}$ represents a DTMC on the state space

$$E = \{(0, F), (0, B), (1, F), (1, B), \ldots, (k, F), (k, B)\}.$$

The transition probabilities of this chain are as follows:

$$P_{(0,F),(0,F)} = 1/2,$$
$$P_{(0,F),(0,B)} = 1/2,$$
$$P_{(k,B),(k,B)} = 1/2,$$
$$P_{(k,B),(k,F)} = 1/2,$$

and further

$$P_{(0,B),(0,B)} = 1/4,$$
$$P_{(0,B),(0,F)} = 1/4,$$
$$P_{(0,B),(1,B)} = 1/4,$$
$$P_{(0,B),(1,F)} = 1/4,$$
$$P_{(k,F),(k-1,B)} = 1/4,$$
$$P_{(k,F),(k-1,F)} = 1/4,$$
$$P_{(k,F),(k,B)} = 1/4,$$

$$P_{(k,F),(k,F)} = 1/4,$$

and finnally, $\forall\ j \in [k-1]$,

$$\begin{aligned}
P_{(j,B),(j,B)} &= 1/4,\\
P_{(j,B),(j+1,F)} &= 1/4,\\
P_{(j,B),(j+1,B)} &= 1/4,\\
P_{(j,B),(j+1,F)} &= 1/4,\\
P_{(j,F),(j-1,B)} &= 1/4,\\
P_{(j,F),(j-1,F)} &= 1/4,\\
P_{(j,F),(j,B)} &= 1/4,\\
P_{(j,F),(j,F)} &= 1/4.
\end{aligned}$$

This chain is irreducible, and its state space is finite, hence all states are positive recurrent, and this implies that the chain has a unique stationary distribution π. Moreover, one can check this chain is also doubly stochastic, so the long-run fraction of time the jogger runs barefoot is given by

$$\begin{aligned}
\lim_{n\to\infty} \frac{1}{n} \sum_{k=1}^{n} \left[\mathbb{1}_{\{(X_k,D_k)=(0,F)\}} + \mathbb{1}_{\{(X_k,D_k)=(k,B)\}}\right] &= \pi_{(0,F)} + \pi_{(k,B)}\\
&= \frac{1}{2(k+1)} + \frac{1}{2(k+1)}\\
&= \frac{1}{k+1}.
\end{aligned}$$

Example 4.69. Three out of every four trucks on the road are followed by a car, while only one out of every five cars is followed by a truck. What fraction of vehicles on the road are trucks?

Imagine sitting on the side of the road watching vehicles go by. If a truck goes by, the next vehicle will be car with probability 3/4 and will be a truck with probability 1/4. If a car goes by, the next vehicle will be a car with probability 4/5 and will be a truck with probability 1/5.

Let X_n be 0 if the nth vehicle is a car and letting it be 1 if the vehivle is a truck gives rise to a two-state MC with

$$P = \begin{pmatrix} \frac{4}{5} & \frac{1}{5} \\ \frac{3}{4} & \frac{1}{4} \end{pmatrix}.$$

Then

$$\pi_0 = \frac{15}{19}, \quad \pi_1 = \frac{4}{19}.$$

That is, 4 out of every 19 cars is a truck.

Example 4.70. A professor continually gives exams to her students. She can give three possible types of exams, and her class is graded as either having done well or badly. Let p_i denote the probability that the class does well on a type i exam, then the next exam is equally likely to be any of the three types. If the class does badly, then the next exam is always type 1. What proportion of exams are type i,$i = 1, 2, 3$?

A professor continually gives exams to her students. She can give three possible types of exams, and her class is graded as either having done well or badly. Let p_i denote the probability that the

class does well on a type i exam, and suppose that $p_1 = 0.3,\ p_2 = 0.6$ and $p_3 = 0.9$. If the class does well on an exam, then the next exam is equally likely to be any of the three types. If the class does badly, then the next exam is always type 1. What proportion of exams are type i, $i = 1, 2, 3$?

Consider the MC whose state at time n is the type of the exam. The transition probabilities of this MC are obtained by conditioning on the performance of the class. Then

$$P_{11} = 0.3 \cdot 1/3 + 0.7 \cdot 1 = 0.8,$$

$$P_{12} = P_{13} = 0.3 \cdot 1/3 = 0.1,$$

$$P_{21} = 0.6 \cdot 1/3 + 0.4 \cdot = 0.6,$$

$$P_{22} = P_{23} = 0.6 \cdot 1/3 = 0.2,$$

$$P_{31} = 0.9 \cdot 1/3 + 0.1 \cdot 1 = 0.4,$$

$$P_{32} = P_{33} = 0.9 \cdot 1/3 = 0.3.$$

Since $P_{i2} = P_{i3},\ \forall\ i$, we have $\pi_2 = \pi_3$. As a result,

$$\pi_1 = 5/7, \quad \pi_2 = \pi_3 = 1/7.$$

Example 4.71. Show that the stationary probabilities for the Markov chain having transition probabilities $P_{i,j}$ are also the stationary probabilities for the Markov chain whose transition probabilities Q_{ij} are given by

$$Q_{ij} = P_{ij}^k.$$

Proof. We must show

$$\pi_j = \sum_i \pi_i P_{ij}^k,$$

which follows since the right-hand side is equal to the probability that the MC with transition probabilities P_{ij} will be in state j at time k when its initial state is chosen according to its stationary probabilities, which is equal to its stationary probability being in state j.

Example 4.72. An individual possesses r umbrellAs that he employs in going from his hom to office, and vice versa. If he is at home (the office) at the beginning (end) of a day and it is rainning, then he will take an umbrellA with him to the office (home), provided there is one to be taken. If it is not raining, then he never takes an umbrellA. Assume that, independent of the past, it rains at the beginning (end) of a day with probability p. Find the fraction of time our man gets wet.
Let the state be the number of umbrellAs he has at his present location. Then

$$P_{0r} = 1,$$

$$P_{i,r-i} = 1 - p,\ \forall\ i = 1, \ldots, r,$$

$$P_{i,r-i+1} = p,\ \forall\ i = 1, \ldots, r,$$

Example 4.73. Let $\{X_n, n \geq 0\}$ denote an ergodic MC with limiting probabilities π_i. Define the process $\{Y_n, n \geq 1\}$ by $Y_n = (X_{n-1}, X_n)$. That is, Y_n keeps track of the last two states of the original chain. Is $\{Y_n, n \geq 1\}$ a MC? If so, determine its transition probabilities and find

$$\lim_{n\to\infty} P\left(Y_n = (i, j)\right).$$

$\{Y_n, n \geq 1\}$ is a Markov chain with states (i, j).

$$P_{(i,j)(k,l)} = \begin{cases} 0, & j \neq k \\ P_{jl}, & j = k \end{cases}.$$

$$\lim_{n\to\infty} P\left(Y_n = (i,j)\right) = \lim_{n\to\infty} P(X_n = i, X_{n+1} = j) = \lim_{n\to\infty} P(X_n = i)P_{ij} = \pi_i P_{ij}.$$

Example 4.74. Consider a MC in steady state. Say that a k length run of zeroes ends at time m if

$$X_{m-k-1} \neq 0,\ X_{m-k} = X_{m-k+1} = \cdots = X_{m-1} = 0,\ X_m \neq 0.$$

Show that the probability of this event is $\pi_0 (P_{00})^{k-1}(1 - P_{00})^2$.

$$\begin{aligned} P &= \sum_{i\neq 0} P(X_{m-k} = X_{m-k+1} = \cdots = X_{m-1} = 0, X_m \neq 0 | X_{m-k-1} = i)\pi_i \\ &= \sum_{i\neq 0} P_{i0}(P_{00})^{k-1}(1 - P_{00})\pi_i \\ &= (P_{00})^{k-1}(1 - P_{00}) \sum_{i\neq 0} \pi_i P_{i0} \\ &= (P_{00})^{k-1}(1 - P_{00}) \left(\sum_i \pi_i P_{i0} - \pi_0 P_{00} \right) \\ &= (P_{00})^{k-1}(1 - P_{00})(\pi_0 - \pi_0 P_{00}). \end{aligned}$$

Example 4.75. Let $P^{(1)}$ and $P^{(2)}$ denote transition probability matrices for ergodic MCs having the same state space. Let π^1 and π^2 denote the stationary (limiting) probability vectors for the two chains. Consider a process defined as follows:

(a) $X_0 = 1$. A coin is then flipped and if it comes up heads, then the remaining states $X_1, \cdots$ are obtained from the transition probability matrix $P^{(1)}$ and if tails from the matrix $P^{(2)}$. Is $\{X_n, n \geq 0\}$ a Markov chain? If $p = P(\text{coin comes up heads})$, what is $\lim_{n\to\infty} P(X_n = i)$?

No.

$$\lim_{n\to\infty} P(X_n = i) = p\pi^1(i) + (1 - p)\pi^2(i).$$

(b) $X_0 = 1$. At each state the coin is flipped and if it comes up heads, then the next state is chosen according to $P^{(1)}$ and if tails comes up, then it is chosen according to P^2. In this case do the successive states constitute a Markov chain? If so, determine the transition probabilities.

Yes.

$$P_{ij} = pP_{ij}^{(1)} + (1 - p)P_{ij}^{(2)}.$$

Example 4.76. A taxi driver provides service in two zones of a city. Fares picked up in zone A will have destinations in zone A with probability 0.6 or in zone B with probability 0.4. Fares picked up in zone B will have destinations in zone A with probability 0.3 or in zone B with probability 0.7. The driver's expected profit for a trip entirely in zone A is 6; for a trip entirely in zone B is 8; and for a trip that involves both zones is 12. Find the taxi driver's average profit per trip.

Let the state be successive zonal pickup locations. Then

$$P = \begin{pmatrix} 0.6 & 0.4 \\ 0.3 & 0.7 \end{pmatrix}.$$

Then

$$\pi_A = \frac{3}{7}, \quad \pi_B = \frac{4}{7}.$$

Let X note the profit in a trip. Then

$$\begin{aligned} E[X] &= \frac{3}{7}E[X|A] + \frac{4}{7}E[X|B] \\ &= \frac{3}{7}(0.6 \cdot 6 + 0.4 \cdot 12) + \frac{4}{7}(0.3 \cdot 12 + 0.7 \cdot 8) \\ &= \frac{62}{7}. \end{aligned}$$

Example 4.77. Consider the Ehrenfest urn model in which M molecules are distributed between two urns, and at each time point one of the molecules is chosen at random and is then removed from its urn and placed in the other one. Let X_n denote the number of molecules in urn 1 after the nth switch and let $\mu_n = E[X_n]$. Show that

$$\mu_{n+1} = 1 + (1 - 2/M)\mu_n.$$

Given X_n,

$$X_{n+1} = \begin{cases} X_n + 1, & \text{with probability } \frac{M - X_n}{M} \\ X_n - 1, & \text{with probability } \frac{X_n}{M} \end{cases}.$$

Hence

$$E[X_{n+1}|X_n] = X_n + \frac{M - X_n}{M} - \frac{X_n}{M} = X_n + 1 - \frac{2X_n}{M}.$$

Then

$$E[X_{n+1}] = E\left[E[X_{n+1}|X_n]\right] = 1 + (1 - 2/M)E[X_n].$$

Example 4.78. A particle moves among $n+1$ vertices that are situated on a circle in the following manner. At each step it moves one step either in the clockwise direction with probability p or the counterclockwise direction with probability $q = 1 - p$. Starting at a specific state, call it state 0, let T be the time of the first return to state 0. Find the probability that all states have been visited by time T.

Let A be the event that all states have been visited by time T. Then conditioning on the direction of the first step gives

$$\begin{aligned} P(A) &= P(A|\text{clockwise})p + P(A|\text{counterclockwise})q \\ &= p\frac{1 - q/p}{1 - (q/p)^n} + q\frac{1 - p/q}{1 - (p/q)^n}. \end{aligned}$$

Chapter 5

The Exponential Distribution and the Poisson Process

Let

$$T \sim \exp(1/\alpha).$$

Then

$$\begin{aligned} E\left[\int_0^T R(x)dx\right] &= E\left[\int_0^\infty R(x)\mathbb{1}_{\{T\geq x\}}\right]dx = \int_0^\infty E\left[R(x)\mathbb{1}_{\{T\geq x\}}\right]dx \\ &= \int_0^\infty E[R(x)]E\left[\mathbb{1}_{\{T\geq x\}}\right]dx = \int_0^\infty E[R(x)]P(T\geq x)dx \\ &= \int_0^\infty e^{-\alpha x}E[R(x)]dx. \end{aligned}$$

Remark. If the lifetime X is exponentially distributed, then the distribution of the remaining amount of time that it survives is the same as the original lifetime distribution, i.e.,

$$P(X > s+t) = P(X > s)P(X > t).$$

Example 5.1. The dollar amount of damage involved in an automobile accident is an exponential random variable with mean 1000. Of this, the insurance company only pays that amount exceeding (the deductible amount of) 400. Find the expected value and the standard deviation of the amount the insurance company pays per accident.

If X is the dollar amount of damage resulting from an accident, then the amount paid by the insurance company is $(X-400)^+$. Let $Y = (X-400)^+$ and $I = \mathbb{1}_{\{X>400\}}$, then by the memoryless of exponential distribution,

$$E\left[Y|I=1\right] = 1000,$$

$$E\left[Y|I=0\right] = 0,$$

$$\mathrm{Var}\left(Y|I=1\right) = 1000^2,$$

$$\mathrm{Var}\left(Y|I=0\right) = 0,$$

which can be conveniently written as

$$E\left[Y|I\right] = 1000I,$$
$$\mathrm{Var}\left(Y|I\right) = 1000000I.$$

Thus,

$$E[Y] = E\left[E[Y|I]\right] = 10^3 E[I] = 1000e^{-0.4} \approx 670.32,$$
$$\mathrm{Var}(Y) = E\left[\mathrm{Var}(Y|I)\right] + \mathrm{Var}\left(E[Y|I]\right) = 1000000e^{-0.4} + 1000000e^{-0.4}\left(1 - e^{-0.4}\right).$$

5.0.1 Minimum of exponential distribution

Theorem 5.2. *Suppose that X_1 and X_2 are independent exponential random variables with respective means $1/\lambda_1$ and $1/\lambda_2$; then*

$$P(\min(X_1, X_2) = X_1) = P(X_1 < X_2) = \int_0^\infty P(X_1 < X_2 | X_1 = x)\lambda_1 e^{-\lambda_1 x} dx = \frac{\lambda_1}{\lambda_1 + \lambda_2}.$$

Corollary 5.3. Let $X_1, \ldots, X_n$ be independent exponential random variables, with respective rates $\lambda_1, \ldots, \lambda_n$, then

$$P\left(X_i = \min_{1 \le j \le n} X_j\right) = P\left(X_i < \min_{j \ne i} X_j\right) = \frac{\lambda_i}{\sum_{j=1}^n \lambda_j}.$$

Theorem 5.4. *If $X_1, \ldots, X_n$ are independent exponential random varaibles with respective rates $\lambda_1, \ldots, \lambda_n$, then $\min_i X_i$ and the rank ordering of the X_i are independent.*

Proof. Consider the conditional probability that $X_{i_1} < \cdots < X_{i_n}$ given that the minimal value is greater than t. Because $\min_i X_i > t$ means that all the X_i are greater than t, it follows from the lack of memory property of exponential random variables that their remaining lives beyond t remain independent exponential random variables with their original rates. Consequently,

$$\begin{aligned} P\left(X_{i_1} < \cdots < X_{i_n} | \min_i X_i > t\right) &= P\left(X_{i_1} - t < \cdots < X_{i_n} - t | \min_i X_i > t\right) \\ &= P(X_{i_1} < \cdots < X_{i_n}). \end{aligned}$$

□

Example 5.5. Let

$$X \sim \exp(1/\lambda),$$
$$Y \sim \exp(1/\mu).$$

Let $t \ge 0$.

$$\begin{aligned} P\left(\min(X, Y) > t | X < Y\right) &= \frac{P(t < X < Y)}{P(X < Y)} = \frac{\int_0^\infty P(t < x < Y)\lambda e^{-\lambda x} dx}{P(X < Y)} \\ &= \frac{\int_t^\infty e^{-\mu x}\lambda e^{-\lambda x} dx}{P(X < Y)} = e^{-(\mu+\lambda)t}. \end{aligned}$$

Alternative:

$$\begin{aligned} P\left(\min(X, Y) > t | X < Y\right) &= \frac{P(t < X < Y)}{P(X < Y)} = \frac{\int_t^\infty P(t < X < y)\mu e^{-\mu y} dy}{P(X < Y)} \\ &= \frac{\int_t^\infty \left(e^{-\lambda t} - e^{-\lambda y}\right)\mu e^{-\mu y} dy}{P(X < Y)} = e^{-(\mu+\lambda)t}. \end{aligned}$$

Example 5.6. Suppose you arrive at a post office having two clerks at a moment when both are busy but there is no one else waiting in line. You will enter service when either clerk becomes free. If service time for clerk i are exponential with rate $\lambda_i, i = 1, 2$, find $E[T]$, where T is the amount of time that you spend in the post office.

Let R_i denote the remaining service time of the customer with cleak $i, i = 1, 2$, and note by the lack of memory property of exponentials, that R_1 and R_2 are independent exponential random variables with respective rates λ_1 and λ_2. Then

$$\begin{aligned} E[T] &= E[T|R_1 < R_2]P(R_1 < R_2) + E[T|R_2 \leq R_1]P(R_2 \leq R_1) \\ &= E[T|R_1 < R_2]\frac{\lambda_1}{\lambda_1 + \lambda_2} + E[T|R_2 \leq R_1]\frac{\lambda_2}{\lambda_1 + \lambda_2}. \end{aligned}$$

Now, with S denoting your service time

$$\begin{aligned} E[T|R_1 < R_2] &= E[R_1 + S|R_1 < R_2] \\ &= E[R_1|R_1 < R_2] + E[S|R_1 < R_2] \\ &= E[\min(R_1, R_2)|R_1 < R_2] + E[S|R_1 < R_2] \\ &= \frac{1}{\lambda_1 + \lambda_2} + \frac{1}{\lambda_1}. \end{aligned}$$

Similarly,

$$E[T|R_2 \leq R_1] = \frac{1}{\lambda_1 + \lambda_2} + \frac{1}{\lambda_2}.$$

Thus,

$$E[T] = \frac{3}{\lambda + \lambda_2}.$$

Alternative:

$$E[T] = E\left[\min(R_1, R_2) + S\right] = E\left[\min(R_1, R_2)\right] + E[S] = \frac{1}{\lambda_1 + \lambda_2} + E[S],$$

where

$$E[S] = E[S|R_1 < R_2]\frac{\lambda_1}{\lambda_1 + \lambda_2} + E[S|R_2 \leq R_1]\frac{\lambda_2}{\lambda_1 + \lambda_2} = \frac{2}{\lambda_1 + \lambda_2}.$$

Example 5.7. Suppose that customers are in line to receive service that is provided sequentially by a server; whenever a service is completed, the next person in line enters the service facility. However, each waiting customer will only wait an exponentially distributed time with rate θ; if its service has not yet begun by this time then it will immediately depart the system. These exponential times, one for each waiting customer, are independent. In addition, the service times are independent exponential random variables with rate μ. Suppose that someone is presently being served and consider the person who is n^{th} in line.

(a) Find P_n, the probability that this customer is eventually served.

(b) Find W_n, the conditional expected amount of time this person spends waiting in line given that she is eventually served.

Consider the $n + 1$ random varibles consisting of the remaning service time of the person in service along with the n additional exponential departure times with rate θ of the first n in line.

(a) Think of yourself being the nth customer in line. Conditioning on whether or not the next event to occur is you departing from the line. Note the probability of this occuring is $\frac{\theta}{n\theta+\mu}$. By the memoryless property of exponential random variables,

$$\begin{aligned} P_n &= \left(1 - \frac{\theta}{n\theta+\mu}\right) P_{n-1} = \frac{(n-1)\theta+\mu}{n\theta+\mu} P_{n-1} \\ &= \frac{(n-2)\theta+\mu}{n\theta+\mu} P_{n-2} = \cdots \\ &= \frac{\theta+\mu}{n\theta+\mu} P_1 = \frac{\mu}{n\theta+\mu}. \end{aligned}$$

(b) By the memoryless property of exponential random variables,

$$W_n = \frac{1}{n\theta+\mu} + W_{n-1} = \cdots = \sum_{i=1}^{n} \frac{1}{i\theta+\mu}.$$

Theorem 5.8. *Let $X_1, \ldots, X_n$ be independent exponential random variables with respective rates $\lambda_1, \ldots, \lambda_n$, where $\lambda_i \neq \lambda_j$ when $i \neq j$. Let T be independent of these random variables and suppose that*

$$\sum_{j=1}^{n} P_j = 1, \text{ where } P_j = P(T = j).$$

The random variable X_T is said to be a hyperexponential random variable.

$$1 - F_{X_T}(t) = P(X_T > t) = \sum_{i=1}^{n} P(X_T > t | T = i) P(T = i) = \sum_{i=1}^{n} P(X_i > t) P(T = i) = \sum_{i=1}^{n} P_i e^{-\lambda_i t}.$$

$$f_T(t) = \sum_{i=1}^{n} \lambda_i P_i e^{-\lambda_i t}.$$

Theorem 5.9. *Let $X_i, i = 1, \ldots, n$ be independent exponential random variables with respective rates $\lambda_i, i = 1, \ldots, n$, and suppose that $\lambda_i \neq \lambda_j$ for $i \neq j$. Then the random variable $\sum_{i=1}^{n} X_i$ is said to be a hypoexponential random variable and*

$$f_{\sum_{i=1}^{n} X_i}(t) = \sum_{i=1}^{n} C_{i,n} \lambda_i e^{-\lambda_i t},$$

where

$$C_{i,n} = \prod_{j \neq i} \frac{\lambda_j}{\lambda_j - \lambda_i}.$$

Then

$$P\left(\sum_{i=1}^{n} X_i > t\right) = \sum_{i=1}^{n} C_{i,n} e^{-\lambda_i t}.$$

Example 5.10. Let $X_1, \ldots, X_m$ be independent exponential random variables with respective rates $\lambda_1, \ldots, \lambda_m$, where $\lambda_i \neq \lambda_j$ when $i \neq j$. Let N be independent of these random variabless and suppose that $\sum_{n=1}^{m} P_n = 1$, where $P_n = P(N = n)$. The random variable

$$Y = \sum_{j=1}^{N} X_j$$

is said to be a Coxian random variable.

$$f_Y(t) = \sum_{n=1}^{m} f_{Y|N=n}(t) P_n = \sum_{n=1}^{m} f_{\sum_{i=1}^{n} X_i}(t) P_n = \sum_{n=1}^{m} P_n \sum_{i=1}^{n} C_{i,n} \lambda_i e^{-\lambda_i t}.$$

5.1 The Poisson process

5.1.1 Counting processes

Definition 5.11. A stochastic process $\{N(t), t \geq 0\}$ is said to be a *counting process* if $N(t)$ represents the total number of "events" that occur in the interval $[0, t]$ consisting of random variables or points $\{T_n\}_{n \in \mathbb{Z}^+}$, where $0 \leq T_1 \leq T_2 \leq \cdots$, i.e.,

$$N(t) = \sum_{n=1}^{\infty} \mathbb{1}_{\{T_n \leq t\}}.$$

In fact, we could also define, for each Borel set A,

$$N(A) := \sum_{n=1}^{\infty} \mathbb{1}_{\{T_n \in A\}}.$$

Remark. It is clear that if all of the points $\{T_n\}_{n \geq 0}$ are known, then the entire sample path of $\{N(t); t \geq 0\}$ is known too. At the same time, one can easily show that if the entire sample path of $\{N(t); t \geq 0\}$ is known, then each points $\{T_n\}_{n \in \mathbb{Z}^+}$ is known as well.

Furthermore,

$$\{T_n \leq t\} = \{N(t) \geq n\}, \ \forall \ n \in \mathbb{Z}^+, t \geq 0.$$

From the simple fact, we can immediately conclude that the marginal distribution of the counting process determine the distribution of each T_n, and vice versa.

Example 5.12. If we had let $N(t)$ equal the number of persons in the store at time t, then $\{N(t); t \geq 0\}$ would not be a counting process.

Definition 5.13. From the definition, we see that for a *counting process* $N(t)$ must satisfy:

(a)
$$N(t) \geq 0.$$

(b) $N(t)$ is integer valued.

(c) If $s < t$, then $N(s) \leq N(t)$.

(d) For $s < t$, $N(t) - N(s)$ equals the number of events that occurs in the interval $(s, t]$.

Definition 5.14. A counting process is said to posses *independent increments* if the numbers of events that occur in disjoint time intervals are independent.

Definition 5.15. A counting process is said to possess *stationary increments* if the distribution if the number of events that occur in any interval of time depends only on the length of the time interval. In other words, the process has stationary increments if the number of events in the interval $(s, s + t)$ has the same distribution for all s.

5.1.2 Definition of the Poisson process

Definition 5.16. The function $f(\cdot)$ is said to be $o(h)$ if

$$\lim_{h \to 0} \frac{f(h)}{h} = 0,$$

namely, for h small, $f(h)$ must be small compared with h.

Remark. If X is continuous with density f and failure rate function $\lambda(t)$, then the approximate statements

$$P(t < X < t + h) \approx f(t)h,$$
$$P(t < X < t + h | X > t) \approx \lambda(t)h,$$

can be precisely expressed as

$$P(t < X < t + h) = f(t)h + o(h),$$
$$P(t < X < t + h | X > t) = \lambda(t)h + o(h).$$

Definition 5.17. The counting process $\{N(t), t \geq 0\}$ is said to be a *Poisson process* with rate $\lambda > 0$ if the following axioms hold:

(a)
$$N(0) = 0,$$

(b) $\{N(t), t \geq 0\}$ has independent increments,

(c)
$$P\left(N(t + h) - N(t) = 1\right) = P\left(N(t, t + h] = 1\right) = \lambda h + o(h),$$

(d)
$$P\left(N(t + h) - N(t) \geq 2\right) = o(h).$$

The preceding is called a *homogeneous Poisson process* because the number of events in any interval of length t is Poisson distributed with mean λt.

Remark. (c) and (d) is equivalent to

$$N(t) \sim \text{Poi}(\lambda t).$$

Remark. One interesting consequence of (b), (c) and (d) is that Poisson processes have stationary increments, namely, for $0 \leq s < t$, we have

$$N(s, t] \stackrel{d}{=} N(t - s).$$

5.1.3 Interarrival and waiting time distributions

Definition 5.18. Consider a Poisson process, and let us denote the time of the first event by X_1. Further, for $n \in \mathbb{Z}^{\geq 2}$, let X_n denote the elapsed time between the $(n-1)^{\text{th}}$ and the nth event. The sequence $\{X_n, n = 1, 2 \cdots\}$ is called the sequence of *interarrival times*.

The key to studying the joint distribution of $T_1, \ldots, T_n$ for $n \in \mathbb{Z}^+$ is to instead study the joint distribution of $X_1, \ldots, X_n$. Letting $T_0 = 0$,

$$X_n = T_n - T_{n-1}, \ \forall \ n \in \mathbb{Z}^+.$$

Proposition 5.19. $\{X_n\}_{n \in \mathbb{Z}^+}$ are independent identically distributed exponential random variables having mean $1/\lambda$.

Proof.

$$P(X_1 > t) = P\left(N(t) = 0\right) = e^{-\lambda t}, \ \forall \ t > 0.$$

So

$$T_1 \sim \exp(1/\lambda).$$

Now,

$$P(X_2 > t) = E\left[P(X_2 > t | X_1)\right], \ \forall \ t > 0$$

However, $\forall \ s > 0$,

$$P\left(X_2 > t | X_1 = s\right) = P\left(N(s, s+t] = 0 | \ X_1 = s\right) = P\left(N(s, s+t] = 0\right) = e^{-\lambda t},$$

where the last two equations follows from independent and stationary increments. Thus,

$$X_2 \sim \exp(1/\lambda).$$

Furthermore, X_2 is independent of X_1.

Alternatively, $\forall \ t_1, t_2 \geq 0$,

$$\begin{aligned}
P\left(X_1 \leq t_1, X_2 \leq t_2\right) &= \int_0^{t_1} P\left(X_2 \leq t_2 | X_1 = s\right) \lambda e^{-\lambda s} ds \\
&= \int_0^{t_1} P\left(N(s, s+t_2] \geq 1 | X_1 = s\right) \lambda e^{-\lambda s} ds \\
&= \int_0^{t_1} P\left(N(s, s+t_2] \geq 1\right) \lambda e^{-\lambda s} ds \\
&= \int_0^{t_1} \left(1 - e^{-\lambda t_2}\right) \lambda e^{-\lambda s} ds \\
&= \left(1 - e^{-\lambda t_1}\right)\left(1 - e^{-\lambda t_2}\right),
\end{aligned}$$

which proves X_1 and X_2 are iid $\exp(1/\lambda)$.

Same logic can be used to establish $\{X_n\}_{n \in \mathbb{Z}^+}$ are iid $\exp(1/\lambda)$. □

Remark. The proposition should not surprise us. The assumpton of stationary and independent increments is basically equivalent to asserting that, at any point in time, the process probabilistically restarts itself. That is, the process from any point on is independent of all that has previously occurred and also has the same distribution as the original process. In other words, the process has no memory, and hence exponential interarrival times are to be expected.

Definition 5.20. Another quantity of interest is S_n, the *arrival time* of the nth event, also called the waiting time until the nth event. It is easily seen that

$$S_n = \sum_{i=1}^{n} T_i \sim \text{Gamma}(n, 1/\lambda),\ n \geq 1.$$

Definition 5.21. Suppose we start with a sequence $\{X_n, n \geq 1\}$ of independent identically distributed exponential random variables each having mean $1/\lambda$. Now let us define a counting process by saying that the nth event of this process occurs at time

$$T_n = X_1 + \cdots + X_n.$$

The resulant counting process $\{N(t), t \geq 0\}$ will be Poisson with rate λ. A formal definition of $N(t)$ is given by $N(t) = \max\{n : T_n \leq t\}$ when $T_0 = 0$.

Definition 5.22.

$$T_{N(t)} = \sup\{T_n : T_n \leq t\}.$$
$$T_{N(t)+1} = \inf\{T_n : T_n > t\}.$$

Then for $0 \leq x < t$, $y \geq 0$, we get

$$\begin{aligned} P\left(t - T_{N(t)} \leq x, T_{N(t)+1} - t \leq y\right) &= P\left(T_{N(t)} \geq t - x, T_{N(t)+1} \leq t + y\right) \\ &= P\left(N(t) - N(t-x) \geq 1, N(t+y) - N(t) \geq 1\right) \\ &= P\left(N(t) - N(t-x) \geq 1\right) P\left(N(t+y) - N(t) \geq 1\right) \\ &= \left(1 - e^{-\lambda x}\right)\left(1 - e^{-\lambda y}\right). \end{aligned}$$

Letting $x = t$, we have $\forall\ y \geq 0$,

$$P\left(t - T_{N(t)} \leq t, T_{N(t)+1} - t \leq y\right) = P\left(T_{N(t)+1} - t \leq y\right) = P\left(N(t+y) - N(t) \geq 1\right) = 1 - e^{-\lambda y}.$$

5.1.4 Thinning of a Poisson processes

Again, suppose $\{N(t); t \geq 0\}$ is a Poisson process having rate λ and points $\{T_n\}$, but further asssume that associated with each point T_n is a label L_n. We assume $\forall\ n \in \mathbb{Z}^+$, L_n is independent of all points T_n, as well as all other labels L_k, $\forall\ k \neq n$.

In fact, each L_n is an integer-valued random variable having range $\{0, \ldots, m\}$ and pmf

$$P(L_n = k) = p_k,\ \forall\ 0 \leq k \leq m,$$

which of course implies the labels $\{L_n\}_{n \geq 1}$ form an iid sequence.

Using this sequence of labels, we construct $m+1$ counting processes $\{N_0(t); t \geq 0\}, \ldots, \{N_m(t); t \geq 0\}$, where for $t \geq 0$,

$$N_j(t) = \sum_{n=1}^{\infty} \mathbb{1}_{\{T_n \leq t, L_n = j\}},$$

represents the number of points in $[0, t]$ having label j.

Theorem 5.23. *The counting processes $\{N_0(t); t \geq 0\}, \ldots, \{N_m(t); t \geq 0\}$ are independent Poisson processes, where $\{N_j(t); t \geq 0\}$ has rate $\lambda p_j, 0 \leq j \leq m$.*

Example 5.24. If immigrants to area A arrive at a Poisson rate of ten per week, and if each immigrant is of English descent with probability $\frac{1}{12}$, then what is the probability that no people of English descent will emigrate to area A during the month of February?

The number of Englishmen emigrating to area A during the month of February is Poisson distributed with mean $4 \cdot 10 \cdot \frac{1}{12} = \frac{10}{3}$. Hence the desired probability is $e^{-10/3}$.

The result easily generalizes to the case where the classification is into any one of r possible groups, we have the following application to a model of employees moving about in an organization.

Example 5.25. Consider a system in which individuals at any time are classified as being in one of r possible states, and assume that an individual changes states in accordance with a Markov chain having transition probabilities $P_{ij}, i, j = 1, \ldots, r$. That is, if an individual is in state j during a time period then, independently of its previous states, it will be in state j during the next time period with probability P_{ij}. The individuals are assumed to move through the system independently of each other. Suppose that the number of people initially in states $1, \ldots, r$ are independent Poisson random varaibles with respective means $\lambda_1, \ldots, \lambda_r$. We are interested in determing the joint distribution of the numbers of individuals in states $1, \ldots, r$ at some time n.

For fixed i, let $N_j(i), j = 1, \ldots, r$ denote the number of those individuals, intially in state i, that are in state j at time n. Now each of the (Poisson distributed) number of people initially in state i will, independently of each other, be in state j at time n with probability P_{ij}^n. Hence, the $N_j(i), j = 1, \ldots, r$ will be independent Poisson random variables with respective means $\lambda_i P_{ij}^n, j = 1, \ldots, r$. Because the sum of independent Poisson random variables is itself a Poisson random variable, it follows that the number of individuals in state j at time $n-$namely $\sum_{i=1}^r N_j(i)-$will be independent Poisson random variables with respective means $\sum_i \lambda_i P_{ij}^n$ for $j = 1, \ldots, r$.

Example 5.26 (The coupon collecting problem)**.** There are m different types of coupons. Each time a person collects a coupon it is, independently of ones previously obtained, a type j coupon with probability p_j, $\sum_{j=1}^m p_j = 1$. Let N denote the number of coupons one needs to collect in order to have a complete collection of at least one of each type. Find $E[N]$.

Thinking of each L_n as the type of coupon collected during the nth trival, $\forall\ n \in \mathbb{Z}^+$. Let

$$N_j = \min\{n \geq 1 : L_n = j\},\ \forall\ j \in [m].$$

Then

$$N = \max_{1 \leq j \leq m} N_j.$$

However, event though each N_j is geometric with parameter p_j, the foregoing representation of N is not that useful since the random variables N_j are not independent. Fortunately, there is a clever way for us to get around this dilemma. Think of collecting coupons at the arrivals of a Poisson process.

Introduce a Poisson process $\{N(t); t \geq 0\}$ with rate $\lambda = 1$, and assign to each point T_n of $\{N(t); t \geq 0\}$ the label L_n, the nth coupon collected variable from our iid sequence of interest.

Setting $\forall\ t \geq 0$,

$$N_k(t) = \sum_{n=1}^{\infty} \mathbb{1}_{\{T_n \leq t, L_n = k\}}.$$

Then $\{N_1(t); t \geq 0\}, \ldots, \{N_m(t); t \geq 0\}$ are all independent Poisson processes.

In fact, letting $\{T_n^k\}_{n\in\mathbb{Z}^+}$ represent the points of $\{N_k(t); t \geq 0\}$, we see that $\{T_n^1\}_{n\in\mathbb{Z}^+}, \ldots, \{T_n^m\}_{n\in\mathbb{Z}^+}$ are all independent of each other as well. Let's now compute the expected value of

$$T = \max_{1\leq j\leq m} T_1^j.$$

Since $T_1^1, \ldots, T_1^m$ are all independent and $T_1^j \sim \exp(1/p_j)$, we have

$$P(T < t) = P\left(\max_{1\leq j\leq m} X_j < t\right) = P(X_j < t,\ \forall\ j \in [m]) = \prod_{j=1}^{m}\left(1 - e^{-p_j t}\right).$$

Therefore,

$$E[T] = \int_0^\infty P(T > t)dt = \int_0^\infty \left[1 - \prod_{j=1}^{m}\left(1 - e^{-p_j t}\right)\right] dt.$$

Also note

$$T = \sum_{i=1}^{N} T_i.$$

Since N and $\{T_n\}_{n\in\mathbb{Z}^+}$ are independent,

$$E[T] = E[N]E[T_1] = E[N].$$

The next probability calculation related to Poisson processes that we shall determine is the probability that n events occur in one Poisson process before m events have occured in a second and independent Poisson process. More formally let $\{N_1(t), t \geq 0\}$ and $\{N_2(t), t \geq 0\}$ be two independent Poisson processes having respective rates λ_1 and λ_2. Also, let S_n^1 denote the time of the nth event of the first process, and S_m^2 the time of the mth event of the second process. We seek

$$P(S_n^1 < S_m^2).$$

Since $S_1^1 \sim \exp(1/\lambda_1)$ and $S_2 \sim \exp(1/\lambda_2)$,

$$P(S_1^1 < S_1^2) = \frac{\lambda_1}{\lambda_1 + \lambda_2}.$$

By the memoryless property of Poisson process,

$$P(S_2^1 < S_1^2) = \left(\frac{\lambda_1}{\lambda_1 + \lambda_2}\right)^2.$$

In fact, each event that occurs is going to be an event of the $N_1(t)$ process with probability $\frac{\lambda_1}{\lambda_1+\lambda_2}$ or an event of the $N_2(t)$ process with probability $\frac{\lambda_2}{\lambda_1+\lambda_2}$, independent of all that has previously occured. In other words, the probability that the $N_1(t)$ process reaches n before the $N_2(t)$ process reaches m is just the probability that n heads will appear before m tails if one flip a coin having probability $p = \frac{\lambda_1}{\lambda_1+\lambda_2}$ of a head appearing. But by noting that this event will occur if and only if the first $n + m - 1$ tosses result in n or more heads, we see that

$$P\left(S_n^1 < S_m^2\right) = \sum_{k=n}^{n+m-1} \binom{n+m-1}{k} \left(\frac{\lambda_1}{\lambda_1 + \lambda_2}\right)^k \left(\frac{\lambda_2}{\lambda_1 + \lambda_2}\right)^{n+m-1-k}.$$

Theorem 5.27. *Let $A_1, \ldots, A_n$ be disjoint intervals with union B, let $a_1, \ldots, a_n$ be their respective lengths, and set $b = a_1 + \cdots + a_n$. Then for $k_1 + \cdots + k_n = k,\ k_1, \ldots, k_n \in \mathbb{N}$,*

$$P\left(N_{A_1} = k_1, \ldots, N_{A_n} = k_n \mid N_B = k\right) = \frac{k!}{k_1! \cdots k_n!} \left(\frac{a_1}{b}\right)^{k_1} \cdots \left(\frac{a_n}{b}\right)^{k_n}.$$

Corollary 5.28. Since $T_{N(t)} \leq t \leq T_{N(t)+1}$,

$$E[T_{N(t)+1} - T_{N(t)}] = \frac{1}{\lambda} + E[t - T_{N(t)}].$$

5.1.5 Conditional distribution of points

Suppose we are told that exactly one event of a Poisson process has taken place by time t, and we are asked to determine the distribution of the time at which the event occured. Now, since a Poisson process possesses stationary and independent increments it seems reasonable that each interval in $[0, t]$ of equal length should have the same probability of containing the event. In other words, the time of the event should be uniformly distributed over $[0, t]$. This is easily checked since, for $s \leq t$,

$$P(T_1 < s | N(t) = 1) = \frac{P\left(T_1 < s, N(t) = 1\right)}{P\left(N(t) = 1\right)} = \frac{P\left(\text{1 event in [0,s),0 event in [s,t])}\right.}{P\left(N(t) = 1\right)} = \frac{s}{t}.$$

Theorem 5.29. *If $Y_i, i = 1, \ldots, n$ are uniformly distributed over $(0, t)$, then we obtain from the preceding that the joint density function of the order statistic $Y_{(1)}, \ldots, Y_{(n)}$ is*

$$f(y_1, \ldots, y_n) = \frac{n!}{t^n},\ 0 < y_1 < \cdots < y_n < t.$$

Theorem 5.30. *Given that $N(t) = n$, the n arrival times $T_1, \ldots, T_n$ have the same distribution as the order statistic corresponding to n independent random variables uniformly distributed on the interval $(0, t)$, i.e.,*

$$f_{T_1,\ldots,T_n|N(t)=n}(t_1, \ldots, t_n) = \begin{cases} \frac{n!}{t^n}, & 0 < t_1 < \cdots < t_n; \\ 0, & \textit{otherwise} \end{cases}.$$

Proof. Note that for $0 < t_1 < \cdots < r_n < t$, the event that $T_1 = t_1, \ldots, T_n = t_n, N(t) = n$ is equivalent to the event that the first $n+1$ interarrival times satisfy $X_1 = t_1, X_2 = t_2 - t_1, \ldots, X_n = t_n - t_{n-1}, X_{n+1} > t - t_n$. Then

$$\begin{aligned} f_{T_1,\ldots,T_n|N(t)=n}(t_1, \ldots, t_n) &= \frac{f_{T_1,\ldots,T_n,N(t)}(t_1, \ldots, t_n, n)}{P\left(N(t) = n\right)} \\ &= \frac{\lambda e^{-\lambda t_1} \lambda e^{-\lambda(t_2 - t_1)} \cdots \lambda e^{-\lambda(t_n - t_{n-1})} e^{-\lambda(t - t_n)}}{e^{-\lambda t}(\lambda t)^n / n!} \\ &= \frac{n!}{t^n},\ 0 < t_1 < \cdots < t_n < t. \end{aligned}$$

□

5.1.6 Another thinning result

Still assume the range of a label L_n, $\forall\, n \in \mathbb{Z}^+$ is $\{0, \ldots, m\}$, and whose distribution is characterized as follows: $\forall\, j \in \{0, \ldots, m\}$,

$$P(L_n = j | T_1 = t_1, \ldots, T_n = t_n, T_{n+1} = t_{n+1}) = P(L_n = j | T_n = t_n) = P_j(t_n).$$

A simple way of explaning this thinning property is as follows: a point occuring at location s is labeled j with probability $P_j(s)$, independent of everything else. Then

$$\sum_{j=1}^{m} P_j(s) = 1, \ \forall\, s \geq 0.$$

When $P_j(s) = p_j$, $\forall\, s \geq 0$, the thinning rules reduces to the simple thinning rule previously discussed in this chapter.

Next, $\forall\, t \geq 0, j \in \{0, \ldots, m\}$, define

$$N_j(t) = \sum_{n=1}^{\infty} \mathbb{1}_{\{T_n \leq t, L_n = j\}}.$$

Theorem 5.31. *If $N_0(t), \ldots, N_m(t)$ are independent Poisson random variables having means*

$$E[N_j(t)] = \lambda \int_0^t P_j(s) ds, \ \forall\, j = 0, \ldots, m.$$

5.2 M/G/∞ Queue

Suppose that customer arrive at a service station in accordance with a Poisson process with rate λ. Upon arrival the customer is immediately served by one of an infinite number of possible servers, and service times $\{B_n\}_{n \in \mathbb{Z}^+}$ (the amount of work brought by the nth customer is B_n) are assumed to be independent with a common distribution G. What is the distribution of $X(t)$, the number of customers that have complete service by time t?

Our goal is to study the joint distribution of the random variables $Q(t)$ and $D(t), t > 0$, where $Q(t)$ denotes the number of customers present in the system during the interval $[0, t]$.

Given a customer arrives at some time s, the probability he/she is still in the system at time t (Type I customer), where $s < t$, is given by

$$P_{\text{I}}(s) = \overline{G}(t - s) := P(B_1 > t - s) = 1 - G(t - s),$$

independently of everything else. Similarly, the probability this arrival departs from the system at or before time t (Type II customer) is

$$P_{\text{II}}(s) = G(t - s) = P(B_1 \leq t - s),$$

independently over everything else.

Hence, both $Q(t)$ and $D(t)$ can be thought of as thinned random variables relative to $N(t)$, meaning they are independent and Poisson distributed. More specifically,

$$\begin{aligned} &P\left(Q(k) = k_0, D(t) = k_1\right) \\ &= \frac{\left(\lambda \int_0^t \left[1 - G(t-s)\right] ds\right)^{k_0} e^{-\lambda \int_0^t [1 - G(t-s)] ds}}{k_0!} \frac{\left(\lambda \int_0^t G(t-s) ds\right)^{k_1} e^{-\lambda \int_0^t G(t-s) ds}}{k_1!}. \end{aligned}$$

5.3 Examples

Example 5.32. If $X \sim \exp(1/\lambda)$,

$$E[X^2|X > 1] = E\left[(X+1)^2\right].$$

Example 5.33. If X_1 and X_2 are independent nonnegative continuous random variables, show that

$$P\left(X_1 < X_2 | \min(X_1, X_2) = t\right) = \frac{r_1(t)}{r_1(t) + r_2(t)},$$

where $r_i(t)$ is the failure rate function of X_i.
Proof.

$$\begin{aligned} P\left(X_1 < X_2 | \min(X_1, X_2) = t\right) &= \frac{P(X_1 < X_2, \min(X_1, X_2) = t)}{P\left(\min(X_1, X_2) = t\right)} \\ &= \frac{P(X_1 = t, X_2 > t)}{P(X_1 = t, X_2 > t)) + P(X_2 = t, X_1 > t))} \\ &= \frac{f_1(t)\overline{F}_2(t)}{f_1(t)\overline{F}_2(t) + f_2(t)\overline{F}_1(t)} \\ &= \frac{r_1(t)}{r_1(t) + r_2(t)}. \end{aligned}$$

Example 5.34. Let $X, Y_1, \ldots, Y_n$ be independent exponential random variables; X having rate λ, and Y_i having rate μ. Let A_j be the event that the jth smallest of these $n+1$ random variables is one of the Y_i. Find $p = P(X > \max_i Y_i)$ by using the identity

$$p = P(A_1 \cdots A_n) = P(A_1)P(A_2|A_1) \cdots P(A_n|A_1 \cdots A_{n-1}).$$

Note

$$P(A_1) = \frac{n\mu}{\lambda + n\mu}.$$

For $j > 1$,

$$P(A_j|A_1 \cdots A_{j-1}) = \frac{(n-j+1)\mu}{\lambda + (n-j+1)\mu}.$$

Hence,

$$p = \prod_{j=1}^{n} \frac{(n-j+1)\mu}{\lambda + (n-j+1)\mu}.$$

Example 5.35. Let

$$X_i \sim \exp(1/\lambda_i), \ \forall\ i = 1, 2, 3.$$

$$\begin{aligned} P(X_1 < X_2 < X_3) &= P\left(X_1 = \min(X_1, X_2, X_3)\right) P\left(X_2 < X_3 | X_1 = \min(X_1, X_2, X_3)\right) \\ &= \frac{\lambda_1}{\lambda_1 + \lambda_2 + \lambda_3} P\left(X_2 < X_3 | X_1 = \min(X_1, X_2, X_3)\right) \\ &= \frac{\lambda_1}{\lambda_1 + \lambda_2 + \lambda_3} \frac{\lambda_2}{\lambda_2 + \lambda_3}, \end{aligned}$$

where the final equality follows by the lack of memory property.

Example 5.36. One hundred items are simultaneously put on a life test. Suppose the lifetimes of the individual items are independent exponential random variables with mean 200 hours. The test will end when there have been a total of 5 failures. Let T be the time at which the tests end, then letting T_i denote the time between the $(i-1)^{\text{th}}$ and the ith failure, then T_i are independent with T_i being exponential with rate $\frac{101-i}{200}$,

$$E[T] = \sum_{i=1}^{5} \frac{200}{101-i}.$$

$$\text{Var}(T) = \sum_{i=1}^{5} \frac{200^2}{(101-i)^2}.$$

Example 5.37. Let X_1 and X_2 be independent exponential random varaibles, having rate μ_1 and μ_2, respectively. Then

$$\begin{aligned} E\left[X_{(2)}\right] &= E\left[X_{(1)}\right] + E\left[X_{(2)} - X_{(1)}\right] \\ &= \frac{1}{\mu_1+\mu_2} + \frac{1}{\mu_2}\frac{\mu_1}{\mu_1+\mu_2} + \frac{1}{\mu_1}\frac{\mu_2}{\mu_1+\mu_2}. \end{aligned}$$

Example 5.38. Consider a two-server system in which a customer is served first by server 1, then by server 2, and then departs. The service times at server i are exponential random variables with rates $\mu_i, i = 1, 2$. When you arrive, you find server 1 free and two customers at server 2-customer A in service and customer B waiting in line.

(a) Find P_A, the probability that A is still in service when you move over to server 2.

$$P(A) = \frac{\mu_1}{\mu_1+\mu_2}.$$

(b) Find P_B, the probabiliy that B is still in the system when you move over to server 2.

$$P_B = 1 - \left(\frac{\mu_2}{\mu_1+\mu_2}\right)^2,$$

(c) Find $E[T]$, where T is the time that you spend in the system.

$$E[T] = \frac{1}{\mu_1} + \frac{1}{\mu_2} + \frac{P_A}{\mu_2} + \frac{P_B}{\mu_2},$$

where

$$T = S_1 + S_2 + W_A + W_B,$$

where S_i is your service time at server i, W_A is the amount of time you wait in queue while A is being served, and W_B is the amount of time you wait in queue while B is being served.
Alternative: Let T_1 be the time of the first event, and R be the rest of the time, and we compute $E[R]$ by conditioning on what happens first, and compute the rest time after the event of server 1 finshing my job before server 2 finish A by conditioning on what happens first,

$$\begin{aligned} E[T] = E[T_1] + E[R] &= \frac{1}{\mu_1+\mu_2} + E[R|A \text{ before I}]P(A \text{ before I}) + E[R|\text{I before } A]P(\text{I before } A) \\ &= \frac{1}{\mu_1+\mu_2} + \frac{\mu_1}{\mu_1+\mu_2}\frac{3}{\mu_2} + \frac{\mu_2}{\mu_1+\mu_2}\left(\frac{1}{\mu_1+\mu_2} + \frac{\mu_1}{\mu_1+\mu_2}\frac{2}{\mu_2} + \frac{\mu_2}{\mu_1+\mu_2}\left(\frac{1}{\mu_1}+\frac{1}{\mu_2}\right)\right). \end{aligned}$$

Example 5.39. A flashlight needs two batteries to be operational. Consider such a flashlight along with a set of n functional batteries-battery 1, battery 2, $\cdots$, battery n. Initially, battery 1 and 2 are installed. Whenever a battery fails, it is immediatly replaced by the lowest numbered functional battery that has not yet been put in use. Suppose that the lifetimes of the different batteries are independent exponential random variables each having rate μ. At a random time, call it T, a battery will fail and our stockpile will be empty. At that moment exactly one of the batteries-which we call battery X-will not yet have failed. Then

$$P(X = n) = \frac{1}{2}.$$

$$P(X = 1) = \left(\frac{1}{2}\right)^{n-1}.$$

$$P(X = i) = \frac{1}{2}\left(\frac{1}{2}\right)^{n-i}, i > 1.$$

T is the sum of $n-1$ independent exponentials with rate 2μ.

$$E[T] = \frac{n-1}{2\mu}.$$

Example 5.40. There are two servers available to process n jobs. Initially, each server begins work on a job. Whenever a server completes work on a job, that job leaves the system and the server begins processing a new job (provided there are still jobs waiting to be processed). Let T denote the time until all jobs have been processed. If the time that it takes server i to process a job is exponential distributed with rate $\mu_i, i = 1, 2$, find $E[T]$. Let T_i denote the time between the $(i-1)^{\text{th}}$ and the ith job completion. Then the T_i are independent, with $T_i, i = 1, \ldots, n-1$ being exponential with rate $\mu_1 + \mu_2$. Then

$$\begin{aligned} E[T] &= \sum_{i=1}^{n-1} E[T_i] + E[T_n] \\ &= \frac{n-1}{\mu_1 + \mu_2} + \frac{\mu_1}{\mu_1 + \mu_2}\frac{1}{\mu_2} + \frac{\mu_2}{\mu_1 + \mu_2}\frac{1}{\mu_1}. \end{aligned}$$

Example 5.41. Customers can be served by any of three servers, where the service time of server i are exponentially distributed with rate $\mu_i, i = 1, 2, 3$. Whenever a server becomes free, the customer who has been waiting the longest begin service with that server.

(a) If you arrive to find all three servers busy and no one waiting, find the expected time until you depart the system.

$$\frac{1}{\mu_1 + \mu_2 + \mu_3} + \sum_{i=1}^{3} \frac{\mu_i}{\mu_1 + \mu_2 + \mu_3}\frac{1}{\mu_i} = \frac{4}{\mu_1 + \mu_2 + \mu_3}.$$

(b) If you arrive to find all three servers busy and one person waiting, find the expected time until you depart the system.

$$\frac{1}{\mu_1 + \mu_2 + \mu_3} + \frac{4}{\mu_1 + \mu_2 + \mu_3} = \frac{5}{\mu_1 + \mu_2 + \mu_3}.$$

Example 5.42. Consider n components with independent lifetimes, which are such that component i functions for an exponential time with rate λ_i. Suppose that all components are initially in use and remain so until they fail.

(a) Find the probability that component 1 is the second component to fail.
Condition on which item fails first,

$$\sum_{i\neq 1} \frac{\lambda_i}{\sum_{j=1}^n \lambda_j} \frac{\lambda_1}{\sum_{j\neq i} \lambda_j}.$$

(b) Find the expected time of the second failure.

$$\frac{1}{\sum_{i=1}^n \lambda_j} + \sum_{i=1}^n \frac{\lambda_i}{\sum_{j=1}^n \lambda_j} \frac{1}{\sum_{j\neq i} \lambda_j}.$$

Example 5.43. Let X and Y be independent exponential random variables with respective rates λ and μ, where $\lambda > \mu$. Let $c > 0$.

(a) Show that the conditional density function of X, given that $X + Y = c$, is

$$f_{X|X+Y}(x|c) = \frac{(\lambda-\mu)e^{-(\lambda-\mu)x}}{1-e^{-(\lambda-\mu)c}},\ 0 < x < c$$

Proof.

$$\begin{aligned} f_{X|X+Y}(x|c) &= C_1 f_{X,X+Y}(x,c) = C_1 f_{X,Y}(x, c-x) \\ &= C_1 f_X(x) f_Y(c-x) = Ce^{-\lambda x} e^{-\mu(c-x)},\ 0 < x < c \\ &= C_2 e^{-(\lambda-\mu)x},\ 0 < x < c, \end{aligned}$$

where C_1, C_2 does not depend on x. Hence, we can conclude that the conditional distribution is that of an exponential random variable conditioned to be less than c.

(b) Use part (a) to find $E[X|X + Y = c]$. Easy.

(c) Find $E[Y|X + Y = c]$.

$$c = E[X + Y|X + Y = c] = E[X|X + Y = c] + E[Y|X + Y = c],$$

implying that

$$E[Y|X + Y = c] = \cdots .$$

Example 5.44. A doctor has scheduled two appointments, one at 1 P.M. and the other at 1:30 P.M. The amounts of time that appointments last are independent exponential random variables with mean 30 minutes. Assuming that both patients are on time, find the expected amount of time that the 1:30 appointment spends at the doctor's office.

By the memoryless property of exponential distribution,

$$30\left(1 - e^{-\frac{30}{30}}\right) + (30 + 30)e^{-\frac{30}{30}}.$$

Example 5.45. Let X be a uniform random varaible on $(0,1)$, and consider a counting process where events occur at times $X+i$, for $i = 0, 1, 2, \cdots$. Then this counting process has no independent increment but has stationary increment.

Example 5.46. Customers arrive at a two-server service station according to a Poisson process with rate λ. Whenever a new customer arrives, any customer that is in the system immediately departs. A new arrival enters service first with server 1 and then with server 2. If the service times at the servers are independent exponentials with respective rates μ_1 and μ_2, what proportion of entering customers completes their service with server 2?
Let S_i denote the service time at server $i, i = 1, 2$ and let X denote the time until the next arrival. Then, with p denoting the proportion of customers that are served by both servers, we have

$$p = P(X > S_1 + S_2) = P(X > S_1)P(X > S_1 + S_2 | X > S_1) = \frac{\mu_1}{\mu_1 + \lambda}\frac{\mu_2}{\mu_2 + \lambda}.$$

Example 5.47. Consider an n-server parallel queuing system where customers arrive according to a Poisson process with rate λ, where the service times are exponential random variables with rate μ, and where any arrival finding all servers busy immediately departs without receiving any service. If an arrival finds all servers busy, find

(a) the expected number of busy servers found by the next arrival,
Given T, the time until the next arrival, N, the number of busy servers found by the next arrival, is a binomial random variable with parameter n and $p = e^{-\mu T}$.

$$E[N] = \int E[N|T = t]\lambda e^{-\lambda t} dt = \int n e^{-\mu t} \lambda e^{-\lambda t} dt = \frac{n\lambda}{\lambda + \mu}.$$

(b) the probability that the next arrival finds all servers free,

Similar to previous Example,

$$P(N = 0) = \prod_{j=1}^{n} \frac{(n - j + 1)\mu}{\lambda + (n - j + 1)\mu},$$

(c) the probability that the next arrival finds exactly i of the servers free.

$$P(N = n - i) = \frac{\lambda}{\lambda + (n - i)\mu} \prod_{j=1}^{i} \frac{(n - j + 1)\mu}{\lambda + (n - j + 1)\mu}.$$

Example 5.48 (!!!). If an individual has never had a previous automobile accident, then the probability he or she has an accident in the next h time units is $\beta h + o(h)$; on the other hand, if he or she has ever had a previous accident, then the probability is $\alpha h + o(h)$. Find the expected number of accidents an individual has by time t.

Let T_1 be the time of the first accident, then

$$\alpha = E\left[N(t)\right] = \int_0^\infty E[N(t)|T_1 = s]\beta e^{-\beta s} ds = \int_0^t \left[1 + \alpha(t - s)\right] \beta e^{-\beta s} ds.$$

Example 5.49. Teams 1 and 2 are playing a match. The teams score points according to independent Poisson processes with respective rates λ_1 and λ_2. If the match ends when one of the teams has scored k more points than the other, find the probability that team 1 wins.

This is the gambler's ruin probobaility that starting with k, the gambler's fortune reaches $2k$ before 0 when her probability of winning each bet is $\frac{\lambda_1}{\lambda_1+\lambda_2}$. The desired probability is

$$\frac{1-(\lambda_2/\lambda_1)^k}{1-(\lambda_2/\lambda_1)^{2k}}.$$

Example 5.50. Consider a single server queuing system where customers arrive according to a Poisson process with rate λ, service times are exponential with rate μ, and customers are served in the order of their arrival. Suppose that a customer arrives and finds $n-1$ others in the system. Let X denote the number in the system at the moment that customer departs. Find the probability mass function of X.

As long as customers are present to be served, every event (arrival or departure) will, independently of other events, be a departure with probability $p = \frac{\mu}{\lambda+\mu}$. $P(X = m)$ is the probability that there have been a total of m tails at the moment that the nth head occurs, when independent flips of a coin having probability p of coming up heads are made: that is, it is the probability that the nth head occurs on trial number $n + m$. Hence

$$P(X = m) = \binom{n+m-1}{n-1} p^n (1-p)^m.$$

Example 5.51. A system has a random number of flaws that we will suppose is Poisson distributed with mean c. Each of these flaws will, independently, cause the system to fail at at a random time having distribution G. When a system failure occurs, suppose that the flaw causing the failure is immediately located and fixed.

(a) What is the distribution of the number of failures by time t ?
Poisson with $\lambda G(t)$.

(b) What is the distribution of the number of flaws that remain in the system at time t?
Poisson with $\lambda\,(1 - G(t))$.

(c) Are the random variables in parts (a) and (b) dependent or independent?
Yes.

Example 5.52 (!!!)**.** Consider an infinite server queuing system in which customers arrive in accordance with a Poisson process with rate λ, and where the service distribution is exponential with rate μ. Let $X(t)$ denote the number of customers in the system at time t. Find

(a) $E[X(t + s)|X(s) = n]$;

(b) $\text{Var}(X(t + s)|X(s) = n)$.

Let X and Y be respectively the number of customers in the system at time $t+s$ that were present at time s, and the number in the system at $t + s$ that were not in the system at time s. Since there are an infinite number of servers, it follows that X and Y are independent. Since the service distribution is exponential with rate μ, it follows that given $X(s) = n$, $X \sim \text{Binom}(n, e^{-\mu t})$. Also, $Y \stackrel{d}{=} X(t)$, which is independent of $X(s)$.

Therefore, Y is Poisson with mean $\lambda \int_0^t e^{-\mu y} dy = \lambda \left(1 - e^{-\mu t}\right)/\mu$. Then

$$\begin{aligned} E\left[X(t+s)|X(s)=n\right] &= E\left[X|X(s)=n\right] + E\left[Y|X(s)=n\right] \\ &= ne^{-\mu t} + E[Y] \\ &= ne^{-\mu t} + \lambda\left(1 - e^{-\mu t}\right)/\mu. \end{aligned}$$

Example 5.53. Suppose that people arrive at a bus stop accordance with a Poisson process with rate λ. The bus departs at time t. Let X denote the total amount of waiting time of all those who get on the bus at time t. We want to determine $\mathrm{Var}(X)$. Let $N(t)$ denote the number of arrivals by time t.

(a) What is $E\left[X|N(t)\right]$? Let $U_1, U_2, \cdots$ be independent uniform$(0,1)$ random variables. Then

$$\begin{aligned} E\left[X|N(t)=n\right] &= E\left[\sum_{i=1}^{N}(t)(t-T_i)\middle| N(t)=n\right] = E\left[\sum_{i=1}^{n}(t-T_i)\middle| N(t)=n\right] \\ &= nt - E\left[\sum_{i=1}^{n} T_i\middle| N(t)=n\right] = nt - E\left[\sum_{i=1}^{n} tU_{(i)}\right] \\ &= nt - E\left[\sum_{i=1}^{n} tU_i\right] = \frac{nt}{2}. \end{aligned}$$

(b) Argue that $\mathrm{Var}(X|N(t)) = N(t)t^2/12$.

$$\mathrm{Var}(X|N(t)=n) = \mathrm{Var}\left(\sum_{i=1}^{n}(t - tU_{(i)})\right) = \mathrm{Var}\left(\sum_{i=1}^{n}(t - tU_i)\right) = t^2\mathrm{Var}(U_i) = nt^2/12.$$

(c) What is $\mathrm{Var}(X)$?

By (a) and (b),

$$\mathrm{Var}(X) = \mathrm{Var}\left(N(t)t/2\right) + E\left[N(t)t^2/12\right] = \lambda t^3/3.$$

Example 5.54. Satellites are launched into space at times distributed according to a Poisson process with rate λ. Each satellite independently spends a random time (having distribution G) in space before falling to the ground. Find the probability that none of (the statellites in the air at time t) was launched before time s, where $s < t$.

If we count a satellite if it is launched before time s but remains in operation at time t, then the number of items counted is Poisson with mean $m(t) = \lambda \int_0^s \overline{G}(t-y)dy$. The answer is $e^{-m(t)}$.

Example 5.55. For the infinite server queue with Poisson arrivals and general service distribution G,

(a) find the probability that the first customer to arrive is also the first to depart.

Let A be the event that the first to arrive is the first to depart, let S_1 be the first service time, and let $X(t)$ denote the number of departures by time t.

$$P(A) = \int P(A|S_1 = t)g(t)dt = \int P\left(X(t)=0\right)g(t)dt = \int e^{-\lambda\int_0^t G(y)dy} g(t)dt.$$

(b) Let $S(t)$ equal the sum of the remaining service times of all customers in the system at time t.

Argue that $S(t)$ is a compound Poisson random variable.

Given $N(t)$, the number of arrivals by t, the arrival times are iid uniform$(0, t)$. Thus, given $N(t)$, the contribution of each arrival to the total remaining service times are independent with the same distribution, which does not depend on $N(t)$.

Example 5.56. A cable car starts off with n riders. The times between successive stops of the car are independent exponential random variables with rate λ. At each stop one rider gets off. This takes no time, and no additional riders get on. After a rider gets off the car, he or she walks home. Independently of all else, the walk takes an exponential time with rate μ. Then distribution of the time S_n at which the last rider departs the car is Gamma(n, λ).

Suppose the last rider departs the car at time t. What is the probability that all the other riders are home at that time?

Use the result that given $S_n = t$, the set of times at which the first $n-1$ riders departed are independent uniform$(0, t)$ random variables. Therefore, each of these riders will not be walking at time t with probability

$$p = \int_0^t \left(1 - e^{-\mu(t-s)}\right) ds/t = \frac{1 - e^{-\mu t}}{\mu t}.$$

Hence, the probability that none of the riders are walking at time t is p^{n-1}.

Example 5.57. Shocks occur according to a Poisson process with rate λ, and each shock independently causes a certain system to fail with probability p. Let T denote the time at which the system fail and let N denote the number of shocks that it takes.

(a) Find the conditional distriubtion of T given that $N = n$.
It is Gamma(n, λ).

(b) Calculate the conditional distribution of N, given that $T = t$.
For $n \geq 1$,

$$\begin{aligned} P(N = n|T = t) &= \frac{P(T = t|N = n)p(1-p)^{n-1}}{f_T(t)} = C\frac{(\lambda t)^{n-1}}{(n-1)!}(1-p)^{n-1} \\ &= C\frac{(\lambda(1-p)t)^{n-1}}{(n-1)!} = e^{-\lambda(1-p)t}\frac{(\lambda(1-p)t)^{n-1}}{(n-1)!}, \end{aligned}$$

where the last equality follows since the probabilities must sum to 1.

(c) Explain how the result in part (b) could have been obtained without any calculations.

The Poisson events are broken into two classes, those that cause failure and those that do not. Then this results two independent Poisson processes with respective rates λp and $\lambda(1-p)$.

Example 5.58 (Be careful)**.** Suppose that customers arrive to a system according to a Poisson process with rate λ. There are an infinite number of servers in this system so a customer begins service upon arrival. The service times of the arrivals are independent exponential random variables with rate μ, and are independent of the arrival process. Customers depart the system when their service ends. Let N be the total number of arrivals before the first departure.

$$P(N = 1) = \frac{\mu}{\lambda + \mu},$$

$$P(N=2) = \frac{\lambda}{\lambda+\mu}\frac{2\mu}{\lambda+2\mu},$$

$$P(N=j) = \prod_{i=1}^{j-1}\frac{\lambda}{\lambda+i\mu}\frac{j\mu}{\lambda+j\mu},\ j>1.$$

The probability that the first to arrive is the first to depart

$$\sum_{j=1}^{\infty}\frac{1}{j}P(N=j).$$

The expected time of the first departure

$$\sum_{j=1}^{\infty}P(N=j)\sum_{i=0}^{j}\frac{1}{\lambda+i\mu}.$$

5.4 Challenging problems

Example 5.59. Groups arrive to Howie's Rafting Tours according to a Poisson process having rate $\lambda = 10$. Thirty percent of the groups consist of one person and the rest consist of two people. Each group takes a raft down the river (if a group consist of two people, both of them share the same raft) and the amount of time it takes to complete the trip is a random variable with distribution function F.

Let Z_t be the number of people on the river at time t. Assume that $Z_0 = 0$ and the travel times of different groups are independent.

(a) Find the distribution of Z_t.

(b) Does Z_t have a Poisson distribution.

(a) Let X_t be the number of groups with one person and Y_t be the number of groups with two person at time t. Then

$$X_t \sim \text{Poi}\left(0.3\lambda\int_0^t \overline{F}(t-s)ds\right),$$

$$Y_t \sim \text{Poi}\left(0.7\lambda\int_0^t \overline{F}(t-s)\right)ds.$$

So

$$Z_t = X_t + 2Y_t.$$

Then $\forall\ k \in \mathbb{N}$,

$$\begin{aligned}
P\left(Z_t = k\right) &= P\left(X_t + 2Y_t = k\right)\\
&= \sum_{i=0}^{k} P\left(X_t + 2Y_t = k | Y_t = i\right) P\left(Y_t = i\right)\\
&= \sum_{i=0}^{\lfloor k/2\rfloor} P(X_t = k-2i)P(Y_t = i).
\end{aligned}$$

(b) No. The MGF of Z_t is

$$\begin{aligned} M(u) = E\left[e^{uZ_t}\right] &= E\left[e^{uX_t}\right] E\left[e^{2uY_t}\right] \\ &= e^{0.3\lambda \int_0^t \overline{F}(t-s)ds(e^u-1)} e^{0.7\lambda \int_0^t \overline{F}(t-s)ds(e^{2u}-1)} \\ &= e^{\lambda_1(e^u-1)+\lambda_2(e^{2u}-1)}, \end{aligned}$$

which is not a MGF of any Poisson random variable.

Example 5.60. Suppose $\{N(t); t \geq 0\}$ is a Poisson process having rate λ, and let $\{T_n\}_{n\geq 1}$ represent its points, in that for each $t \geq 0$,

$$N(t) = \sum_{k=1}^{\infty} \mathbb{1}_{\{T_k \leq t\}}.$$

(a) For each integer $n \geq 1$, derive a closed-form expression for $E\left[e^{tI_n}\right]$, where

$$I_n = \int_0^{T_n} \frac{1}{N(s)+1} ds.$$

$$M(t) = \prod_{i=1}^{n} \frac{i\lambda}{i\lambda - t}.$$

(b) Find the cdf of I_n, for each integer $n \geq 1$.

Let $T_0 = 0$, then

$$\begin{aligned} I_n &= \sum_{i=i}^{n} \int_{T_{i-1}}^{T_i} \frac{1}{i} ds = \sum_{i=1}^{n} \frac{T_i - T_{i-1}}{i} \\ &= \sum_{i=1}^{n} \frac{X_i}{i} \stackrel{d}{=} \sum_{i=1}^{n} Y_{(i)} - Y_{(i-1)} \\ &\stackrel{d}{=} \max_{1\leq i\leq n} \{Y_i\}, \end{aligned}$$

by assuming there are n components put in a system and Y_i the the lifetime of the nth component and $Y_i \sim \exp(1/\lambda)$.

Example 5.61. Customers and taxis arrive at a taix stand according to two independent Poisson processes $\{N_C(t); t \geq 0\}$ and $\{N_T(t); t \geq 0\}$ with respective rate λ and μ. As soon as both a taxi and customer are available, they leave the stand together, in that the customer rides off in the taxi. Suppose that at time 0 the stand is empty, i.e., there are no taxis or customers.

Let T denote the first time instant at which a customer leaves the stand with a taxi. Then

$$E[T] = \frac{1}{\mu}\frac{\lambda}{\lambda+\mu} + \frac{1}{\lambda}\frac{\mu}{\lambda+\mu}.$$

Example 5.62. Woody is visiting a remote location in South America, in order to fish in the Amazon river. While fishing, Woody catches fish in accordance to a Poisson process with rate λ.

Assume furthermore whenever Woody catches a fish, it is a Piranha with probability $p = \frac{1}{10}$, and not with probability $1 - p = \frac{9}{10}$.

Then the probability Woody catches exactly three non-Piranhas between 8AM and 10AM ($N_2(0,2) = 3$), and exactly one fish between 9AM and 10AM $N(1,2) = 1$ is

$$\begin{aligned}
P\left(N_2(0,2) = 3, N(1,2) = 1\right) &= P\left(N_2(0,2) = 3, N(1,2) = 1\right) \\
&= P\left(N_2(0,2) = 3, N_1(1,2) = 1, N_2(1,2) = 0\right) \\
&\quad + P\left(N_2(0,2) = 3, N_1(1,2) = 0, N_2(1,2) = 1\right) \\
&= P\left(N_2(0,1) = 3, N_1(1,2) = 1, N_2(1,2) = 0\right) \\
&\quad + P\left(N_2(0,1) = 2, N_1(1,2) = 0, N_2(1,2) = 1\right).
\end{aligned}$$

Chapter 6

Renewal Theory and Its Application

In this chapter, we will use renewal theory to establish many types of limit theorems associated with random sequence and processes. Some of our limit theorem will be with respect to convergence 'with probability one', and will be established through creative use of the classical strong law of large numbers (SSLN) for i.i.d. sequences of random variables. The other limit theorems we encounter will address a limit phenomenon known in probability theory as "weak convergence": the central limit theorem (CLT) is arguably the most well-known example of a result involving weakly convergence sequences of random variables. The weak convergence theorems we discuss here will be somewhat different in flavor from the CLT, however, and will be derived using two new tools: Blackwell's Theorem, and the Key Renewal Theorem.

We have seen that a Poisson process is a counting process for which the times between successive events are independent and identically distributed exponential random variables. One possible generalization is to consider a counting process for which the times between successive events are independent and identically distributed with an arbitrary distribution. Such a counting process is called a renewal process.

6.1 Point processes

Definition 6.1. A *point process* is made up of two things:

(a) a collection of nonnegative random variables $\{T_n\}_{n\in\mathbb{Z}^+}$ satisfying

$$0 \le T_1(\omega) \le T_2(\omega) \le \cdots, \quad \forall\, \omega \in \Omega.$$

(b) a counting process $\{N(t); t \ge 0\}$, where $\forall\, \omega \in \Omega$,

$$N(t,\omega) = \sum_{n=1}^{\infty} \mathbb{1}_{\{T_n(\omega)\le t\}}.$$

Remark. we will often refer to the counting process $\{N(t); t \geq 0\}$ as a point process, with $\{T_n\}_{n\in\mathbb{Z}^+}$ being its points and

$$T_n(\omega) = \inf\{t \geq 0 : N(t,\omega) \geq n\},$$

$$N(t,\omega) = \sup\{n \geq 1 : T_n(\omega) \leq t\}.$$

6.1.1 Properties of Point processes

Proposition 6.2. For any $t \geq 0$ and any $n \in \mathbb{Z}^+$, we have

$$\{T_n \leq t\} = \{N(t) \geq n\},$$

which is shorthand of

$$\{\omega : T_n(\omega) \leq t\} = \{\omega : N(t,\omega) \geq n\},$$

Remark. The random variables $T_{N(t)}$ and $T_{N(t)+1}$ can be interpreted as

$$T_{N(t)}(\omega) = T_{N(t,\omega)}(\omega),\ \forall\ \omega \in \Omega,$$

$$T_{N(t)+1}(\omega) = T_{N(t,\omega)+1}(\omega),\ \forall\ \omega \in \Omega.$$

Proposition 6.3.

$$T_{N(t)} = \sup\{T_n : T_n \leq t\}.$$

$$T_{N(t)+1} = \inf\{T_n : T_n > t\}.$$

$$T_{N(t)} \leq t < T_{N(t)+1}.$$

6.2 Convolutions

Convolutions of increasing functions will be used many times in our study of renewal processes and regenerative processes.

Definition 6.4. Suppose $F, G : \mathbb{R} :\to \mathbb{R}$ are two increasing functions. Then the *convolution* of F and G is a function $F * G : \mathbb{R} \to \mathbb{R}$ defined as follows:

$$F * G(t) = \int_{-\infty}^{\infty} G(t-s)dF(s),\ \forall\ t \in \mathbb{R}.$$

Remark. Note the use of the notation $dF(s)$ in the definition of $F * G$: this type of integral is known as a Riemann-Stieltjes integral.

Theorem 6.5. *Suppose $F, G : \mathbb{R} \to \mathbb{R}$ are both increasing functions on $\mathbb{R}$. Then $F * G$ is also increasing.*

Theorem 6.6 (Commutativity)**.** *Given any two increasing functions $F, G : \mathbb{R} \to \mathbb{R}$, we have $F * G(t) = G * F(t),\ \forall\ t \in \mathbb{R}$.*

Proof.

$$\begin{aligned} F * G(t) &= \int_{-\infty}^{\infty} G(t-s)dF(s) = \int_{-\infty}^{\infty}\int_{-\infty}^{t-s} dG(u)dF(s) \\ &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \mathbb{1}_{\{u\le t-s\}} dG(u)dF(s) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \mathbb{1}_{\{s\le t-u\}} dF(u)dG(s) \\ &= \int_{-\infty}^{\infty} F(t-u)dG(u) = G * F(t). \end{aligned}$$

□

Theorem 6.7 (Associativity)**.** *Suppose $F, G, H : \mathbb{R} \to \mathbb{R}$ are increasing functions. Then $(F * G) * H(t) = F * (G * H)(t),\ \forall\ t \in \mathbb{R}$.*

Theorem 6.8. *Suppose $F, G, H : \mathbb{R} \to \mathbb{R}$ are increasing functions. Then*

$$F * (G + H)(t) = (F * G)(t) + (F * H)(t).$$

Remark. Convolutions arise naturally when studying sums of independent random variables. Suppose X and Y are two independent random variables, having cdfs F_X and F_Y. Then

$$\begin{aligned} F_{X+Y}(t) = P(X+Y \le t) &= \int_{-\infty}^{\infty} P(X+Y \le t | X = x) dF_X(x) \\ &= \int_{-\infty}^{\infty} P(Y \le t-x) dF_X(t) = \int_{-\infty}^{\infty} F_Y(t-x) dF_X(x) \\ &= F_X * F_Y(t). \end{aligned}$$

Remark. Noting that in most cases, we will be studying convolutions $F * G$ when F and G are both cdfs of nonnegative random variables, meaning in particular that $F(t) = G(t) = 0,\ \forall\ t < 0$. Then

$$F * G(t) = \int_{-\infty}^{\infty} G(t-s)dF(s) = \int_{0}^{t} G(t-s)dF(s),\ \forall\ t \ge 0.$$

Then we can safely interpret any such integrals as Riemann-Stieltjes integrals. However, we need to be careful with realizing that the integration set includes both the boundaries $\{0\}$ and $\{t\}$. This is an important observation that must be made when F and/or G have jumps at 0 or t. Usually we will emphasize this point by writing

$$\int_{[0,t]} G(t-s)dF(s).$$

6.3 Laplace Transforms

Definition 6.9. Given a random variable X, the Laplace transform is given by a function ϕ_X, where

$$\phi_X(\alpha) := E\left[e^{-\alpha X}\right] = \int_{-\infty}^{\infty} e^{-\alpha x} dF_X(x).$$

Remark. Most of the random variable we will study in this chapter will be nonnegative. Then $\phi_X(\alpha)$ exists for all $\alpha > 0$ since $e^{-\alpha X} \le 1$ w.p.1., implying also that $E\left[e^{-\alpha X}\right] \le 1 < \infty$.

Proposition 6.10.

$$E\left[e^{-\alpha X}\right] = \alpha \int_0^\infty P(X \le x) e^{-\alpha x} dx.$$

Proof.

$$\begin{aligned} E\left[e^{-\alpha X}\right] &= \int_0^\infty e^{-\alpha x} dF_X(x) = \int_0^\infty \int_x^\infty \alpha e^{-\alpha y} dy dF_X(x) \\ &= \int_0^\infty \int_0^y \alpha e^{-\alpha y} dF_X(x) dy = \alpha \int_0^\infty P(X \le y) e^{-\alpha y} dy. \end{aligned}$$

□

6.4 Stopping times

Definition 6.11. The **nonnegative integer** valued random variable τ is said to be a stopping time w.r.t $\{X_n\}_{n=0}^\infty$ if the event $\{\tau \le n\}$ is independent of $X_{n+1}, X_{n+2}, \cdots$ for all $n = 1, 2, \cdots$, or equivalently, there exists a function $h_n : \mathbb{R}^{n+1} \to \{0, 1\}$ satisfying

$$\mathbb{1}_{\{\tau \le n\}} = h_n(X_0, \ldots, X_n), \ \forall\ n \in \mathbb{N}.$$

Theorem 6.12. *The **nonnegative integer** valued random variable τ is a stopping time w.r.t $\{X_n\}_{n=0}^\infty$ if the event $\{\tau = n\}$ is independent of $X_{n+1}, X_{n+2}, \cdots$ for all $n = 1, 2, \cdots$, or equivalently, $\exists$ a function $g_n : \mathbb{R}^{n+1} \to \{0, 1\}$ satisfying*

$$\mathbb{1}_{\{\tau = n\}} = g_n(X_0, \ldots, X_n), \ \forall\ n \in \mathbb{N}.$$

Example 6.13. Fix $x \in \mathbb{R}$, and define

$$\eta_x = \inf\{n \ge 0 : X_n > x\}.$$

Then

$$\mathbb{1}_{\{\eta_x \le n\}} = 1 - \mathbb{1}_{\{\eta_x > n\}} = 1 - \mathbb{1}_{\{X_0 \le x, \ldots, X_n \le x\}}.$$

So η_x is a stopping time w.r.t. $\{X_n\}_{n \in \mathbb{N}}$.

6.5 Renewal processes

Let $\{N(t), t \ge 0\}$ be a point process and let $\{X_n\}_{n \in \mathbb{Z}^+}$ denote the interrenewals and are given by, assuming $T_0 = 0$,

$$X_n = T_n - T_{n-1}, \ \forall\ n \in \mathbb{Z}^+.$$

Also,

$$T_n = \sum_{i=1}^n X_i, \ n \ge 1.$$

Definition 6.14. A point process $\{N(t); t \ge 0\}$ is a *renewal process* if its interrenewal distances $\{X_n\}_{n \in \mathbb{Z}^+}$ are iid random variables, having a common cdf F.

Proposition 6.15. The cdf of T_n is given by $F^{(n)}$, $\forall\ n \in \mathbb{Z}^+$, where

$$F^{(1)} := F,$$

$$F^{(n+1)}(t) = F^{(n)} * F(t), \ \forall\ t \in \mathbb{R}^{\ge 0}.$$

6.5.1 Distribution of $N(t)$

$$\begin{aligned} P\left(N(t)=n\right) &= P(T_n \le t < T_{n+1}) \\ &= P(T_n \le t) - P(T_{n+1} \le t) \\ &= P\left(N(t) \ge n\right) - P\left(N(t) \ge n+1\right) \\ &= F^n(t) - F^{n+1}(t). \end{aligned}$$

$$P\left(N(t)=0\right) = P(T_1 > t) = 1 - F(t).$$

Remark.

$$N(t) = \sum_{n=1}^{\infty} \mathbb{1}_{\{T_n \le t\}} = \max\{n : T_n \le t\}.$$

Remark. Each moment of $N(t)$ can be expressed in terms of convolutions as well. For example,

$$E[N(t)] = \sum_{n=1}^{\infty} P\left(N(t) \ge n\right) = \sum_{n=1}^{\infty} P(T_n \le t) = \sum_{n=1}^{\infty} F^{(n)}(t).$$

Alternatively,

$$E\left[N(t)\right] = E\left[\sum_{n=1}^{\infty} \mathbb{1}_{\{T_n \le t\}}\right] = \sum_{n=1}^{\infty} F^{(n)}(t).$$

6.5.2 Do renewal processes explode?

A renewal process along some realization $\omega \in \Omega$ explode if

$$\lim_{n\to\infty} T_n(\omega) < \infty,$$

or there exists $t \in \mathbb{R}$ such that

$$N_t(\omega) = \infty.$$

Lemma 6.16. If $E[X_1] > 0$, then

$$\lim_{n\to\infty} T_n = \infty \ w.p.1,$$

or

$$P\left(\lim_{n\to\infty} T_n = \infty\right) = 1,$$

or

$$P\left(N(t) < \infty\right) = 1, \ \forall\, t \ge 0.$$

Thus, the renewal process cannot explode w.p.1..

Proof. By SLLN,

$$\lim_{n\to\infty} \frac{T_n}{n} = \lim_{n\to\infty} \frac{\sum_{k=1}^{n} X_k}{n} = E[X_1] > 0, \ w.p.1.$$

So

$$\lim_{n\to\infty} T_n = \infty, \ w.p.1.$$

□

Does these observations by themselves imply $E[N(t)] < \infty,\ \forall\ t \geq 0$? **No!!!** since a random variable can be finite w.p.1., yet have an infinite first moment. Here is an example of such a random variable.

Example 6.17. Suppose Z is a positive, integer-valued random variable, where its pmf is given by

$$P(Z = k) = \frac{6}{\pi^2 k^2},\ k = 1, 2, \cdots.$$

Observe that

$$P(Z < \infty) = \sum_{k=1}^{\infty} P(Z = k) = 1,$$

which follows from the identity

$$\sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6}.$$

However, the expected value of Z is

$$E[Z] = \sum_{k=1}^{\infty} kP(Z = k) = \sum_{k=1}^{\infty} \frac{6}{\pi^2 k} = \infty.$$

6.5.3 Limit theorems and their applications

If $\lim_{t\to\infty} N(t, \omega) = n$ is finite, then $T_k(\omega) = \infty,\ \forall\ k \in \mathbb{Z}^{\geq n+1}$.

Proposition 6.18. Assume $\{N(t)\}$ is a counting process with $0 < T_1 < T_2 < \cdots,\ \forall\ n \in \mathbb{Z}^+$ and $P(\lim_{n\to\infty} T_n = \infty) = 1$.

$$P\left(\lim_{t\to\infty} \frac{N(t)}{t} \to \mu\right) = 1 \Longleftrightarrow P\left(\lim_{t\to\infty} \frac{T_n}{n} = \frac{1}{\mu}\right) = 1.$$

Proof. "$\Leftarrow$". Then $\exists\ A \subseteq \Omega$ such that

$$\lim_{n\to\infty} \frac{T_n(\omega)}{n} = \mu,\ \forall\ \omega \in A,$$

and

$$P(A) = 1.$$

Fix an $\omega \in A$, then $\forall\ t > 0$,

$$T_{N(t)}(\omega) \leq t < T_{N(t)+1}(\omega).$$

Let t large enough s.t. $N(t, \omega) > 0$, then divide the above inequalities by $N(t, \omega)$, we have the result.

"$\Rightarrow$". Similarly,

$$\lim_{n\to\infty} \frac{T_n(\omega)}{n} = \lim_{n\to\infty} \frac{T_n(\omega)}{N(T_n(\omega), \omega)} = \frac{1}{\mu}.$$

□

Corollary 6.19. Let $\{N(t)\}$ be a renewal process with $E[X_1] > 0$, by *SLLN*,

$$P\left(\lim_{n\to\infty} \frac{T_n}{n} = E[X_1]\right) = 1.$$

Then by previous Proposition,

$$P\left(\lim_{t\to\infty} \frac{N(t)}{t} = \frac{1}{E[X_1]}\right) = 1,$$

i.e., the long run rate of arriving is equal to the inverse of the expectation of the interarrival time.

Example 6.20. Suppose that potential customers arrive at a single-server bank in accordance with a Poisson process having rate λ. However, suppose that the potential customer will enter the bank only if the server is free when he arrives. If the amount of time spent in the bank by an entering customer is a random variable having distribution G, then

(a) what is the rate at which customers enter the bank?

(b) what proportion of ponential customers actually enter the bank?

Let's suppose that at time 0, a customer has just entered the bank. By the memoryless property of the Poisson process, the mean time between entering customers is

$$\mu = \mu_G + \frac{1}{\lambda}.$$

Hence, the rate at which customers enter the bank will be given by

$$\frac{1}{\mu} = \frac{\lambda}{1 + \lambda\mu_G}.$$

On the other hand, since potential customers will be arriving at a rate λ, if follows that the proportion of them entering the bank will be given by

$$\frac{1/\mu}{\lambda} = \frac{1}{1 + \lambda\mu_G}.$$

6.5.4 Renewal reward processes

A large number of probability models are special cases of the following model. Consider a renewal process $\{N(t), t \geq 0\}$ having interarrival times $\{X_n\}_{n\in\mathbb{Z}^+}$, and suppose that each time a renewal occurs we receive reward. We denote by $\{R_n\}_{n\geq 1}$, are independent and identically distributed. Here we are not specifying precisely when the reward is received in $(T_{n-1}, T_n]$ while defining R_n; it could be received at the beginning of the interval, the end of the interval, continuously during the interval, etc..

We relate the point process with the rewards by assuming the sequence of random vectors

$$\{(X_n, R_n)\}_{n\in\mathbb{Z}^+}$$

is iid, but with each vector (X_n, R_n), $\forall\, n \in \mathbb{Z}^+$, we do allow there to be some dependency between X_n and R_n. For instance, one may wish to set $R_n = cX_n$ for some constant c, which would imply reward is accumulated at a rate c per unit time.

$$R(t) = \sum_{n=1}^{N(t)} R_n,$$

then $R(t)$ represents the *total reward* earned by time t.

Proposition 6.21. If $E[R_1] < \infty$ and $E[X_1] < \infty$, then

$$P\left(\lim_{t\to\infty} \frac{R(t)}{t} = \frac{E[R_1]}{E[X_1]}\right) = 1.$$

Proof. Note

$$R(t) \approx \sum_{n=1}^{N(t)} R_n.$$

Then w.p.1,

$$\lim_{t\to\infty} \frac{R(t)}{t} = \lim_{t\to\infty} \frac{\sum_{n=1}^{N(t)} R_n}{t} = \frac{\lim_{t\to\infty} \sum_{n=1}^{N(t)} R_n/N(t)}{\lim_{t\to\infty} t/N(t)} = \frac{E[R_1]}{E[X_1]}. \qquad \square$$

Remark. If we say that a cycle is completed every time a renewal occurs, the Proposition states that the long-run average reward per unit time is equal to the expected reward earned during a cycle divided by the expected length of a cycle.

6.6 The age process and excess process

Consider a renewal process having interarrival distribution F and define $A(t)$ to be the time at t since the last renewal. If renewals represent old items failing and being replaced by new ones, then $A(t)$ represents the age of the item in use at time t. Let

$$A(t) = t - T_{N(t)} = t - \sup\{T_n : T_n \le t\}.$$

Then $\{A(t); t \ge 0\}$ is the age process w.r.t. our renewal process $\{N(t); t \ge 0\}$. Let

$$Y(t) = t - T_{N(t)} = t - \sup\{T_n : T_n \le t\}.$$

Then $\{Y(t); t \ge 0\}$ is the access process w.r.t. our renewal process $\{N(t); t \ge 0\}$.

Example 6.22. One may wonder, after a large amount of time has gone by, what proportion of time the age process spends at a level less than or equal to some fixed number x. In other words, does the limit

$$\lim_{t\to\infty} \frac{1}{t} \int_0^t \mathbb{1}_{\{A(s)\le x\}} ds$$

exist, and if it does, how do we compute it?
Let

$$R(t) := \int_0^t \mathbb{1}_{\{A(s)\le x\}} ds.$$

Note $\forall\ n \geq 1$,

$$\begin{aligned} R_n &= \int_{T_{n-1}}^{T_{n-1}+X_n} \mathbb{1}_{\{A(s)\leq x\}} ds = \int_{T_{n-1}}^{T_{n-1}+X_n} \mathbb{1}_{\{s-T_{n-1}\leq x\}} ds \\ &= \int_0^{X_n} \mathbb{1}_{\{s\leq x\}} ds = \min(X_n, x). \end{aligned}$$

From here, we can easily see that the sequence $\{(X_n, R_n)\}_{n\in\mathbb{Z}^+}$ is iid. Also,

$$E[R_1] = \int_0^\infty P\left(\min(T_1, x) > y\right) dy = \int_0^\infty P(T_1 > y, x > y) dy = \int_0^x P(T_1 > y) dy.$$

Hence, by the renewal reward theorem, w.p.1,

$$\lim_{t\to\infty} \frac{1}{t} \int_0^t \mathbb{1}_{\{A(s)\leq x\}ds} = \frac{\int_0^x P(T_1 > y) dy}{E[T_1]}.$$

Next, similarly, since

$$R_1 = \int_0^{X_1} t dt = \frac{X_1^2}{2},$$

we have the average age of a renewal process is

$$\lim_{t\to\infty} \frac{\int_0^t A(s) ds}{t} = \frac{E[X_1^2]}{2E[X_1]},$$

and since

$$R_1 = \int_0^{X_1} (X_1 - t) dt = \frac{X_1^2}{2},$$

Similarly, we have the average access of a renewal process is

$$\lim_{t\to\infty} \frac{\int_0^t Y(t) dt}{t} = \frac{E[X_1^2]}{2E[X_1]}.$$

Example 6.23. Suppose that passengers arrive at a bus stop according to a Poisson process with rate λ. Suppose also that buses arrive according to a renewal process with distribution function F, and that buses pick up all waiting passengers. Assuming that the Poisson process of people arriving and the renewal process of buses arriving are independent, find

(a) the average number of people who are waiting for a bus, averaged over all time;

(b) the average amount of time that a passenger waits, averaged over all passengers.

We will solve this by using renewal reward processes. Say that a new cycle begins each time a bus arrives. Let T be the time of a cycle, and note that T has distribution function F. If we suppose that each passenger pays us oney at a rate of 1 per unit time while that wait for a bus, then the reward rate at any time is the number waiting at that time, and so the average reward per unit

time is the average number of people that are waiting for a bus. Letting R be the reward earned during a cycle, the renewal reward theorem gives

$$\text{Average number waiting} = \frac{E[R]}{E[T]}.$$

Let N be the number of arrivals during a cycle. Then

$$E[R|T = t, N = n] = \frac{nt}{2},$$

which follows because given there are n arrivals by time t, their set of arrival times are distributed as n independent uniform $(0, t)$ random variables. Hence,

$$E[R|T, N] = \frac{NT}{2}.$$

Then

$$E[R] = \frac{1}{2}E[NT].$$

Also,

$$E[NT|T] = TE[N|T] = \lambda T^2.$$

So

$$E[NT] = \frac{\lambda}{2}E[T^2].$$

Hence

$$\text{Average number waiting} = \frac{\lambda E[T^2]}{2E[T]}.$$

Next, because each passenger pays 1 per unit time while waiting for a bus, the total amount paid by a passenger is the amount of time the passenger waits. Because R is the total reward earned in a cycle, it thus follows that, with W_i being the waiting time of passenger i,

$$R = W_1 + \cdots + W_n.$$

Then by the renewal reward theory (discrete time),

$$\lim_{n\to\infty} \frac{W_1 + \cdots + W_n}{n} = \frac{E[W_1 + \cdots + W_N]}{E[N]} = \frac{E[R]}{E[N]}.$$

Note

$$E[N] = E\left[E[N|T]\right] = E[\lambda T] = \lambda E[T].$$

Hence

$$\lim_{n\to\infty} \frac{W_1 + \cdots + W_n}{n} = \frac{E[T^2]}{2E[T]}.$$

Because $\frac{E[T^2]}{2E[T]}$ is the average value of the excess for the renewal process of arriving buses, the preceding equation yields the interesting result that the average waiting time of a passenger is equal to the average time until the next bus arrives when we average over all **time**.

6.6.1 The elementary renewal theorem

Theorem 6.24 (the elementary renewal theorem)**.**

$$\lim_{t\to\infty} \frac{E\left[N(t)\right]}{t} = \frac{1}{E[X_1]}.$$

Remark.

$$\lim_{t\to\infty} E\left[\frac{N(t)}{t}\right] \stackrel{?}{=} E\left[\lim_{t\to\infty} \frac{N(t)}{t}\right].$$

You should be fully aware that it is not always possible to interchange limit and expectation.

Theorem 6.25 (Wald's identity)**.** *If $\{X_n\}_{n=1}^{\infty}$ is a sequence of independent and identically distributed random variables with finite expectation, and if N is a stopping time w.r.t. $\{X_n\}$ such that $E[N] < \infty$, then*

$$E\left[\sum_{n=1}^{N} X_n\right] = E[N]E[X_1].$$

Proof. By Fubini theorem,

$$E\left[\sum_{i=1}^{N} X_i\right] = E\left[\sum_{i=1}^{\infty} X_i \mathbb{1}_{\{N\geq i\}}\right] = \sum_{i=1}^{\infty} E\left[X_i \mathbb{1}_{\{N\geq i\}}\right].$$

Next,

$$\mathbb{1}_{\{N\geq i\}} = 1 - \mathbb{1}_{\{N<i\}} = 1 - \mathbb{1}_{\{N\leq i-1\}}.$$

So $\mathbb{1}_{\{N\geq i\}}$ is a function of $X_0, \ldots, X_{i-1}$, and hence independent of X_i. Thus,

$$\begin{aligned} E\left[\sum_{i=1}^{N} X_i\right] &= \sum_{i=1}^{\infty} E[X_i]P(N \geq i) = E[X_1]\sum_{i=1}^{\infty} P(N \geq i) \\ &= E[X_1]E[N]. \end{aligned}$$

□

Example 6.26. $N(t) + 1$ is a stopping time w.r.t. $\{X_n\}_{n=1}^{\infty}$, since $\forall\ n \geq 1$, we have

$$\mathbb{1}_{\{N(t)+1=n\}} = \mathbb{1}_{\{N(t)=n-1\}} = \mathbb{1}_{\{T_{n-1}\leq t<T_n\}}.$$

6.7 Renewal equations and renewal function

Renwal equations will play a major role in our development of limit theorems that have the flavor of 'weak-convergence'. In applications, each of these limiting results will require an application of the Key Renewal Theorem, and a key part of applying this theorem is to first find a renewal equation, then check whether or not each of its parts satisfy certain properties.

Definition 6.27. Suppose $h : \mathbb{R} \to \mathbb{R}$ is a real-valued function satisfying $h(t) = 0$ for $t < 0$, and let F be a cdf on $\mathbb{R}$ satisfying $F(t) = 0$ for $t < 0$ and $\lim_{t\uparrow\infty} F(t) = 1$. The renewal equation generated by h and F is given by

$$H(t) = h(t) + \int_{[0,t]} H(t-s)dF(s), \quad t \geq 0,$$

where $H : [0, \infty) \to \mathbb{R}$ is a solution of the renewal equation.

Remark. It might help you if you instead think of $H : \mathbb{R} \to \mathbb{R}$, where $H(t) = 0,\ \forall\ t < 0$ and similarly for F. This of course implies

$$\int_{[0,t]} H(t-s)dF(s) = \int_{-\infty}^{\infty} H(t-s)dF(s) = F * H(t),\ \forall\ t \geq 0,$$

meaning we can rewrite the renewal equation as

$$H(t) = h(t) + F * H(t), \quad t \geq 0.$$

Observe that our definition of $F * H$ is still valid, even though we haven't said H is a nondecreasing function. Right now we cannot define $H * F$ since not enough is known about H: however, if for instance, we knew H is of bounded variation on each finite interval contained in $[0, \infty)$, then $H * F$ becomes a legal definition.

Our goal now is to study properties of 'nice' solutions H for a fixed function h and cdf F. We limit our search for solutions to the renewal equation by also assuming h is bounded on finite subintervals of $[0, \infty)$: in other words, we assume $\sup_{a \leq s \leq b} |h(s)| < \infty,\ \forall\ 0 \leq a < b$. This search for solutions will lead us to a function $U : \mathbb{R} \to \mathbb{R}$,

$$U(t) = \sum_{n=0}^{\infty} F^{(n)}(t) = 1 + \sum_{n=1}^{\infty} F^{(n)}(t) = 1 + E\left[N(t)\right], \quad t \geq 0.$$

which is often referred to as the renewal function.

Note $F^{(0)}(t) = \mathbb{1}_{\{t \geq 0\}}$, then $U(t) = 0,\ \forall\ t < 0$.

Proposition 6.28. For the renewal equation

$$H(t) = h(t) + \int_0^t H(t-s)dF(s),\ t \geq 0.$$

If $U * h(t) = 0\ \forall\ t < 0$ and is bounded on finite intervals, i.e.,

$$\sup_{a \leq s \leq b} |U * h(s)| < \infty,\ \forall\ [a, b] \subseteq \mathbb{R}^+.$$

Then the equation has the unique solution $H(t) = U * h(t)$.

The **first** step analysis: An integral equation satisfied by the renewal function can be obtained by conditioning on the time of the first renewal.

Example 6.29.

$$\begin{aligned}
M(t) = E[N(t)] = E\left[E[N(t)|X_1]\right] &= \int_0^{\infty} E\left[N(t)|X_1 = s\right] dF(s) \\
&= \int_{[0,t]} E\left[N(t)|X_1 = s\right] dF(s) = \int_{[0,t]} \left(E\left[1 + N(s,t]\right]|X_1 = s\right) dF(s) \\
&= F(t) + \int_{[0,t]} E\left[N(s,t]|X_1 = s\right] ds = F(t) + \int_{[0,t]} E\left[N(t-s)\right] dF(s) \\
&= F(t) + \int_{[0,t]} M(t-s)dF(s).
\end{aligned}$$

Since $h = F$ is bounded on finite intervals, then we have the unique solution

$$U * F(t) = \left(\sum_{n=0}^{\infty} F^{(n)}\right) * F(t) \ = \sum_{n=0}^{\infty} F^{(n)} * F(t) = \sum_{n=0}^{\infty} F^{(n+1)}(t) = E\left[N(t)\right].$$

Example 6.30. Suppose $\{X(t)\}$ represents a stochastic process that cycles between the states in the state space $E = \{0, 1, 2, \ldots, k\}$ in that it goes through the states in the order

$$0, 1, \ldots, k, 0, 1 \cdots, k, 0, 1 \cdots.$$

Whenever it visits a state i, it stays there for a generally distributed amount of time X_i having cdf F_i with mean μ. Assume that $X(0) = 0$ w.p.1.

Let T_1 represents the first time the process returns to 0, i.e.,

$$T_1 = \inf\{t \geq 0 : X(t^-) \neq X(t) = 0\}.$$

Since the system regenerates when it reaches state 0 again, we have

$$\begin{aligned}
H_i(s) &= P\left(X(t) = i\right) = P\left(X(t) = i, T_1 > t\right) + P\left(X(t) = i, T_1 \leq t\right) \\
&= P\left(X(t) = i, T_1 > t\right) + \int_0^t P\left(X(t) = i | T_1 = s\right) dF(s) \\
&= P\left(X(t) = i, T_1 > t\right) + \int_0^t P\left(X(t-s) = i\right) dF(s) \\
&= P\left(X(t) = i, T_1 > t\right) + \int_0^t H_i(t-s) dF(s).
\end{aligned}$$

6.7.1 Blackwell's theorem and the Key Renewal theorem

The goal of this section is to establish whether or not the limit

$$\lim_{t \to \infty} U * h(t).$$

Definition 6.31. We say that a point $x \in \mathbb{R}$ is a point of increase of a cdf F, if $\forall\ \epsilon > 0$, $F(x + \epsilon) - F(x - \epsilon) > 0$.

Example 6.32. Suppose

$$F(t) = \begin{cases} 0, & t < 1 \\ 1/3, & 1 \leq t < \pi \\ 1, & t > \pi \end{cases}.$$

Clearly, we see that the two points of increase of F are just 1 and π.

Example 6.33. Now suppose F is absolutely continuous and has all of its mass concentrated on $[0, \infty)$, meaning $\forall\ t \geq 0$,

$$F(t) = \int_{-\infty}^{t} f(x) = \int_0^t f(x) dx.$$

Also assume that f is continuous on $(0, \infty)$. If $x_0 > 0$ is a real number satisfying $f(x_0) > 0$, then x_0 must be a point of increase of F: to see why, notice we can choose $\epsilon > 0$ so that, by continuity of f, $|f(x_0) - f(x)| < f(x_0)/2 \ \forall \ x \in (x_0 - \delta, x_0 + \delta)$. Hence, on this set

$$f(x) \geq f(x_0)/2.$$

If $\epsilon \leq \delta$, since $(x_0 - \epsilon, x_0 + \epsilon) \subseteq (x_0 - \delta, x_0 + \delta)$,

$$F(x_0 + \epsilon) - F(x_0 - \epsilon) = \int_{x_0-\epsilon}^{x_0+\epsilon} f(x)d \geq f(x_0)/2 \cdot (2\epsilon) = \epsilon f(x_0) > 0.$$

If $\epsilon > \delta$, since F is nondecreasing,

$$F(x_0 + \epsilon) - F(x_0 - \epsilon) \geq F(x_0 + \delta) - F(x_0 - \delta) = \int_{x_0-\delta}^{x_0+\delta} f(x)dx \geq f(x_0)/2 \cdot (2\delta) = \delta f(x_0) > 0.$$

We will also need to make use of another new concept.

Definition 6.34. Let F be the cdf of a nonnegative random variable X. This function F is arithmetic (or lattice) if there exists some $a > 0$ such that the points of increase of F are contained in the set $\{0, a, 2a, 3a, \cdots\}$,
or

$$\sum_{n=0}^{\infty} P(X = na) = 1,$$

or

$$P\left(X \in \{0, a, 2a, 3a, \cdots\}\right) = 1.$$

Theorem 6.35 (Blackwell's theorem)**.** *Suppose $\{N(t); t \geq 0\}$ is a renewal process whose inter-renewals have cdf F, and let $U : \mathbb{R} \to \mathbb{R}$ be its corresponding renewal function. If F is also nonarithmetic, then*

$$\lim_{t\to\infty} U(t + h) - U(t) = \frac{h}{E[X_1]}, \ \forall \ h > 0.$$

Remark. This theorem is fairly difficult to prove. When F is arithmetic, analogous result is known as the discrete renewal theorem, and is easier to visualize if you are comfortable with using Markov chains.

Definition 6.36. The function $h : [0, T] \to \mathbb{R}$ is *Riemann integrable* if

$$\lim_{\delta\to 0} \delta \sum_{n=1}^{T/\delta} \sup_{(n-1)\delta \leq t < n\delta} h(t) = \lim_{\delta\to 0} \delta \sum_{n=1}^{T/\delta} \inf_{(n-1)\delta \leq t < n\delta} h(t) \in (-\infty, \infty).$$

The function $h : [0, \infty) \to \mathbb{R}$ is *Riemann integrable* if

$$\int_0^{\infty} h(t)dt = \lim_{T\to\infty} \int_0^T h(t) \text{ exists.}$$

Definition 6.37. The function $h : [0, T] \to \mathbb{R}$ is *directly Riemann integrable* if

$$\lim_{\delta\to 0} \delta \sum_{n=1}^{\infty} \sup_{(n-1)\delta \leq t < n\delta} h(t) = \lim_{\delta\to 0} \delta \sum_{n=1}^{\infty} \inf_{(n-1)\delta \leq t < n\delta} h(t) \in (-\infty, \infty).$$

Theorem 6.38 (Key Renewal Theorem)**.** *Suppose F is nonarithmetic cdf that has a finite mean and suppose h is a directly Riemann integrable function on $[0, \infty)$. Then*

$$\lim_{t\to\infty} U * h(t) = \frac{\int_0^\infty h(s)ds}{E[X_1]}.$$

6.8 Application of the key renewal theorem

The goal is to establish $\lim_{t\to\infty} H(t)$ for some given function $H(t), t \geq 0$.

(a) Show that H is a solution that is bounded on finite intervals and satisfy the renewal equation

$$H(t) = h(t) + \int_{[0,t]} H(t-s)dF(s),$$

where $h(t), t \geq 0$ is bounded on fintie intervals and F is a cdf satisfying $F(0) = 1$ and $\lim_{t\to\infty} F(t) = 1$.

(b) Check to see if F is nonarithmetic and h is directly Riemann integrable on $[0, \infty)$.

(c) If all the above conditions are satisfied, then

$$\lim_{t\to\infty} H(t) = \frac{\int_0^\infty h(s)ds}{\int_0^\infty xdF(x)}.$$

Proposition 6.39. A necessary condition for $h(t), t \geq 0$ being directly Riemann integrable is h is bounded and continuous a.e. w.r..t lebesgue measure.

Some sufficient conditions are

(a) h is nonnegative and nonincreasing and Riemann integrable on $[0, \infty)$.

(b) h is continuous a.e. on $[0, \infty)$, and there exists a positive function $b(t), t \geq 0$ such that $|h(t)| \leq b(t),\ \forall\ t \geq 0$ with b being directly Riemann integrable.

Remark. Saying h is continuous a.e. means h is continuous at all points except for those contained in some set having having Lebesgue measure zero.

Example 6.40. Suppose $\{N(t)\}$ is a renewal process with point $\{T_n\}$. We set $T_0 = 0$. We are interested in studying how $P(A(t) \leq x)$ behaves as $t \to \infty$, where F, the cdf of interrenewal is assumed to be nonarithmetic and $E[X_1] < \infty$.

Fix $x > 0$. Then since $A(t)$ restarts at $\{T_n\}_{n\in\mathbb{Z}^+}$,

$$\begin{aligned} P(A(t) \leq x) &= P(A(t) \leq x, T_1 > t) + P(A(t) \leq x, T_1 \leq t) \\ &= P(t \leq x, T_1 > t) + \int_0^t P(A(t) \leq x | T_1 = s)\, dF(s) \\ &= \mathbb{1}_{\{t\leq x\}} P(T_1 > t) + \int_0^t P(A(t-s) \leq x)\, dF(s), \end{aligned}$$

where

$$h(t) = \mathbb{1}_{\{t \le x\}} P(T_1 > t),\ t > 0.$$

Since h is nonnegative and nonincreasing and Riemann integrable on $[0, \infty)$ given $E[X_1] < \infty$, we have h is directly Riemann integrable. Then by Key Renewal Theorem,

$$\lim_{t\to\infty} P\left(A(t) \le x\right) = \frac{\int_0^\infty h(t)dt}{E[X_1]} = \frac{\int_0^x P(T_1 > t)dt}{\int_0^\infty P(T_1 > t)dt}.$$

Example 6.41. Consider the excess life process $\{B(t), t \ge 0\}$ for a renewal process $\{N(t), t \ge 0\}$, where $B(t) = T_{N(t)+1} - t, t \ge 0$.

Fix $x \ge 0$. Then

$$\begin{aligned} P\left(B(t) > x\right) &= P\left(B(t) > x, T_1 > t\right) + P\left(B(t) > x, T_1 \le t\right) \\ &= P(T_1 - t > x, T_1 > x) + \int_0^t P\left(B(t) > x | T_1 = s\right) dF(s) \\ &= P(T_1 > t + x) + \int_0^t P\left(B(t-s) > x\right) dF(s). \end{aligned}$$

Assume F is nonarithmetic. The function $h(t)$ is Riemann integrable if and only if $E[T_1] < \infty$. Further noting that $h(t)$ is nonnegative and nonincreasing, so $h(t)$ is directly Riemann integrable if $E[T_1] < \infty$. Then

$$\lim_{t\to\infty} P\left(B(t) > x\right) = \frac{\int_0^\infty P(X_1 > t + x)dt}{E[X_1]} = \frac{\int_x^\infty P(X_1 > t)dt}{E[X_1]}.$$

Thus,

$$\lim_{t\to\infty} P\left(B(t) \le x\right) = \frac{\int_0^x P(X_1 > t)dt}{E[X_1]}.$$

6.9 Regenerative processes

The Key Renewal Theorem can often be used to study the limiting behavior of regenerative processes.

Consider a stochastic process $\{X(t), t \ge 0\}$ with state space $0, 1, 2, \cdots$ having the property that there exist time points at which the process (probabilistically) restarts itself. That is, suppose that with probability 1, there exists a time T_1, such that the continuation of the process beyond T_1 is a probabilistic replica of the whole process starting at 0. Note that this property implies the existence of further times $T_2, T_3, \ldots,$ having the same property as T_1. Such a stochastic process is known as a regenerative process.

From the preceding, it follows that $T_1, T_2, \cdots,$ constitute the arrival times of a renewal process, and we shall say that a cycle is completed every time a renewal occurs.

Definition 6.42. A stochastic process $\{X(t), t \ge 0\}$ is said to be a *regenerative process* w.r.t. the sequence of random times $\{\tau_n\}_{n\in\mathbb{Z}^+}$ if the random blocks $\{(\tau_{n+1} - \tau_n, \{X(t); \tau_n \le t < \tau_{n+1}\})\}_{n\in\mathbb{N}}$ are iid.

Example 6.43. (a) A renewal process is regenerative, and T_1 represents the time of the first renewal.

(b) A recurrent Markov chain is regenerative, and τ_1^i represents the time of the first transition into the initial state i.

Assume $X_0 = i$ w.p.1. Then

$$\tau_0^i = 0,$$

$$\tau_n^i = \inf\{m \geq \tau_{n-1}^i + 1 : X_m = i\}, \ \forall\ n \geq 1.$$

(c) By definition, the excess process $\{B(t)\}$ is regenerative w.r.t $\{T_n\}$.

Example 6.44 (M/G/1 workload process)**.** Suppose customer arrive to a queueing system according to a Poisson process with rate λ, where the kth arrival brings with it an amount of work B_k for processing, where B_k is generally distributed and $\{B_k\}_{k\geq 1}$ is iid and is independent of the arrival process.

Let $W(t)$ be the amount of unfinished work in the system at time t and assume that $W(0) = 0$. Let

$$\tau_0 = 0,$$

$$\tau_n = \inf\{t > \tau_{n-1} : W(t^-) \neq 0, W(t) = 0\}, \ n \geq 1.$$

Then $\{W(t); t \geq 0\}$ is a regenerative process w.r.t. $\{\tau_n\}$.

We are interested in determining the long-run proportion of time that a regenerative process spends in state j. To obtain this quantity, let us imagine that we earn a reward at a rate 1 per unit time when the process is in state j and at rate 0 otherwise, and total reward earned by t is

$$R(t) = \int_0^t \mathbb{1}_{\{X(s)=j\}} ds.$$

As the preceding is clearly a renewal reward process that starts over again at the cycle time T_1, we see

$$\lim_{t\to\infty} \frac{R(t)}{t} = \frac{E\left[\int_0^{T_1} \mathbb{1}_{\{X(s)=j\}} ds\right]}{E[T_1]}.$$

However, the average reward per unit is just equal to the proportion of time that the process is in state j. That is, we have the following

Proposition 6.45. For a regenerative process, the long-run

$$\textit{proportion of time in state } j = \frac{E[\textit{amount of time in } j \textit{ during a cycle}]}{E[\textit{time of a cycle}]}.$$

Remark. If the cycle time T_1 is a **continuous** random variable, then it can be shown by Key Renewal Theorem that the preceding is equal also to the limiting probability that the system in state j at time t. That is, if T_1 is continuous, then

$$\lim_{t\to\infty} P\left(X(t) = j\right) = \frac{E[\text{amount of time in } j \text{ during a cycle}]}{E[\text{time of a cycle}]}.$$

Example 6.46. Consider a positive recurrent CTMC that is initially in state i. By the Markovian property, each time the process reenters state i it starts over again. Thus returns to state i are renewals and constitute the beginnings of new cycles. It follows that the long-run

$$\text{proportion of time in state } j = \frac{E[\text{amount of time in } j \text{ during an } i-i \text{ cycle}]}{\mu_{ii}},$$

where μ_{ii} represents the mean time to return to state i. If we take j to equal i, then we obtain

$$\text{proportion of time in state } i = \frac{1/\nu_i}{\mu_{ii}}.$$

Theorem 6.47. *Consider a real-valued regenerative process $\{X(t)\}$ with regenerative times $\{\tau_n\}_{n\geq 1}$. Then for $x \in \mathbb{R}$, assume $E[\tau_1]$ is nonzero and fintie, and τ_1 is nonarithmetic, then*

$$\lim_{t\to\infty} P\left(X(t) \leq x\right) = \frac{E\left[\int_0^{\tau_1} \mathbb{1}_{\{X(s)\leq x\}} ds\right]}{E[\tau_1]}.$$

Proof. Note

$$\begin{aligned}
P\left(X(t) \leq x\right) &= P\left(X(t) \leq x, \tau_1 > t\right) + P\left(X(t) \leq x, \tau_1 \leq t\right) \\
&= P\left(X(t) \leq x, \tau_1 > t\right) + \int_0^t P\left(X(t) \leq x | \tau_1 = s\right) dP(\tau_1 \leq s) \\
&= P\left(X(t) \leq x, \tau_1 > t\right) + \int_0^t P\left(X(t-s) \leq x\right) dP(\tau_1 \leq s).
\end{aligned}$$

Since $P(\tau_1 > t)$ is nonnegative and nonincreasing and Riemann integrable on $[0, \infty)$ since $E[\tau_1] > \infty$, we have $P(\tau_1 > t)$ is directly Riemann integrable. Since $h(t) = P\left(X(t) \leq x, \tau_1 > t\right) \leq P(\tau_1 > t)$, we have $h(t)$ is direcly Riemann integrable.
Hence, by the Key Renewal theorem, we have

$$\begin{aligned}
\lim_{t\to\infty} P\left(X(t) \leq x\right) &= \frac{1}{E[\tau_1]} \int_0^\infty P\left(X(s) \leq x, \tau_1 > s\right) ds \\
&= \frac{1}{E[\tau_1]} \int_0^{\tau_1} P\left(X(s) \leq x\right) ds \\
&= \frac{1}{E[\tau_1]} E\left[\int_0^{\tau_1} \mathbb{1}_{\{X(s)\leq x\}} ds\right].
\end{aligned}$$

□

Example 6.48. Suppose $\{N(t)\}$ is a renewal process with increments $\{X_n\}$ that have cdf F satisfying $F(0) = 0$ and $\lim_{t\to\infty} F(t) = 1$. Show that its renewal function u satisfies

$$u(x+y) \leq u(x) + u(y), \ x, y \geq 0.$$

Proof.

$$u(t) = 1 + E[N(t)], \ t \geq 0.$$

$$\begin{aligned}
u(t+a)-u(t) &= E[N(t+a)-N(t)] \\
&= \int_0^\infty E\left[N(t+a)-N(t)|Y(t)=y\right] dG_t(y) \\
&= \int_0^a E\left[N(t+a)-N(t)|Y(t)=y\right] dG_t(y) \\
&= \int_0^a E\left[1+N(a-y)\right] dG_t(y) \\
&= \int_0^a u(a-y) dG_t(y) \\
&\le u(a)\int_0^a G_t(y) \\
&\le u(a).
\end{aligned}$$

Example 6.49. For a renewal process $\{N(t)\}$ with points $\{T_n\}$ and interrenewal $\{X_n\}$. Show that

$$E\left[T_{N(t)}\right] = E[X_1]E\left[N(t)+1\right] - E\left[X_{N(t)+1}\right].$$

Proof. It is equivalent to show that

$$E\left[T_{N(t)+1}\right] = E[X_1]E\left[N(t)+1\right].$$

Note that $N(t)+1$ is a stopping time w.r.t $\{X_n\}$. Then by Wald's identity, we have

$$E\left[\sum_{i=1}^{N(t)+1} X_i\right] = E[X_1]E[N(t)+1].$$

Example 6.50. Each time a certain machine breaks down, it is replaced by a new one of the same type. In the long run, what percentage of time is the machine in use less than one year old if the life distribution of a machine is

(a) Uni$(0,2)$;

(b) Exp(1).

Since $A(t)$ is a regenerative process,

$$\frac{1}{t}\int_0^t \mathbb{1}_{\{A(s)\le 1\}} ds = \frac{E\left[\int_0^{T_1} \mathbb{1}_{\{A(s)\le 1\}} ds\right]}{E[T_1]}.$$

Example 6.51. In a server and rally competition involving players A and B, each rally that beginning with a server by A is won by A w.p. P_a and is won by B w.p. $1-P_a$, whereas each rally that begins with a server by B is won by A w.p.1 P_b, and is won by B w.p. $1-P_b$. The winner of the rally earns a point and becomes the server of the next rally. What's the proportion of points are won by A in the long run is

$$\frac{E[T_1]}{E[S_1]} = \frac{\frac{1}{1-P_a}}{\frac{1}{1-P_a}+\frac{1}{P_b}} = \frac{P_b}{P_b+1-P_a}.$$

Example 6.52. Suppose days walk through a "dogg-door" in accordance to a Poisson process $\{A(t); t \geq 0\}$ with rate λ. Furthermore, suppose an unruly child checks the door according to a renewal process $\{N(t); t \geq 0\}$ with points $\{T_n\}$ and interrenewals having cdf F. If upon arrival of the child, the door is unlocked, he lockes it. Similarly, he unlocks it if upon arrival, the door is locked. Suppose at time 0, the door is open. Let $S_n = T_{2n}$ be the times at which the child open the door. Let $\{O(t)\}$ be the renewal process with points $\{S_n\}$. Let $Q(t)$ be the # of dogs that arrive at the door and are denoyed entrance in the interval $(S_{O(t)}, t]$. Compute

$$\lim_{t\to\infty} P\left(Q(t) \geq k\right).$$

Note

$$\begin{aligned} P\left(Q(t)\right) &= P\left(Q(t) \geq k, S_1 \leq t\right) + P\left(Q(t) \geq k, S_1 > t\right) \\ &= \int_0^t P\left(Q(t-s) \geq k\right) dF(s) + P\left(A(t - T_1) \geq k\right). \end{aligned}$$

6.10 Examples

Example 6.53. Suppose that the interarrival distribution for a renewal process is Poisson distributed with mean μ. That is, suppose

$$P(X_n = k) = e^{-\mu}\frac{\mu^k}{k!},\ k = 0, 1, \cdots$$

(a) Find the distribution of T_n.
T_n is Poisson with mean $n\mu$.

(b) Calculate $P\left(N(t) = n\right)$.

$$\begin{aligned} P\left(N(t) = n\right) &= P\left(N(t) \geq n\right) - P\left(N(t) \geq n+1\right) \\ &= P(T_n \leq t) - P(T_{n+1} \leq t) \\ &= \sum_{k=0}^{\lfloor t \rfloor} e^{-n\mu}(n\mu)^k/k! - \sum_{k=0}^{\lfloor t \rfloor} e^{-(n+1)\mu}\left[(n+1)\mu\right]^k/k!. \end{aligned}$$

Example 6.54. Let $\{N_1(t), t \geq 0\}$ and $\{N_2(t), t \geq 0\}$ be independent renewal processes. Let $N(t) = N_1(t) + N_2(t)$.

(a) Are the interarrival times of $\{N(t), t \geq 0\}$ independent?

No, suppose, for instance, that the interarrival times of the first renewal process are identically equal to 1. Let the second be a Poisson process. If the first interarrival time of the process $\{N(t), t \geq 0\}$ is the equal to 3/4, then we can be certain that the next one is **less** than or equal to 1/4.

(b) Are they identically distributed?

No. Use the same processes as in (a). For instance, the first interarrival will equal to 1 with probability $e^{-\lambda}$, where λ is the rate of the Poisson process. The probability will be different for the next interarrival.

(c) Is $\{N(t), t \geq 0\}$ a renewal process?

No, because of (a) and (b).

Example 6.55. A machine in use is replaced by a new machine either when it fails or when it reaches the age of T years. If the lifetimes of successive machines are independent with a common distribution F having density f, show that

(a) the long-run rate at which machines are replaced equals

$$\frac{1}{\int_0^T xf(x)dx + T\left[1 - F(T)\right]}.$$

The number of replaced machines by time t constitutes a renewal process. The time between replacements equals T if lifetime of new machine is $\geq T$, x if lifetime of new machine is $x, x < T$. Hence,

$$E[\text{time between replacements}] = \int_0^T xf(x)dx + T\left[1 - F(T)\right].$$

(b) the long-run rate at which machines in use fail equals

$$\frac{F(T)}{\int_0^T xf(x)dx + T\left[1 - F(T)\right]}.$$

The number of machines that have failed in use by time t constitutes a renewal process. The mean time between in-use failure

$$E[F] = E\left[E\left[F|\text{lifetime of initial machine}\right]\right].$$

Now

$$E[F|\text{lifetime of initial machine is } x] = \begin{cases} x, & x \leq T \\ T + E[F], & x > T \end{cases}.$$

Hence,

$$E[F] = \int_0^T xf(x)dx + (T + E[F])\left[1 - F(T)\right],$$

or

$$E[F] = \frac{\int_0^T xf(x)dx + T\left[1 - F(T)\right]}{F(T)}.$$

Example 6.56. Consider a miner trapped in a room that contains three doors. Door 1 leads him to freedom after two days of travel; door 2 returns him to his room after a four-day journey; and door 3 returns him to his room after a six-day journey. Suppose at all times he is equally likely to choose any of the three doors, and let T denote the time it takes the miner to become free.

(a) Define a sequence of independent and identically distributed random variables $X_1, X_2, \cdots$ and a stopping time N such that

$$T = \sum_{i=1}^{N} X_i.$$

Note you may have to imagine that the miner continues to randomly choose doors even after he reaches safety. Let X_i denote the amount of time he has to travel after his i^{th} choice (we will assume that he keeps on making choices even after becoming free). N is the number of choices he makes until becoming free.

(b) Use Walds's equation to find $E[T]$.

$$N \sim \text{Geo}(1/3).$$

$$E[X] = \frac{1}{3}(2+4+6) = 4.$$

Hence,

$$E[T] = 12.$$

(c) Compute $E\left[\sum_{i=1}^{N} X_i \middle| N = n\right]$ and note that it is not equal to $E\left[\sum_{i=1}^{n} X_i\right]$. Given $N = n$, $X_1, \ldots, X_{n-1}$ are equally likely to be either 4 or 6, $X_n = 2$. Then

$$E\left[\sum_{i=1}^{N} X_i \middle| N = n\right] = (n-1)\frac{1}{2}(4+6) + 2 = 5n - 3.$$

(d) Use part (c) for a second derivation of $E[T]$.

$$E[T] = E[5N - 3] = 15 - 3 = 12.$$

Example 6.57. If we want to use Wald's identity, X_i's must be independent.

Example 6.58. For a renewal process, let $A(t)$ be the age at time t. Prove that if $\mu < \infty$, then with probability 1

$$\frac{A(t)}{t} \to 0 \text{ as } t \to \infty.$$

Proof.

$$\frac{A(t)}{t} = \frac{t - T_{N(t)}}{t} = 1 - \frac{T_{N(t)}}{t} = 1 - \frac{T_{N(t)}}{N(t)}\frac{N(t)}{t},$$

the result follows since $\frac{T_{N(t)}}{N(t)} \to \mu$ by SLLN and $\frac{N(t)}{t} \to \frac{1}{\mu}$.

Example 6.59. If $A(t)$ and $B(t)$ are, respectively, the age and the excess at time t of a renewal process having an interarrival distribution F, calculate

$$P\left(B(t) > x | A(t) = s\right).$$

$$\begin{aligned}
P\left(B(t) > x | A(t) = s\right) &= P\left(0 \text{ renewals in } (t, t+x] | A(t) = s\right) \\
&= P\left(\text{interarrival} > x + s | A(t) = s\right) \\
&= P\left(\text{interarrival} > x + s | \text{interarrival} > s\right) \\
&= \frac{1 - F(x+s)}{1 - F(s)}.
\end{aligned}$$

Example 6.60. Satellites are launched according to a Poisson process with rate λ. Each satellite will, independently, orbit the earth for a random time having distribution F. Let $X(t)$ denote the number of satellites orbiting at time t. Determine $P(X(t) = k)$.

We can view this an $M/G/\infty$ system where a satellite launching corresponds to an arrival and F si the service distribution. Hence,

$$P(X(t) = k) = e^{-\lambda(t)}\lambda^k(t)/k!,$$

where

$$\lambda(t) = \lambda \int_0^t [1 - F(s)]\, ds.$$

Example 6.61 (!!!)**.** Consider a system that can be in either state 1 or 2 or 3. Each time the system enters state i, it remains there for a random amount of time having mean μ_i and the makes a transition into state j with probability P_{ij}. Suppose

$$P_{12} = 1,\ P_{21} = P_{23} = \frac{1}{2},\ P_{31} = 1.$$

(a) What proprtion of transitions takes the system into state 1?

$$P = \begin{pmatrix} 0 & 1 & 0 \\ 1/2 & 0 & 1/2 \\ 1 & 0 & 1 \end{pmatrix}.$$

We have the limiting probabilities for the MC are

$$\pi_1 = \pi_2 = 2/5, \quad \pi_3 = 1/5.$$

So

$$2/5.$$

(b) If $\mu_1 = 1$, $\mu_2 = 2, \mu_3 = 3$, then what proportion of time does the system spend in each state? Using

$$\pi_i = \frac{\pi_i \mu_i}{\sum_{i=1}^3 \pi_i \mu_i},$$

we have the limiting probabilities for the CTMC are

$$\gamma_1 = 2/9,\ \gamma_2 = 4/9,\ \pi_3 = 3/9.$$

6.11 Challenging problems

Example 6.62. Farmer Bill has a very large field of tomatoes near Edisto Beach. Each time he spends an amount of time x (measured in hours) in the field, the number of tomatoes he finds is Poisson distributed with rate λx, where $\lambda > 0$ is a fixed rate. Whenever he returns home from the field, he places however many tomatoes he has picked in the back of his truck. Assume the amount of time spent in the field is generally distributed with cdf F, and is independent of everything else. Assume further that once he has made two trips in the field, he drives for a fixed amount of time

D (in hours) until he arrives home, where he instantaneously unloads the tomatoes from his truck. He then stays home for a uniformly distributed amount of time in the interval $[6, 8]$, measured in hours. Finally, after spending some time at home, he drives back to the field for a fixed amount of time D, and this process repeats itself indefinitely.

(a) Suppose at time $t = 0$, Farmer Bill has just arrived to the field from his house to pick tomatoes. Define $Z(t)$ to be the number of tomatoes in his truck at time t. Derive a renewal equation whose solution is $E[Z(t)]$.

(b) Compute $\lim_{t\to\infty} E[Z(t)]$. Clearly state whatever further assumptions you need to make in order to ensure this limit exists.

(a) Let T_1 be the first time to return to the field from his house.

$$E[Z(t)] = \int_0^t E[Z(t-s)]dF_{T_1}(s) + \int_t^\infty E[Z(t)|T_1 = s]dF_{T_1}(s).$$

(b) First note F_{T_1} is nonarithmetic due to the uniformly distributed amount of time he spends at home.

Let N be the number of tomatoes caught in the first cycle given $T_1 > t$, then

$$Z(t) \le N \text{ when } T_1 > t.$$

Since

$$h(t) \le \int_t^\infty E[N|T_1 = s]dF_{T_1}(s) = \int_t^\infty E[N]dF_{T_1}(s) = E[N]P(T_1 > t).$$

$$\int_0^\infty h(t)dt = \int_0^\infty E[N]P(T_1 > t)dt = E[N]E[T_1] < \infty.$$

So $h(t)$ is directly Riemman integrable.

$$\lim_{t\to\infty} E[Z(t)] = \frac{\int_0^\infty h(s)ds}{E[T_1]},$$

where

$$E[T_1] = 2\mu_F + 2D + 7,$$

and

$$\begin{aligned}
\int_0^\infty h(t)dt &= \int_0^\infty \int_t^\infty E[Z(t)|T_1 = s]dF_{T_1}(s)dt \\
&= \int_0^\infty \int_0^s E[Z(t)|T_1 = s]dtdF_{T_1}(s) \\
&= \int_0^\infty E\left[\int_0^s Z(t)|T_1 = sdt\right] dF_{T_1}(s) \\
&= \int_0^\infty E\left[\int_0^{X_2} \lambda X_1 dt + \int_{X_2}^{X_2+D} \lambda(X_1 + X_2)dt\right] dF_{T_1}(s) \\
&= \int_0^\infty E\left[\int_0^{X_2} \lambda X_1 dt + \int_{X_2}^{X_2+D} \lambda(X_1 + X_2)dt\right] dF_{T_1}(s) \\
&= \int_0^\infty E\left[\lambda X_1 X_2 + \lambda D(X_1 + X_2)\right] dF_{T_1}(s) \\
&= \int_0^\infty \left(\lambda E[X_1]E[X_2] + \lambda D\left(E[X_1] + E[X_2]\right)\right) dF_{T_1}(s) \\
&= \lambda E[X_1]E[X_2] + \lambda D\left(E[X_1] + E[X_2]\right) \\
&= \lambda\mu F^2 + 2\lambda\mu_F D.
\end{aligned}$$

Example 6.63. Kegs of beer at the OK Saloon hold 30 quarts and at any time only one keg is in use. Customers arrive according to a renewal process with mean $\mu < \infty$ interarrival times. Each customer, independently of other customers, orders a pint of beer with probability $p, 0 < p < 1$, or nothing with probability $1 - p$. When the keg is empty, it is instantaneously replaced with a full keg. Let $X(t)$ be the level of keg in use at time t. You may assume that the state space is $\{0.5, 1, 1.5, \ldots, 30\}$.

(a) Argue that $\{X(t); t \geq 0\}$ is a regenerative process.

(b) Compute

$$\lim_{t\to\infty} P\left(X(t) = k\right).$$

(c) Suppose the OK Saloon makes \$0.75 on every beer sold. What is the long run average reward?

(a) Let τ_n be the time the nth keg is replaced.
Since $\{\tau_n - \tau_{n-1}, \{X(t) : \tau_{n-1} \leq t \leq t_n\}\}_{n\in\mathbb{Z}^+}$ are iid blocks. Customers arrive according to a renewal process and order beers with probability p, independently of everyone else. The process regenerates every time the keg is replaced.

(b)

$$\lim_{t\to\infty} P\left(X(t) = k\right) = \frac{E\left[\int_0^{\tau_1} \mathbb{1}_{\{X(s)=k\}} ds\right]}{E[\tau_1]} = \frac{\frac{\mu}{p}}{60\frac{\mu}{p}} = \frac{1}{60}.$$

Alternative: Consider the DTMC on the transition times. The transition matrix is

$$P = \begin{pmatrix} 0 & 1 & 0 & 0 & \cdots \\ 0 & 0 & 1 & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & 0 & 0 & 0 & 0 \end{pmatrix},$$

which is doubly stochastic.
Then

$$\pi_j = \frac{1}{60}, \ \forall \ j.$$

Then

$$\lim_{t\to\infty} P\left(X(t) = k\right) = \pi_k^* = \frac{\pi_k \mu}{\sum_{j=1}^{60} \pi_j \mu} = \frac{1}{60}.$$

(c) Let $R(t)$ be the reward by time t, by Renewal reward theorem,

$$\lim_{\to\infty} \frac{R(t)}{t} = \frac{E[R_1]}{E[\tau_1]} = \frac{60 * 0.75}{60\frac{\mu}{p}} = \frac{3p}{4\mu}.$$

Example 6.64. Suppose parts have lifetimes $L_1, L_2, \cdots$ which are iid and have distribution ϕ. Suppose that the nth item is replaced whenever it fails or has been in use Y_n units of time, whichever comes first. Here $Y_1, Y_2, \cdots$ are iid with distribution ψ and are independent of the lifetimes.

(a) Show that the times $S_n, n \geq 1$ of successive replacements form a renewal process and compute the distribution function F of the times between replacements.

(b) Show that the times of failures T_n form a renewal process and show that the distribution function G of the time between two failures satisfyies, for each $t > 0$,

$$1 - G(t) = 1 - F(t) + \int_0^t \left(1 - G(t-x)\right)\left(1 - \phi(x)\right)\psi(dx).$$

(c) Compute the expected value m_f of the time between two failures directly from (b) to find

$$m_f = \frac{\int_0^\infty \left(1 - \phi(x)\right)\left(1 - \psi(x)\right) dx}{\int_0^\infty \phi(x)\psi(dx)}.$$

(a) Because $\{\min(L_n, Y_n)\}_{n\in\mathbb{Z}^+}$ are iid.

$$F(t) = 1 - \left(1 - \phi(t)\right)\left(1 - \psi(t)\right).$$

(b)

$$
\begin{aligned}
1-G(t) &= P\left(T_1>t\right)=P\left(T_1>t, S_1\leq t\right)+P\left(T_1>t, S_1>t\right)\\
&= P\left(T_1>t, S_1\leq t\right)+P\left(T_1>t, S_1>t\right)\\
&= P(T_1>t, S_1\leq t)+P(T_1>t)\\
&= 1-F(t)+P(T_1>t, S_1\leq t)\\
&= 1-F(t)+\int_0^\infty P(T_1>t, S_1\leq t|Y_1=x)\psi(dx)\\
&= 1-F(t)+\int_0^\infty P(T_1>t, Y_1\leq t, Y_1\leq L_1|Y_1=x)\psi(dx)\\
&= 1-F(t)+\int_0^t P(T_1>t, x\leq L_1|Y_1=x)\psi(dx)\\
&= 1-F(t)+\int_0^t P(T_1>t-x, x\leq L_1)\psi(dx)\\
&= 1-F(t)+\int_0^t P(T_1>t-x)P(x\leq L_1)\psi(dx)\\
&= 1-F(t)+\int_0^t \left(1-G(t-x)\right)\left(1-\phi(x)\right)\psi(dx).
\end{aligned}
$$

(c)

$$
\begin{aligned}
m_f &= \int_0^\infty \left(1-G(t)\right)dt\\
&= \int_0^\infty \left(1-\phi(t)\right)\left(1-\psi(t)\right)dt+\int_0^\infty\int_0^t \left(1-G(t-x)\right)\left(1-\phi(x)\right)\psi(dx)dt\\
&= \int_0^\infty \left(1-\phi(t)\right)\left(1-\psi(t)\right)dt+\int_0^\infty\int_x^\infty \left(1-G(t-x)\right)\left(1-\phi(x)\right)dt\psi(dx)\\
&= \int_0^\infty \left(1-\phi(t)\right)\left(1-\psi(t)\right)dt+\int_0^\infty\int_0^\infty \left(1-G(u)\right)\left(1-\phi(x)\right)du\psi(dx)\\
&= \int_0^\infty \left(1-\phi(t)\right)\left(1-\psi(t)\right)dt+\int_0^\infty \left(1-G(t)\right)dt\int_0^\infty \left(1-\phi(x)\right)\psi(dx).
\end{aligned}
$$

So

$$
m_f=\frac{\int_0^\infty \left(1-\phi(t)\right)\left(1-\psi(t)\right)dt}{1-\int_0^\infty \left(1-\phi(x)\right)\psi(dx)}=\frac{\int_0^\infty \left(1-\phi(x)\right)\left(1-\psi(x)\right)dx}{\int_0^\infty \phi(x)\psi(dx)}.
$$

Chapter 7

Continuous-Time Markov Chains

Continuous-time Markov chains are stochastic processes that are often used to model random phenomena that evolve in a continuous-time setting. Such processes are typically thought of as being very tractable, in that often many performance measures associated with processes can be computed either analytically or numerically without an excessive amount of difficulty. Continuous-time Markov chains appear in the research literature associated with many fields of study, such as Biology, Computer Science, Electrical Engineering, Industrial Engineering, Operations Research, and Statistics.

One example of a CTMC is Poisson process $\{N(t), t \geq 0\}$, having states $0, 1, 2, \cdots$ that always proceeds from state n to state $n+1$, where $n \geq 0$. Such a process is known as a pure birth process.

7.1 Continuous-Time Markov Chains

Definition 7.1. A continuous-time stochastic process $\{X(t); t \geq 0\}$ with discrete state space. We say that the process $\{X(t), t \geq 0\}$ is a CTMC if $\forall\ s, t \geq 0$ and $\forall\ i, j, x(u) \in E,\ 0 \leq u < s$,

$$P\left(X(t+s) = j | X(s) = i, X(u) = x(u), 0 \leq u < s\right) = P\left(X(t+s) = j | X(s) = i\right).$$

If in addition, $P\left(X(t+s) = j | X(s) = i\right)$ is independent of s, then the CTMC is said to have stationary or **time-homogeneous** transition probabilities, i.e.,

$$P\left(X(s+t) = j | X(s) = i\right) = P\left(X(t) = j | X(0) = i\right) := P_{ij}(t).$$

All Markov chains considered in this text will be assumed to have stationary transition probabilities.

Remark. Suppose that a CTMC enters state i at some time, say, time 0, and suppose that the process does not leave state i during the next ten minutes. What is the probability that the process will not leave state i during the following five minutes? Since the process is in state i at time 10, it follows, by the Markovian property, that the probability that it remains in that state during the interval $[10, 15]$ is just the (unconditional) probability that it stays in state i for at least five miutes. That is, if we let T_i denote the amount of time that the process stays in state i before making a transition into a different state, then

$$P(T_i > 15 | T_i > 10) = P(T_i > 5).$$

Or, in general, by the same reasoning,

$$P(T_i > s + t | T_i > s) = P(T_i > t), \ \forall \ s, t \geq 0.$$

Hence, the random variable T_i is memoryless and must thus be exponentially distributed.

Definition 7.2. In fact, the preceding gives us another way of defining a CTMC. Namely, it is a stochastic process having the properties that each time it enters state i

(a) the amount of time it spends in that state before making a transition into a different state is exponentially distributed with mean, say, $\frac{1}{\nu_i}$, and

(b) when the process leaves state i, it next enters state j with probability, say, P_{ij}. Of course, the P_{ij} must satisfy

$$P_{ii} = 0, \ \forall \ i$$

$$\sum_j P_{ij} = 1, \ \forall \ i.$$

Remark. In other words, a CTMC is a stochastic process that moves from state to state in accordance with a (discrete time) Markov chain, but is such that the amount of time it spends in each state, before proceeding to the next state, is exponentially distributed (since the remaining holding time must only depend on t by the Markov property). In addition, the amount of time the process spends in state i, and the **next** state visited, must be independent random variables. For if the next state visited were dependent on T_i, then information as to how long the process has already been in state i would be relevant to the prediction of the next state−and this contradicts the Markovian assumption.

Remark. A CTMC can simply be described by a transition matrix $P = (P_{ij})_{i,j \in E}$, describing the chain changes state step-by-step at transition time, together with a set of rates $\nu_i, i \in E$, the holding time rates. Each time state i is visited, the chain spends $\text{Exp}(1/\nu_i)$ unit of time there before moving on.

7.2 Path Structure

Definition 7.3. A stochastic process $\{X(t); t \geq 0\}$ on a countable state space E is said to be a *jump process* if its sample path are right continuous and piecewise-constant with finite length w.p.1., and the number of transitions in any finite interval is finite w.p.1.. That is,

$$X(t) = X_n = X(\tau_n) \text{ if } t \in [\tau_n, \tau_{n+1}) \text{ for some } n \in \mathbb{N},$$

where $0 = \tau_0 < \tau_1 < \tau_2 < \cdots$ are the jump times of the process.

Remark. The properties of jump process imply that

(a) $T_n \to \infty$ w.p.1. as $n \to \infty$: this is a consequence of the fact that the number of transitions in any finite time interval is finite, w.p.1..

(b) $X_n \neq X_{n+1}$ for each $n \geq 0$ since the process jumps at each time τ_n.

We define the sojourn time of the process in state X_{n-1} as $T_n = \tau_n - \tau_{n-1}$, $\forall\ n \in \mathbb{Z}^+$: for notational convenience, we set $T_0 = 0$.

Definition 7.4. Let τ_n denote the time at which the nth change of state (transition) occurs, and define $X_n = X(\tau_n)$, the state right after the nth transition. Then $\{X_n; n \in \mathbb{N}\}$ defines the underlying DTMC, called the embedded MC. Then the transition matrix $P = (P_{ij})_{i,j\in E}$, where $P_{ij} = P(X_{n+1} = j|X_n = i)$, $\forall\ i \neq j$.

Theorem 7.5. *A jump process $\{X(t); t \geq 0\}$ on E with embedded process $\{X_n\}_{n\geq 0}$ and sojourn times (holding time) $\{T_n\}_{n\geq 1}$ is a CTMC if and only if the following are satisfied:*

(a) $\{X_n\}_{n\geq 0}$ is a DTMC on E with transition matrix $P = (P_{ij})_{i,j\in E}$, where $P_{ii} = 0$, $\forall\ i \in E$.

(b) For $m \in \mathbb{Z}^+$ and $t_1, \ldots, t_m \geq 0$, we have

$$\begin{aligned} P\left(T_1 \leq t_1, \ldots, T_m \leq t_m | X_n, n \in \mathbb{N}\right) &= \prod_{n=1}^{m} P(T_n \leq t_n | X_{n-1}) \\ &= \prod_{n=1}^{m} \left(1 - e^{-\nu_{X_{n-1}} t}\right), \end{aligned}$$

which says that the sojourn times $\{T_n\}_{n\in\mathbb{Z}^+}$ are conditionally independent, given the entire $\{X_n\}_{n\in\mathbb{N}}$ and such a sojourn time in state i is exponentilly distributed with rate ν_i.

Remark. A CTMC makes transition from state to state, independently of the past according to the embedded DTMC. On entering a state, it remains in that state independently of the past for an exponentially distributed amount of time before changing state again.

Theorem 7.6. *A jump process $\{X(t)\}$ with embedded points $\{(X_n, T_n)\}$ is a CTMC if and only if*

$$P\left(X_{n+1} = j, T_{n+1} \leq t | X_n = i, (X_j, T_j), j \leq n\right) = P_{ij}\left(1 - e^{-\nu_i t}\right).$$

In particular, $\{X(t); t \geq 0\}$ is a CTMC *if and only if $\{(X_n, T_n)\}$ is a* DTMC*, where $X_n = X(\tau_n)$ and*

$$P(X_{n+1} = j, T_{n+1} \leq t | X_n = i, T_n = s) = P_{ij}\left(1 - e^{-\nu_i t}\right).$$

The defining parameters for a CTMC $\{X(t); t \geq 0\}$ are its initial distribution $\{\alpha_i\}_{i\in E}$, the transition matrix P for the embedded DTMC and sojourn rates $\{\nu_i\}_{i\in E}$.

Proposition 7.7. Two CTMC's have the same distribution if and onlt if their defining parameters are equal.

Example 7.8. A set of m jobs are processed in parallel until they are all completed. Upon completion, another m jobs instantaneously enter the system and are processed. This is repeated independently. Assume the times to process jobs are independently exponentially distributed with rate λ. Let $X(t)$ denote the number of jobs being processed at time t. Then $E = \{1, \ldots, m\}$. Let $0 < m$, then

$$P\left(X_{n+1} = i - 1, T_{n+1} > t | X_n = i, (X_j, T_j), j \leq n\right) = e^{-i\lambda t},\ \forall\ i \in [m].$$

Example 7.9 ($M/M/s$)**.**

$$P\left(X_{n+1}=i+1, T_{n+1}>t|X_n=i,(X_j,T_j), j\leq n\right)=e^{-[\min(i,s)\mu+\lambda]t}\frac{\lambda}{\lambda+\min(i,s)\mu},\ \forall\ i\in\mathbb{Z}^+,$$

$$P\left(X_{n+1}=i-1, T_{n+1}>t|X_n=i,(X_j,T_j), j\leq n\right)=e^{-[\min(i,s)\mu+\lambda]t}\frac{\min(i,s)\mu}{\lambda+\min(i,s)\mu},\ \forall\ i\in\mathbb{Z}^+.$$

Example 7.10 (A Shoe Shine Shop)**.** Consider a shoe shine establishment consisting of two chairs−chair 1 and chair 2. A customer upon arrival goes initially to chair 1 where his shoes are cleaned and polish is applied. After this is done the customer moves on to chair 2 where the polish is buffed. The service times at the two chairs are assumed to be independent random variables that are exponentially distributed with respective rates μ_1 and μ_2. Suppose that potential customers arrive in accordance with a Poisson process having rate λ, and that a potential customer will enter the system only if both chairs are empty.

The preceding model can be analyzed as a continuous-time Markov chain, but first we must decide upon an appropriate state space.

0	system is empty
1	a customer is in chair 1
2	a customer is in chair 2

Then

$$\nu_0=\lambda,\quad \nu_1=\mu_1,\quad \nu_2=\mu_2,$$
$$P_{01}=P_{12}=P_{20}=1.$$

7.3 The transition probability function $P_{ij}(t)$

Definition 7.11. Define the transition matrix for a CTMC,

$$P(t)=\left(P_{ij}(t)\right)_{i,j\in E}.$$

Theorem 7.12 (Chapman-Kolmogorov equations)**.**

$$P(s+t)=P(s)P(t),\ \forall\ s,t\geq 0.$$

$$P_{ij}(s+t)=\sum_{k\in E}P_{ik}(s)P_{kj}(t),\ \forall\ s,t\geq 0.$$

Lemma 7.13.

$$\lim_{t\to 0}\frac{1-P_{ii}(t)}{t}=\nu_i,$$

$$\lim_{t\to 0}\frac{P_{ij}(t)}{t}=q_{ij},\ \forall\ i\neq j.$$

Proof.

$$P_{ii}(t)=P\left(X(t)=i|X(0)=i\right),$$
$$1-P_{ii}(t)=P\left(X(t)\neq i|X(0)=i\right).$$

Let $\{N_i(t), t \geq 0\}$ be a Poisson process with rate ν_i. Then

$$P\left(X(t) \neq i | X(0) = i\right) = P\left(N_i(t) \geq 1\right) = \nu_i t + o(t), \text{ as } t \to 0.$$

Thus,

$$\lim_{t \to 0} \frac{1 - P_{ii}(t)}{t} = \nu_i.$$

Next,

$$\begin{aligned}
P_{ij}(t) &= P\left(X(t) = j | X(0) = i\right) \\
&= P\left(X(t) = j, T_1 \leq t | X(0) = i\right) + P\left(X(t) = j, T_1 > t | X(0) = i\right) \\
&= \sum_{k \neq i} P\left(X(t) = j, T_1 \leq t, X(T_1) = k | X(0) = i\right) + \mathbb{1}_{\{i=j\}} e^{-\nu_i t} \\
&= \sum_{k \neq i} P\left(X(t) = j, T_1 \leq t | X(T_1) = k, X(0) = i\right) P_{ik} + \mathbb{1}_{\{i=j\}} e^{-\nu_i t} \\
&= \sum_{k \neq i} \int_0^t P\left(X(t) = j | T_1 = s, X(T_1) = k, X(0) = i\right) \nu_i e^{-\nu_i s} ds P_{ik} + \mathbb{1}_{\{i=j\}} e^{-\nu_i t} \\
&= \sum_{k \neq i} \int_0^t P\left(X(t) = j | X(s) = k\right) \nu_i e^{-\nu_i s} ds P_{ik} + \mathbb{1}_{\{i=j\}} e^{-\nu_i t} \\
&= \sum_{k \neq i} \int_0^t P_{kj}(t-s) \nu_i e^{-\nu_i s} ds P_{ik} + \mathbb{1}_{\{i=j\}} e^{-\nu_i t} \\
&= \int_0^t \sum_{k \neq i} P_{kj}(t-s) \nu_i e^{-\nu_i s} ds P_{ik} + \mathbb{1}_{\{i=j\}} e^{-\nu_i t}.
\end{aligned}$$

Then for $i \neq j$,

$$\begin{aligned}
\lim_{t \to 0} \frac{P_{ij}(t)}{t} &= \lim_{t \to 0} \frac{1}{t} \int_0^t \sum_{k \neq i} P_{kj}(t-s) \nu_i e^{-\nu_i s} P_{ik} ds \\
&= \lim_{t \to 0} \frac{-1}{t} \int_0^t \sum_{k \neq i} P_{kj}(s) \nu_i e^{-\nu_i (t-s)} P_{ik} ds \\
&= \sum_{k \neq i} P_{kj}(0) \nu_i P_{ik} \\
&= \nu_i P_{ij}.
\end{aligned}$$

□

Corollary 7.14.

$$P'_{ij}(0) = \lim_{t \downarrow 0} \frac{P_{ij}(t) - P_{ij}(0)}{t} = \nu_i P_{ij}, \ \forall\ i \neq j,$$

$$P'_{ii}(0) = \lim_{t \downarrow 0} \frac{P_{ii}(t) - P_{ii}(0)}{t} = -\nu_i, \ \forall\ i \neq j,$$

Definition 7.15. The matrix $Q = \left(P'_{ij}(0)\right)_{i,j\in E}$ is called the transition rate matrix or *infinitesmal generator* of the CTMC, where

$$P'_{ij}(0) = \lim_{h\downarrow 0} \frac{P_{ij}(h) - \mathbb{1}_{\{i=j\}}}{h} = q_{ij},$$

where $q_{ij} = Q_{ij}$.

Remark. ν_i can be interpreted as the transition rate out of state i given that $X(t) = i$.

$q_{ij} = \nu_i P_{ij}$, $\forall\, i \neq j$, which can be interpreted as the transition rate from state i to state j given that $X(t) = i$. Then

$$\nu_i = \sum_{j\neq i} \nu_i P_{ij} = \sum_{j\neq i} q_{ij},$$

$$P_{ij} = \frac{q_{ij}}{\nu_i} = \frac{q_{ij}}{\sum_{j\neq i} q_{ij}},$$

it follows that specifying the instantaneous transition rates determines the parameters of the CTMC.

Theorem 7.16 (Kolmogorov's Backward Equations)**.**

$$P'_{ij}(t) = \sum_{k\in E} q_{ik} P_{kj}(t),\ \forall\, t \geq 0,$$

or

$$P'(t) = QP(t).$$

Remark. (a) Both the foward and backward equations satisfy the initial condition $P(0) = I$.

(b) When E is finite, both of these equations have a unique solution. Moreover,

$$P(t) = e^{Qt} = \sum_{k=0}^{\infty} Q^k \frac{t^k}{k!}, \quad t \geq 0.$$

Example 7.17. The backward equations for the pure birth process become

$$P'_{ij}(t) = \lambda_i P_{i+1,j}(t) - \lambda_i P_{ij}(t).$$

Example 7.18. The backward equations for the birth and death process become

$$P'_{0j}(t) = \lambda_0 P_{1,j}(t) - \lambda_i P_{0j}(t),$$

$$P'_{ij}(t) = \lambda_i P_{i+1,j}(t) + \mu_i P_{i-1,j}(t) - (\lambda_i + \mu_i) P_{ij}(t),\ \forall\, i > 0.$$

Example 7.19 (A CTMC consisting of two states)**.** Consider a machine that works for an exponential amount of time with rate λ before break down; and suppose that it takes an exponential amount of time with mean μ repair the machine. If the machine is in working condition at time 0, then what is the probability that it will be working at time $t = 10$?

Note that the process is a birth and death process with state 0 meaning that the machine is working and state 1 being repaired and

$$\lambda_0 = \lambda,\ \mu_1 = \mu,$$

$$\lambda_1 = 0,\ \mu_0 = 0.$$

Note

$$Q = \begin{pmatrix} -\lambda & \lambda \\ \mu & -\mu \end{pmatrix}.$$

Then

$$P'_{00}(t) = \lambda P_{10}(t) - \lambda P_{00}(t),$$
$$P'_{10}(t) = \mu P_{00}(t) - \mu P_{10}(t).$$

Then

$$\mu P'_{00}(t) + \lambda P'_{10}(t) = 0.$$

So

$$\mu P_{00}(t) + \lambda P_{10}(t) = c.$$

Since $P_{00}(0) = 1$ and $P_{10}(0) = 0$, we obtain $c = \mu$. Then

$$\lambda P_{10}(t) = \mu \left[1 - P_{00}(t)\right].$$

As a result,

$$P'_{00}(t) = \mu - (\mu + \lambda) P_{00}(t).$$

Then

$$P_{00}(t) = \frac{\lambda}{\mu + \lambda} e^{-(\mu+\lambda)t} + \frac{\mu}{\mu + \lambda}.$$

So

$$P_{10} = \frac{\mu}{\mu + \lambda} - \frac{\mu}{\mu + \lambda} e^{-(\mu+\lambda)t}.$$

Theorem 7.20 (Kolmogorov's Forward Equations)**.**

$$P'_{ij}(t) = \sum_{k \in E} P_{ik}(t) q_{kj},$$

or

$$P'(t) = P(t)Q.$$

Proof. Assuem E is finite. Then

$$\begin{aligned}
P'_{ij}(t) &= \lim_{s \downarrow 0} \frac{P_{ij}(t+s) - P_{ij}(t)}{s} \\
&= \lim_{s \downarrow 0} \frac{\sum_{k \in E} P_{ik}(t) P_{kj}(s) - \sum_{k \in E} P_{ik}(t) P_{kj}(0)}{s} \\
&= \sum_{k \in E} P_{ik}(t) \lim_{s \downarrow 0} \frac{P_{kj}(s) - P_{kj}(0)}{s} \\
&= \sum_{k \in E} P_{ik}(t) P'_{kj}(0) \\
&= \sum_{k \in E} P_{ik}(t) q_{kj}.
\end{aligned}$$

□

Example 7.21. The forward equations for the pure birth process become

$$P'_{ij}(t) = \lambda_{j-1}P_{i,j-1}(t) - \lambda_j P_{ij}(t).$$

Example 7.22. The forward equations for the birth and death process become

$$P'_{i0}(t) = \mu_1 P_{i1}(t) - \lambda_0 P_{i0}(t),$$

$$P'_{ij}(t) = \lambda_{j-1}P_{i,j-1}(t) + \mu_{j+1}P_{i,j+1}(t) - (\lambda_j + \mu_j)P_{ij}(t).$$

7.3.1 Spectral theorey used to compute $P(t)$

In our two-state CTMC example, we have

$$\alpha I - Q = \begin{pmatrix} \alpha+\lambda & -\lambda \\ -\mu & \alpha+\mu \end{pmatrix}.$$

Then

$$\det(\alpha I - Q) = \alpha^2 + (\lambda+\mu)\alpha.$$

Hence, the two eigenvalues of Q are $-(\lambda+\mu)$ and 0. The eigenvetor associated with eigenvalue $\alpha_1 = 0$ is simply $x_1 = (1,1)^T$ since $Q(1,1)^T = \vec{0}$. Furthermore, one can show that an eigenvector corresponding to the eigenvalue $\alpha_2 = -(\lambda+\mu)$ is the vector $x_2 = (1, -\mu/\lambda)^T$. Since $Q^n\vec{x}_i = \alpha_i^n \vec{x}_i,\ \forall\ i = 1,2$, we have

$$Q^n(\vec{x}_1, \vec{x}_2) = (\vec{x}_1, \vec{x}_2)\mathrm{diag}(\alpha_1^n, \alpha_2^n),$$

i.e.,

$$Q^n S = S\Lambda^n.$$

Then

$$Q^n = S\Lambda^n S^{-1}.$$

Hence,

$$P(t) = \sum_{k=0}^{\infty} Q^k \frac{t^k}{k!} = \cdots,\ t \geq 0.$$

7.4 Stationary and limiting distributions

Definition 7.23. We say $\{X(t); t \geq 0\}$ is *irreducible* if and only if $\{X_n; n \geq 0\}$ is irreducible.

Definition 7.24. Define

$$\xi_i = \inf\{t > 0 : X(t^-) \neq X(t) = i\},\ \forall\ i \in E.$$

Definition 7.25. A state $i \in E$ is said to be *recurrent* if

$$P_i(\xi_i < \infty) = 1.$$

Furthermore, a recurrent state i is said to be *positive recurrent* if

$$E_i[\xi_i] < \infty.$$

Remark. It is not difficult to see that i is recurrent w.r.t. the CTMC $\{X(t)\}$ if and only if it is recurrent w.r.t. the embedded DTMC $\{X_n\}$. The same can not be said with regards to positive recurrence, it is entirely possible that for a state $i \in E$ to satisfy $E_i[\eta_i] < \infty$, but $E_i[\xi_i] = \infty$, or vice verse.

Definition 7.26. A measure $\gamma = (\gamma_i)_{i\in E}$ on E is an *invariant measure* of X if for each $t \geq 0$, we have

$$\gamma = \gamma P(t).$$

Definition 7.27. A invariant measure that also satisfies $\sum_{i\in E} \gamma_i = 1$ is called a *stationary distribution.*

Definition 7.28. We say that X is *stationary* if for any integer $n \geq 1$, any collection of real number $0 \leq s_1 < \cdots < s_n, t \geq 0$

$$(X(s_1 + t), \ldots, X(s_n + t)) \stackrel{d}{=} (X(s_1), \ldots, X(s_n)).$$

Theorem 7.29. *Suppose $\gamma = \{\gamma_i\}_{i\in E}$ is a nonnegative measure on E. Then the following statements are equivalent.*

(a) γ is an invariant measure of $\{X(t); t \geq 0\}$.

(b) $\{\gamma_i \nu_i\}_{i\in E}$ is an invariant measure of $\{X_n\}_{n\geq 0}$.

(c) γ satisfies the balance equations $\gamma Q = 0$.

Lemma 7.30.

$$E_i\left[\int_0^{\xi_i} \mathbb{1}_{\{X(t)=j\}}\right] = \frac{1}{\nu_i} E_i\left[\sum_{n=0}^{\eta_i - 1} \mathbb{1}_{\{X_n=j\}}\right], \ \forall\, i, j \in E.$$

Proposition 7.31. Suppose $i \in E$ is a recurrent state of X. Then the measure $\gamma^{(i)}$ defined by

$$\gamma_i^{(i)} = E_i\left[\int_0^{\xi_i} \mathbb{1}_{\{X(t)=j\}} dt\right],$$

is a positive invariant measure for X. Furthermore, if X is both irreducible and recurrent, then all other invariant measures are scalar multiples of γ.

Theorem 7.32. *Suppose X is irreducible, and let π be a positive measure on E. Then the following statements are equivalent.*

(a) $\{X(t); t \geq 0\}$ is ergodic, i.e., irreducible, and positive recurrent, with stationary distribution π.

(b) π satisfies the balance equation $\pi Q = 0$, $\sum_{i\in E} \pi_i = 1$.

(c) For a fixed $i \in E$, $E_i[\xi_i] < \infty$ and

$$\pi_j = \frac{E_i\left[\int_0^{\xi_i} \mathbb{1}_{\{X(t)=j\}} dt\right]}{E_i[\xi_i]}, \ \forall\, j \in E.$$

Remark. In practise, given an irreducible CTMC $\{X(t); t \geq 0\}$, to check whether X is positive recurrent or not, we can try to solve balance equation for π.

Remark. (a) For an ergodic CTMC, it has a unique stationary distribution.

(b)

$$\pi_i = \frac{E_i\left[\int_0^{\xi_i} \mathbb{1}_{\{X(t)=i\}} dt\right]}{E_i[\xi_i]} = \frac{\frac{1}{\nu_i} E_i\left[\sum_{n=0}^{\eta_i - 1} \mathbb{1}_{\{X_n = i\}}\right]}{E_i[\xi_i]} = \frac{1}{\nu_i E_i[\xi_i]}.$$

Thus,

$$E_i[\xi_i] = \frac{1}{\pi_i \nu_i}.$$

Alternative: By renewal reward theorem,

$$\pi_i = \frac{1/\nu_i}{E_i[\xi_i]}.$$

Remark. (a) π_j is also the long-run proportion of time that the process is in state j.

(b) In any interval $(0, t)$, the number of transitions into state j must equal to within the number of transitions out of state j. Hence, in the long run, the rate at whcih transitions into state j occur must equal the rate at whihc transitions out of state j occur. When the process is in state j, it leaves at rate ν_j, and, as π_j is the proportion of time it is in state j, it thus follows that

$$\nu_j \pi_j = \text{rate at which the process leaves state } j.$$

Similarly, when the process is in state k, it enters j at a rate q_{kj} . Hence, as π_k is the proportion of time in state k, we see that the rate at which transitions from k to j occur is just $q_{kj}\pi_k$; thus

$$\sum_{k \neq j} q_{kj} \pi_k = \text{rate at which the process enters state } j.$$

Since it balances these rates,

$$\nu_j \pi_j = \sum_{k \neq j} q_{kj} \pi_k$$

is sometimes referred to as a set of "balance equations".

Theorem 7.33. *Suppose X is ergodic, with stationary distribution π. Then for each $i, j \in E$, we have*

$$\lim_{t \to \infty} \frac{1}{t} \int_0^t P_{ij}(s) ds = \pi_j,$$

$$\lim_{t \to \infty} P_{ij}(t) = \pi_j.$$

Proof. By the Key Renewal Theorem. □

Remark. In case you missed it, we need to assume irreducibility to show that the limit does not depend on i, and assuming irreducibility guarantees X has a unique stationary distribution. Without this assumption, a limiting distribution may still exist, but the limit will likely depend on the initial state i.

7.5 Birth and Death Processes

Consider a system whose state at any time is represented by the number of people in the system at that time. Suppose that whenever there are n people in the system, then (i) new arrivals enter the system at an exponential rate λ_n, and (ii) people leave the system at an exponential rate μ_n. Such a system is called a birth and death process. The parameters $\{\lambda_n\}_{n\in\mathbb{N}}$ and $\{\mu_n\}_{n\in\mathbb{Z}^+}$ are called, respectively, the arrival (or birth) and departure (or death) rates.

Definition 7.34. A CTMC is called a birth and death process if it can only change its state by increasing by one or decreasing by one, i.e.,

$$P_{i,i+1} + P_{i,i-1} = 1.$$

$$E = \{0, 1, 2, \cdots\}.$$

For each state $i \in \mathbb{N}$, we have a birth rate λ_i and a death rate μ_i. Then

$$\nu_0 = \lambda_0,$$

$$\nu_i = \lambda_i + \mu_i, \ \forall\ i > 0,$$

$$P_{0,1} = 1,$$

$$P_{i,i+1} = \frac{\lambda_i}{\lambda_i + \mu_i}, \ \forall\ i > 0,$$

$$P_{i,i-1} = \frac{\mu_i}{\lambda_i + \mu_i}, \ \forall\ i > 0,$$

$$q_{01} = \nu_0 P_{01} = \lambda_0,$$

$$q_{i,i+1} = \nu_i P_{i,i+1} = \lambda_i, \ \forall\ i > 0,$$

$$q_{i,i-1} = \nu_i P_{i,i-1} = \mu_i, \ \forall\ i > 0.$$

These type of processes often appear when studying random phenomena governed primarily by exponentially distributed random variables.

Example 7.35 (The Poisson process)**.** Consider a birth and death process for which

$$\mu_n = 0, \ \forall\ n \geq 0,$$

$$\lambda_n = \lambda, \ \forall\ n \geq 0,$$

$$\nu_i = \lambda,$$

$$P_{i,i+1} = 1.$$

This is just the Poisson process, which is also a pure death process.

Example 7.36 (A birth process with linear birth rate)**.** Consider a population whose members can give birth to new members but cannot die. If each member acts independently of the others and takes an exponentially distributed amount of time, with mean $\frac{1}{\lambda}$, to give birth, then if $X(t)$ is the population size at time t, then $\{X(t), t \geq 0\}$ is a pure birth process with $\lambda_n = n\lambda, n \geq 0$. This follows since if the population consists of n persons and each gives birth at an exponential rate λ, then the total rate at which births occur is $n\lambda$. This pure birth process is known as a Yule process after G. Yule, who used in his mathematical theory of evolution.

Example 7.37 (A linear growth with immigration)**.** A model in which

$$\mu_n = n\mu, \ \forall \ n \geq 1,$$

$$\lambda_n = n\lambda + \theta, \ \forall \ n \geq 0,$$

is called a linear growth process with immigration. Each individual in the population is assumed to give birth at an exponential rate λ; in addition, there is an exponential rate of increase θ of the population due to an extenal source such as immigration. Deaths are assumed to occur at an exponential rate μ for each member of the population, so $\mu_n = n\mu$.

Example 7.38 (The Queueing System M/M/1)**.** Suppose that customers arrive at a single-server service station in accordance with a Poisson process having rate λ. Upon arrival, each customer goes directly into service if the server is free; if not, then the customer joins the queue. When the server finishes serving a customer, the customer leaves the system and the next customer in line, if there are any waiting, enters the service. The successive service times are assumed to be independent exponential random variables having with rate μ.

The preceding is known as the $M/M/1$ queueing system. The first M refers to the fact that the interarrival process is Markovian (since it is a Poisson process) and the second to the fact that the service distribution is exponential (and hence Markovian).

If we let $X(t)$ denote the number in the **system** at time t, then $\{X(t), t \geq 0\}$ is a birth and death process with

$$\mu_n = \mu, \ n \geq 1,$$

$$\lambda_n = \lambda, \ n \geq 0.$$

Example 7.39 (M/M/s)**.** Consider an exponential queueing system in which there are s servers available, each serving at rate μ. An entering customer first waits in line and then goes to the first free server. This is a birth and death process with parameters

$$\mu_n = \begin{cases} n\mu, & 1 \leq n \leq s \\ s\mu, & n > s \end{cases}.$$

$$\lambda_n = \lambda, \ n \geq 0.$$

Consider now a general birth and death process with birth rates $\{\lambda_n\}$ and death rates $\{\mu_n\}$, where $\mu_0 = 0$, and let T_i denote the time, starting from state i, it takes for the process to enter state $i+1, i \geq 0$. Clearly,

$$E[T_0] = \frac{1}{\lambda_0}.$$

Let

$$I_i = \begin{cases} 1, & \text{the first transition from } i \text{ is to } i+1 \\ 0, & \text{the first transition from } i \text{ is to } i-1 \end{cases}.$$

Note that

$$E[T_i | I_i = 1] = \frac{1}{\lambda_i + \mu_i},$$

$$E[T_i | I_i = 0] = \frac{1}{\lambda_i + \mu_i} + E[T_{i-1}] + E[T_i].$$

This follows since, independent if whether the first transition is from a birth or death, the time until it occurs is exponential with rate $\lambda_i + \mu_i$. Then

$$E[T_i] = \frac{1}{\lambda_i + \mu_i} + \frac{\mu_i}{\lambda_i + \mu_i} \left(E[T_{i-1}] + E[T_i] \right),$$

or

$$E[T_i] = \frac{1}{\lambda_i} + \frac{\mu_i}{\lambda_i} E[T_{i-1}], \ \forall \ i \geq 1.$$

7.6 The limiting probabilities for the birth and death process

State	Rate at which leave = rate at which enter
0	$\lambda_0 \pi_0 = \mu_1 \pi_1$
1	$(\lambda_1 + \mu_1)\pi_1 = \mu_2 \pi_2 + \lambda_0 \pi_0$
2	$(\lambda_2 + \mu_2)\pi_2 = \mu_3 \pi_3 + \lambda_1 \pi_1$
$n \geq 1$	$(\lambda_n + \mu_n)\pi_n = \mu_{n+1}\pi_{n+1} + \lambda_{n-1}\pi_{n-1}$.

Then

$$\pi_0 = \frac{1}{1 + \sum_{n=1}^{\infty} \frac{\lambda_0 \cdots \lambda_{n-1}}{\mu_1 \cdots \mu_n}},$$

$$\pi_n = \frac{\lambda_0 \cdots \lambda_{n-1}}{\mu_1 \cdots \mu_n \left[1 + \sum_{n=1}^{\infty} \frac{\lambda_0 \cdots \lambda_{n-1}}{\mu_1 \cdots \mu_n} \right]}, \ n \geq 1.$$

It is necessary for these limiting probabilities to exist that

$$\sum_{n=1}^{\infty} \frac{\lambda_0 \cdots \lambda_{n-1}}{\mu_1 \cdots \mu_n} < \infty.$$

In the multiserver exponential queueing system, the condition reduces to

$$\sum_{n+1} \frac{\lambda^n}{(s\mu)^n} < \infty,$$

which is equivalent to $\lambda / s\mu < 1$.

Example 7.40 (A machine repair model)**.** Consider a job shop that consists of M machines and one serviceman. Suppose that the amount of time each machine runs before breaking down is exponentially distributed with rate λ, and suppose that the amount of time that it takes for the serviceman to fix a machine is exponentially distributed with rate μ. Then

(a) What is the average number of machines not in use?

(b) What proportion of time is each machine in use?

If we say that the system is in state n whenever n machine are not in use, then the preceding is a birth and death process having parameters

$$\mu_n = \mu,\ n \geq 1,$$

$$\lambda_n = \begin{cases} (M-n)\lambda, & n \leq M \\ 0, & n > M \end{cases}.$$

This is so in the sense that a failing machine is regarded as an arrival and a fixed machine as a departure. From the equation for the limiting probabilities for the birth and death process, we have

$$\pi_0 = \frac{1}{1 + \sum_{n=1}^{M} (\lambda/\mu)^n M!/(M-n)!},$$

$$\pi_n = \frac{(\lambda/\mu)^n M!/(M-n)!}{1 + \sum_{n=1}^{M} (\lambda/\mu)^n M!/(M-n)},\ n = 0, \ldots, M.$$

Thus, the average number of machines not in use is given by

$$\sum_{n=0}^{M} n\pi_n.$$

To obtain the long-run proportion of time that a given machine is working we will compute the **equivalent** limiting probability of the of the machine working.

$$\begin{aligned} P(\text{machine is working}) &= \sum_{n=0}^{M} P(\text{machine is working} \mid n \text{ machines not working})\pi_n \\ &= \sum_{n=0}^{M} \frac{M-n}{M}\pi_n \\ &= 1 - \frac{1}{M}\sum_{n=0}^{M} n\pi_n. \end{aligned}$$

Example 7.41 ($M/M/1$ queue)**.** In the $M/M/1$ queue, $\lambda_n = \lambda$, $\mu_n = \mu$, and thus

$$\pi_n = \frac{(\lambda/\mu)^n}{1 + \sum_{n=1}^{\infty} (\lambda/\mu)^n} = (\lambda/\mu)^n(1 - \lambda/\mu),\ n \geq 0,$$

provided that $\lambda/\mu < 1$.

Example 7.42 (A shoe shine shop)**.** This is not a birth and death process since it can go directly from state 2 to state 0.

State	Rate at which leave = rate at which enter
0	$\lambda\pi_0 = \mu_2\pi_2$
1	$\mu_1\pi_1 = \lambda\pi_0$
2	$\mu_2\pi_2 = \mu_1\pi_1$

7.7 First step analysis

The idea behind the technique involves conditioning on the random variable X_1 from the embedded DTMC $\{X_n\}_{n\in\mathbb{Z}^+}$.

Example 7.43. Suppose $\{X(t); t \geq 0\}$ is a birth-death process on $E = \{0, 1, 2, \cdots\}$ with birth rate λ and death rate μ. Define

$$\tau_j = \inf\{t > 0; X(t^-) \neq X(t) = j\}.$$

Find

$$E_1[\tau_0] = E\left[\tau(0)|X(0) = 1\right].$$

$$\begin{aligned}
E_1[\tau_0] &= E[\tau_0|X_1 = 0, X_0 = 1]P(X_1 = 0|X_0 = 1) + E[\tau_0|X_1 = 2, X_0 = 1]P(X_1 = 2|X_0 = 1) \\
&= E[\tau_0|X_1 = 0, X_0 = 1]\frac{\mu}{\lambda+\mu} + E[\tau_0|X_1 = 2, X_0 = 1]\frac{\lambda}{\lambda+\mu} \\
&= \frac{1}{\lambda+\mu}\frac{\mu}{\lambda+\mu} + E[\tau_0|X_1 = 2, X_0 = 1]\frac{\lambda}{\lambda+\mu}.
\end{aligned}$$

Let T_1 be the first transition time, then

$$\begin{aligned}
E[\tau_0|X_1 = 2, X_0 = 1] &= E[T_1 + (\tau_0 - T_1)|X_1 = 2, X_0 = 1] \\
&= E[T_1|X_1 = 2, X_0 = 1] + E[\tau_0 - T_1|X_1 = 2, X_0 = 1] \\
&= \frac{1}{\lambda+\mu} + E[\tau_0|X_0 = 2] \\
&= \frac{1}{\lambda+\mu} + E[\tau_1 + (\tau_0 - \tau_1)|X_0 = 2] \\
&= \frac{1}{\lambda+\mu} + E[\tau_1|X_0 = 2] + E[\tau_0 - \tau_1|X_0 = 2] \\
&= \frac{1}{\lambda+\mu} + E[\tau_0|X_0 = 1] + E[\tau_0|X_0 = 1] \\
&= \frac{1}{\lambda+\mu} + 2E[\tau_0|X_0 = 1].
\end{aligned}$$

Alternatively: it is clearly that we have the conclusion. Thus,

$$E[\tau_0|X_0 = 1] = \frac{1}{\mu - \lambda},$$

where

$$\lambda < \mu.$$

Example 7.44. Suppose we have n iid exponential(λ) random variables $X_1, \ldots, X_n$, for convinience, we can think of them as being the lifetime of n batteries. Let $W(t)$ represents the number of batteries working at time t. Assume $W(0) = n$.

(a) Is $\{W(t)\}$ a CTMC? Yes.

$$q_{i,i-1} = i\lambda,\ i \in [n].$$
$$q_{00} = 0,\ q_{0i} = 0,\ \forall\ i \in [n],$$

where 0 is absorbing state and mean time is ∞.

(b) Use the fact $\{W(t); t \geq 0\}$ is a CTMC, show that

$$\max_{1 \leq i \leq n} X_i \stackrel{d}{=} \sum_{k=1}^{n} \frac{X_k}{k}.$$

The LHS is the nth transition time. For the RHS, we note that

$$\frac{X_k}{k} \sim \exp(k\lambda),$$

which has the same distribution as the $(n-k+1)^{\text{th}}$ inter-transition time.

Theorem 7.45. *Consider an ergodic CTMC $\{X(t); t \geq 0\}$ with stationary distribution π. Assume that the embedded DTMC $\{X_n\}_{n \geq 1}$ also admits a unique stationary distribution $\tilde{\pi}$. Then*

$$\pi_j = \frac{\tilde{\pi}_j/\nu_j}{\sum_{k \in E} \tilde{\pi}_k/\nu_k}.$$

7.8 Constructing CTMCs

Is it possible to construct a CTMC $\{X(t); t \geq 0\}$ on a countable state space E for any given initial distribution $\{\alpha_i\}_{i \in E}$, the transition matrix $[P_{ij}]_{i,j \in E}$ and the sojourn rates $\{q_i\}_{i \in E}$ as the defining parameters? Granted, assuming the sojourn rates are "nice enough" so that X is P-regular, the answer to this question os indeed yes.

7.8.1 Constructing discrete random variables

WLOG, assume E represents the set of nonnegative integers.

Suppoe that $\{\alpha_i\}_{i \in E}$ is a probability mass function on E. Our goal is to construct a random variable with pmf α. Define, $\forall\ j \in E$,

$$I_j = \left[\sum_{k=0}^{j-1} \alpha_k, \sum_{k=0}^{j} \alpha_k\right).$$

Let $U \sim \text{Uni}(0,1)$ and

$$\begin{aligned} h : (0,1) &\to E \\ x &\mapsto \textstyle\sum_{j=0}^{\infty} j \mathbb{1}_{\{x \in I_j\}}. \end{aligned}$$

Now $h(U)$ is a random variable, taking values in E, and its pmf is of the form

$$P\left(h(U) = j\right) = P(U \in I_j) = \sum_{k=0}^{j} \alpha_k - \sum_{k=0}^{j-1} \alpha_k = \alpha_j.$$

Therefore, $h(U)$ is a random variable with pmf α.

7.8.2 Constructing DTMCs

The same idea can easily be extended to construct a DTMC $\{X_n\}_{n\in\mathbb{N}}$, having initial distribution α and transition matrix P. Let $f : E \times (0,1) \to E$, where

$$f(i,u) = \sum_{j=0}^{\infty} j \mathbb{1}_{\{u \in I_{i,j}\}},$$

where

$$I_{i,j} = \left[\sum_{k=0}^{j-1} P_{ik}, \sum_{k=0}^{j} P_{ik}\right).$$

Let $\{U_n\}_{n\in\mathbb{N}}$ be a sequence of iid Uni$(0,1)$.
Set

$$X_0 = h(U_0),\ X_1 = f(X_0, U_1),\ X_2 = f(X_1, U_2), \ldots, X_n = f(X_{n-1}, U_n),\ \forall\ n \in \mathbb{Z}^+.$$

It is easy to show that $\{X_n\}_{n\in\mathbb{N}}$ is a DTMC. Here, X_0 has distribution α, and $\forall\ i, j \in E$,

$$P\left(X_{n+1} = j | X_n = i\right) = P\left(f(i, U_{n+1}) = j | X_n = i\right) = P\left(U_{n+1} \in I_{i,j}\right) = P_{ij}.$$

7.8.3 Constructing CTMCs

To do this, we will need two sequence of iid random variables that are independent of each other, $\{U_n\}_{n\in\mathbb{N}}$ and $\{V_n\}_{n\in\mathbb{N}}$, where $U_0 \sim$ Uni$(0,1)$ and $V_0 \sim \exp(1)$.
Set for each $n \geq 1$,

$$\xi_n = \frac{V_n}{q_{X_{n-1}}}.$$

Define $\{X(t); t \geq 0\}$ to be the jump process corresponding to the embedded process $\{(X_n, \xi_n)\}_{n\in\mathbb{N}}$.
Now we will verify that $\{X(t); t \geq 0\}$ is indeed a CTMC.
For each $m \in \mathbb{Z}^+$, each m-tuple $(t_1, \ldots, t_m) \in \mathbb{R}^+$, and each $i_1, i_2, \cdots \in E$, we see that

$$\begin{aligned}
P\left(\xi_1 \leq t_1, \ldots, \xi_m \leq t_m | X_k = i_k, k \in \mathbb{N}\right) &= P\left(\frac{V_1}{q_{X_0}} \leq t_1, \ldots, \frac{V_m}{q_{X_m}} \leq t_m \middle| X_k = i_k, 0 \leq k \leq m\right) \\
&= P\left(\frac{V_1}{q_{i_0}} \leq t_1, \ldots, \frac{V_m}{q_{i_k}} \leq t_m \middle| X_k = i_k, 0 \leq k \leq m\right) \\
&= \prod_{k=1}^{m} \left(1 - e^{-q_{i_{k-1}} t_{k-1}}\right).
\end{aligned}$$

7.9 Examples

Example 7.46. Suppose that a one-celled organism can be in one of two states-either A or B. An individual in state A will change to state B at an exponential rate α; an individual in state B divides into new individuals of type A at an exponential rate β. Define an appropriate continuous MC for a population of such organisms and determine the appropriate parameters for this model.

Let $N_A(t)$ be the number of organisms in state A and let $N_B(t)$ be the number of organisms in state B. Then $\{N_A(t); N_B(t)\}$ is a CTMC with

$$\nu_{n,m} = \alpha n + \beta m,$$

$$P_{(n,m),(n-1,m+1)} = \frac{\alpha n}{\alpha n + \beta m},$$

$$P_{(n,m),(n+2,m-1)} = \frac{\beta m}{\alpha n + \beta m}.$$

Example 7.47. There are N individuals in a population, some of whom have a certain infection that spreads as follows. Contacts between two members of this population occur in accordance with a Poisson process having rate λ. When a contact occurs, it is equally likely to involve any of the $\binom{N}{2}$ pairs of iiduals in the population. If a contact involves an infected and a noninfected individual, then with probability p, the noninfected individual becomes infected. Once infected, an individual remains infected throughout. Let $X(t)$ denote the number of infected members of the population at time t. Then $\{X(t), t \geq 0\}$ is a CTMC of pure birth process. Starting with a single infected individual, what is the expected time until all members are infected?
If there are i infected individuals then since a contact will involve an infected and an uninfected with probability $i(n-i)/\binom{n}{2}$, it follows that the birth rates are

$$\lambda_i = \lambda i(n-i)/\binom{n}{2}, \ \forall\ i \in [n].$$

Hence,

$$E[\text{time all infected}] = \frac{n(n-1)}{2\lambda p} \sum_{i=1}^{n} \frac{1}{i(n-i)}.$$

Example 7.48. The birth and death process with parameters $\lambda_n = 0$ and $\mu_n = \mu, n \geq 0$ is called a pure death process. Find $P_{ij}(t)$.
Since the death rate is constant, it follows that as long as the system is nonempty, the number of deaths in any interval of length t will be a Poisson random variable with mean μt. Hence,

$$P_{ij}(t) = e^{-\mu t}(\mu t)^{i-j}/(i-j)!, \ 0 < j \leq i,$$

$$P_{i0}(t) = \sum_{k=i}^{\infty} e^{-\mu t}(\mu t)^k/k?$$

Example 7.49. A small barbershop, operated by a single barber, has room for at most two customers. Potential customers arrive at a Poisson rate of 3 per hour, and the successive service times are independent exponential random variables with mean 1/4 hour.

(a) What is the average number of customers in the shop?

(b) What is the proportion of potential customers that enter the shop?

(c) If the barber could work twice as fast, how much more business would he do?

With the number of customers in the shop as the state, we get a birth and death process with

$$\lambda_0 = \lambda_1 = 3, \ \mu_1 = \mu_2 = 4.$$

Then

$$\pi_0 = \frac{16}{37}.$$

(a) The average number of customers in the shop is

$$\pi_1 + 2\pi_2 = \frac{30}{37}.$$

(b) The proportion of customers that enter the shop is (!!!)

$$\frac{\lambda(1-\pi_2)}{\lambda} = 1 - \pi_2 = \frac{28}{37}.$$

(c) Now $\mu = 8$ and so

$$\pi_0 = \frac{64}{97}.$$

So the proportion of customers who now enter the shop is

$$1 - \pi_2 = \frac{88}{97}.$$

The rate of added customers is therefore

$$\lambda\frac{88}{97} - \lambda\frac{28}{37} = 0.45.$$

The business he does would improved by 0.45 customers per hour.

Example 7.50. A single repairperson looks after both machines 1 and 2. Each time it is repaired, machine i stays up for an exponential time with rate $\lambda_i, i = 1, 2$. When machine i fails, it requires an exponentially distributed amount of work with rate μ_i to complete its repair. The repairperson will always service machine 1 when it is down. For instace, if machine 1 fails while 2 is being repaird, then the repairperson will immediately stop work on machine 2 and start on 1. What proportion of time is machine 2 down?

Let state 0 mean that no machines are down, 1 that machine 1 is down and 2 is up, 2 that machine 1 is up and 2 is down, 3 that both machines are down. Then

$$(\lambda_1 + \lambda_2)\pi_0 = \mu_1\pi_1 + \mu_2\pi_2,$$

$$(\lambda_2 + \mu_1)\pi_1 = \pi_0,$$

$$(\lambda + \mu_2)\pi_2 = \lambda_2\pi_0 + \mu_1\pi_3,$$

$$\mu_1\pi_3 = \lambda_2\pi_1 + \lambda_1\pi_2.$$

The proportion of time machine 2 is down is $\pi_2 + \pi_3$.

Example 7.51. Customers arrive at a single-server queue in accordance with a Poisson process having rate λ. However, an arrival that finds n customers already in the system will only join the system with probability $1/(n+1)$. Show that the limiting distribution of the number of customers in the system is Poisson with mean λ/μ.

The number in the system is a birth and death process with parameters

$$\lambda_n = \lambda/(n+1),\ n \geq 0,$$

$$\mu_n = \mu,\ n \geq 1.$$

By the formula,

$$\pi_0 = e^{-\lambda/\mu},$$

$$\pi_n = \pi_0(\lambda/\mu)^n/n! = e^{-\lambda/\mu}(\lambda/\mu)^n/n!,\ \forall\ n \geq 0.$$

Example 7.52. Four workers share an office that contains four telephone. At any time, each workers is either "working" or "on the phone". Each "working" period of worker i lasts for an exponentially distributed time with rate λ_i, and each "on the phone" period lasts for an exponentially distributed time with rate $\mu_i, i = 1, 2, 3, 4$.

(a) What proportion of time are all workers "working"?

Let $X_i(t) = 1$ if worker i is working at time t, and $X_i(t) = 0$. The process $\{X_i(t)\}$ is a two state CTMC and

$$\lim_{t\to\infty} P\left(X_i(t) = 1\right) = \frac{1/\lambda_i}{1/\mu_i + 1/\lambda_i},\ i = 1, \ldots, 4.$$

by independence, proportion of time all working is

$$\prod_{i=1}^{4} \frac{\mu_i}{\mu_i + \lambda_i}.$$

(b) Let $\vec{X}(t) = (X_1(t), X_2(t), X_3(t), X_4(t))$. Argue that $\{\vec{X}(t), t \geq 0\}$ is a continuous-time Markov chain.

It is a CTMC since the processes $\{X_i(t)\}$ are independent with each being a CTMC.

(c) Suppose now that one of the phones has broken down. Suppose that a worker who is about to use a phone but finds them all being used begins a new "working" period. What proportion of time are all workers "working"?

$$P(\text{all working/truncated}) = \frac{P(\text{all working})}{1 - P(0,0,0,0)} = \frac{\prod_{i=1}^{4} \frac{\mu_i}{\mu_i+\lambda_i}}{1 - \prod_{i=1}^{4} \frac{\lambda_i}{\lambda_i+\mu_i}}.$$

7.10 Challenging problems

Example 7.53. Consider a tandem queue with two servers labeled A and B. A customer must first be served by A, and then by B. Customers arrive according to a Poisson process with rate λ, and the service time at servers A and B are exponential with rates μ_A and μ_B, respectively. We assume the service times are independent of each other as well as the arrival process.

Upon completion of service at server A, a customer enters server B and starts receiving service if B is free. Otherwise, if B is busy, the customer waits until B becomes available: however, there is no waiting room ar server B, and so the customer will stay at server A, preventing A from serving any new customers. In such a situation, we say that A is blocked. If A is blocked, and B becomes available, the customer at A starts receiving service from B immediately, and frees up A. Upon completion of service at B, a customer leaves the system.

(a) Assume first that an arrival finding server A busy/blocked leaves immediately.

(1) Construct a CTMC to describe the state of the system, and find its stationary distribution.

(2) What is the long run proportion of time server A is blocked.

(b) Now assume that an arrival finding server A busy/blocked waits in line for the server.

(1) Construct a CTMC to describe the state of the system, and derive its infinitesimal generator. You may use a rate diagram to display the infinitesimal generator.

(2) Suppose that a customer arrives and finds that there is one customer in the system and that customer is being served by server A. What is the expected total amount of time the new arrival spends in the system?

(a) (1) Define at time t,

$$X(t) = \begin{cases} 0, & \text{server } A \text{ is empty} \\ 1, & \text{server } A \text{ is busy} \\ 2, & \text{server } A \text{ is blocked} \end{cases},$$

and

$$Y(t) = \begin{cases} 0, & \text{server } B \text{ is empty} \\ 1, & \text{server } B \text{ is busy} \end{cases}.$$

Then $\{(X(t), Y(t)); t \geq 0\}$ is a CTMC with state space

$$E = \{(0,0), (0,1), (1,0), (1,1), (2,1)\}.$$

The infinitesimal generator is

$$Q = \begin{pmatrix} -\lambda & 0 & \lambda & 0 & 0 \\ \mu_B & -(\mu_B + \lambda) & 0 & \lambda & 0 \\ 0 & \mu_A & -\mu_A & 0 & 0 \\ 0 & 0 & \mu_B & -(\mu_A + \mu_B) & \mu_A \\ 0 & \mu_B & 0 & 0 & -\mu_B \end{pmatrix}.$$

The stationary distribution is

$$\pi_{(0,0)} = \frac{\mu_A \mu_B^2 (\mu_A + \mu_B)}{\mu_A \mu_B^2 (\mu_A + \mu_B) + \lambda \mu_A \mu_B (\mu_A + \mu_B) + \lambda \mu_B (\mu_B + \lambda)(\mu_A + \mu_B) + \lambda^2 \mu_A^2}.$$

$$\pi_{(0,1)} = \frac{\lambda \mu_A \mu_B (\mu_A + \mu_B)}{\mu_A \mu_B^2 (\mu_A + \mu_B) + \lambda \mu_A \mu_B (\mu_A + \mu_B) + \lambda \mu_B (\mu_B + \lambda)(\mu_A + \mu_B) + \lambda^2 \mu_A^2}.$$

$$\pi_{(1,0)} = \frac{\lambda\mu_A\mu_B^2 + \lambda^2\mu_B^2 + \lambda\mu_B^3}{\mu_A\mu_B^2(\mu_A+\mu_B) + \lambda\mu_A\mu_B(\mu_A+\mu_B) + \lambda\mu_B(\mu_B+\lambda)(\mu_A+\mu_B) + \lambda^2\mu_A^2}.$$

$$\pi_{(1,1)} = \frac{\lambda^2\mu_A\mu_B}{\mu_A\mu_B^2(\mu_A+\mu_B) + \lambda\mu_A\mu_B(\mu_A+\mu_B) + \lambda\mu_B(\mu_B+\lambda)(\mu_A+\mu_B) + \lambda^2\mu_A^2}.$$

$$\pi_{(2,1)} = \frac{\lambda^2\mu_A^2}{\mu_A\mu_B^2(\mu_A+\mu_B) + \lambda\mu_A\mu_B(\mu_A+\mu_B) + \lambda\mu_B(\mu_B+\lambda)(\mu_A+\mu_B) + \lambda^2\mu_A^2}.$$

(2) The long run proportion of time server A is blocked is π_{21}.

(b) (1) Let $W(t)$ denote the number of waiting people in the system.
Then $\{(X(t), Y(t), W(t))\}$ is a CTMC with state space

$$\{(0,0,0), (0,1,0), (1,0,i), (1,1,j), (2,1,k)\},$$

where

$$i, j, k \in \mathbb{N}.$$

Then for $i = j = k = 0$, we have

$$q_{(a,b,0)(x,y,0)} = Q_{(a,b)(x,y)}.$$

Furthermore,

$$q_{(1,0,i)(1,0,i+1)} = \lambda, \ \forall\ i \in \mathbb{N}.$$

$$q_{(1,0,i)(1,0,i-1)} = \mu_A, \ \forall\ i \in \mathbb{Z}^+.$$

$$q_{(1,1,i)(1,1,i+1)} = \lambda, \ \forall\ i \in \mathbb{N}.$$

$$q_{(2,1,i)(2,1,i+1)} = \lambda, \ \forall\ i \in \mathbb{N}.$$

$$q_{(2,1,i)(1,1,i-1)} = \mu_B, \ \forall\ i \in \mathbb{Z}^+.$$

Besides, !!!

$$q_{(1,1,i),(2,1,i)} = \mu_A.$$

(2)

$$E[T] = \frac{1}{\mu_A} + \frac{1}{\mu_A+\mu_B} + \frac{\mu_A}{\mu_A+\mu_B}\frac{2}{\mu_B} + \frac{\mu_B}{\mu_A+\mu_B}\left(\frac{1}{\mu_A} + \frac{1}{\mu_B}\right).$$

Example 7.54. Suppose organisms arrive to a region in accordance to a Poisson process with rate λ, and each organism stays in the region for an exponentially distributed amount of time having rate μ, independently of everything else. However, assume that every third potential arrival to the region is 'eaten' by a larger organism with probability one.

(a) Model this probabilistic phenomena with an appropriately-chosen CTMC, and either describe its transition matrix or draw a rate diagram that describes how transitions are made. Hint: You need to somehow keep track of the fact that each third arrival gets eaten.

(b) Let $Q(t)$ represent the number of organisms present in the system at time t, and let $N(t)$ represent the total number of potential arrivals (counting those that enter the region, and those eaten) that occur in $(0, t]$. Compute

$$P\left(Q(t) = k | N(t) = 10\right)$$

for all nonnegative integers for which this probability is nonzero.

(a) Note $X(t) = \{(Q(t), N(t))\}$ is a CTMC with state space

$$E = \{(i, j)\},$$

where $i, j \in \mathbb{N}$ and $i \leq j - \lfloor j/3 \rfloor$. The infinitesimal generator is

$$q_{(i,j)(i-1,j)} = \mu, \ \forall \ i \in \mathbb{Z}^+, i \leq j - \lfloor j/3 \rfloor,$$

$$q_{(i,j)(i,j+1)} = \lambda, \ \forall \ i \in \mathbb{Z}^+, i \leq j - \lfloor j/3 \rfloor.$$

(b) Let $D(t)$ denote the number of organisms departure at time t, let Q_1 denote the number of organisms arrived at t including the ones got eaten. Then

$$D(t) \sim \text{Poi}\left(\lambda \int_0^t \left(1 - e^{-u(t-s)}\right) ds\right).$$

$$Q_1(t) \sim \text{Poi}\left(\lambda \int_0^t e^{-u(t-s)} ds\right).$$

Also,

$$N(t) \sim \text{Poi}(\lambda t).$$

For $0 \leq k \leq 7$,

$$\begin{aligned} P\left(Q(t) = k | N(t) = 10\right) &= O\left(Q_1(t) = k + 3 | N(t) = 10\right) \\ &= \frac{P\left(D(t) = 7 - k\right) P\left(Q_1(t) = k + 3\right)}{P\left(N(t) = 10\right)}. \end{aligned}$$

Example 7.55. Suppose customers arrive in batches to a service station in accordance to a Poisson process $\{N(t); t \geq 0\}$ having rate λ, where the size of each batch is independent of everything else. Assume the distribution of a typical batch size $B \sim \text{Geo}(p)$.
Assume further that each person brings with it an exponentially amount of work having rate μ, independent of everything else, to receive processing from a single server.

(a) Suppose $Q(t)$ denotes the number of customers in the system at time $t, t \geq 0$. Briefly explain why $\{Q(t); t \geq 0\}$ is a continuous-time Markov chain (CTMC) and either describe the elements of its rate matrix, or draw its transition diagram.

(b) Compute

$$\lim_{t \to \infty} \frac{1}{t} \int_0^t \mathbb{1}_{\{Q(s)=0\}} ds,$$

where the limit should be interpreted as existing with probability one. Hint: You may want to make use of the expected amount of time it takes the chain to return to state 0, given it started in state 0.

(a) This is a CTMC since

(1) it is a jump process on $\{0, 1, \cdots\}$;

(2) holding times are iid $\exp(\lambda)$;

(3) A DTMC can be defined on transition times.

The generators can be defines as follows

$$q_{i,i-1} = \mu, \ \forall \ i \in \mathbb{Z}^+;$$

$$q_{i,j} = \lambda(1-p)^{j-i}, \ \forall \ j > i, i, j \in \mathbb{N},$$

since the arriving rate is $\frac{\lambda}{p}$.

(b) Assume state 0 is positive recurrent, by renewal reward theorem and the definition of π_0,

$$\lim_{t\to\infty} \frac{1}{t}\int_0^t \mathbb{1}_{\{Q(s)=0\}} ds = \frac{E\left[\int_0^{\xi_0} \mathbb{1}_{\{Q(s)=0\}} ds\right]}{E_0[\xi_0]} = \pi_0 = \frac{1}{\nu_0 E_0[\xi_0]} = \frac{p}{\lambda E_0[\xi_0]}.$$

$$\begin{aligned}
E_0[\xi_0] &= \sum_{k=1}^{\infty} E_0[\tau_0|B=k]P_0(B=k) \\
&= \sum_{k=1}^{\infty} \left(\frac{1}{\lambda} + E_k[\tau_0]\right) P(B=k) \\
&= \sum_{k=1}^{\infty} \left(\frac{1}{\lambda} + kE_1[\tau_0]\right) P(B=k) \\
&= \frac{1}{\lambda} + \frac{E_1[\tau_0]}{p}.
\end{aligned}$$

Next, letting X_1 be the state of the first transition starting from 1,

$$\begin{aligned}
E_1[\tau_0] &= \frac{u}{\lambda+\mu} E_1[\tau_0|X_1=0] + \sum_{k=1}^{\infty} E_1[\tau_0|X_1=k+1]P(B=k)\frac{\lambda}{\lambda+\mu} \\
&= \frac{u}{(\lambda+\mu)^2} + \sum_{k=1}^{\infty} \left(\frac{1}{\lambda+\mu} + E_{k+1}[\tau_0]\right) P(B=k)\frac{\lambda}{\lambda+\mu} \\
&= \frac{u}{(\lambda+\mu)^2} + \sum_{k=1}^{\infty} \left(\frac{1}{\lambda+\mu} + (k+1)E_1[\tau_0]\right) P(B=k)\frac{\lambda}{\lambda+\mu} \\
&= \frac{1}{\lambda+\mu} + \frac{\lambda p+\lambda}{\lambda p+\mu p} E_1[\tau_0].
\end{aligned}$$

Example 7.56. Consider an environment consisting of five organisms. Each organism, independently of all other organisms, alternates indefinitely between being a type-1 organism and a type-2 organism. The amount of time an organism is of type-1 is exponentially distributed with rate λ_1,

at the end of which time it switches to a type-2 organism. It then stays a type-2 organism for an exponential amount of time with rate λ_2, at the end of which it becomes a type-1 particle, and the process repeats itself indefinitely.

Let $X_1(t)$ represent the type of particle i at time $t, 1 \leq i \leq 5$. Then $\{X_i(t); t \geq 0\}$ is a CTMC and

$$P_{21}(t) = P\left(X_i(t) = 1 | X_i(0) = 2\right) = \frac{\lambda_2}{\lambda_1 + \lambda_2} - \frac{\lambda_2}{\lambda_1 + \lambda_2} e^{-(\lambda_1+\lambda_2)t}.$$

Let Y_1 represents the total number of type-1 organisms present in the environment at time t. Then $Y_1(t); t \geq 0$ is a CTMC and

$$Y_1(t) \sim \text{Binom}\left(5, P_{21}(t)\right).$$

Example 7.57. A particle moves among positions $0, \ldots, J$ according to a Markov chain whose one-step transition matrix is doubly stochastic. Each time the particle visits state i it spends an amount of time there having distribution function F_i and expected value m_i indepdendent of previous states visited and times spent in those states. Let $X(t)$ be the position of the particle at time t.

(a) Develop a renewal equation for $P\left(X(t) = i\right)$.

(b) Find

$$\lim_{t\to\infty} P\left(X(t) = i\right).$$

(a) Let $0 \leq l \leq n$ be the initial state. Let T_1 be the time first return to l. !Be careful, T_1 is not the transition time and the chain may return to i before T_1.

$$\begin{aligned} P\left(X(t) = i\right) &= P\left(X(t) = i, T_1 \leq t\right) + P\left(X(t) = i, T_1 \geq t\right) \\ &= \int_0^t P\left(X(t) = i | T_1 = s\right) dF_l(s) + P\left(X(t) = i, T_1 \geq t\right) \\ &= \int_0^t P\left(X(t-s) = i\right) dF_l(s) + P\left(X(t) = i, T_1 \geq t\right). \end{aligned}$$

(b) The embedded DTMC is doubly stochastic,

$$\pi_j = \frac{1}{j+1}, \ \forall\, j.$$

It is easy to find it is irreducible, then positive recurrent since finite states. Furthermore,

$$\pi_j = \frac{E_i\left[\sum_{n=0}^{\eta_l - 1} \mathbb{1}_{\{X_n = j\}}\right]}{E_i[\eta_l]} = \frac{1}{J+1}, \ \forall\, j.$$

$$\pi_j^* = \frac{E_l\left[\int_0^{\xi_l} \mathbb{1}_{\{X(t)=j\}} dt\right]}{E_l[\xi_l]}, \ \forall\, j.$$

In particular,

$$E_l\left[\int_0^{\xi_l} \mathbb{1}_{\{X(t)=j\}} dt\right] = \pi_i m_i E[\text{number of steps in a cycle}].$$

Thus,

$$\pi_i^* = \frac{m_i}{\sum_{j=0}^{J} m_i.}$$

Chapter 8

Brownian Motion

8.1 Introdiction

Definition 8.1. We say that a stochastic process $\{B(t); t \geq 0\}$ is a *standard Brownian Motion* if it satisfies the following condition:

(a) $B(0) = 0$ w.p.1;

(b) $\{B(t); t \geq 0\}$ has stationary increments; i.e., for each $s, t \geq 0$, the distribution of $B(t+s) - B(t)$ depends only on s;

(c) $\{B(t); t \geq 0\}$ has independent increment; i.e., for $t_1 \leq \cdots \leq t_n \in \mathbb{R}^+$, the random variables $B(t_1), B(t_2) - B(t_1), \ldots, B(t_n) - B(t_{n-1})$ are independent of each other;

(d) $B(t) \sim N(0, t),\ \forall\ t > 0$.

Remark. (a) The state space of $\{B(t)\}$ is $\mathbb{R}$.

(b) A standard BM is a continuous time, continuous state space time-homogeneous Markov process since for any $s, t \geq 0$ and $x, y \in \mathbb{R}$,

$$\begin{aligned}
P\left(B(s+t) \leq y | B(t) = x, B(u), 0 \leq u < t\right) &= P\left(B(s+t) - B(t) \leq y - B(t) | B(t) = x, B(u), 0 \leq u < t\right) \\
&= P\left(B(s+t) - B(t) \leq y - x | B(t) = x, B(u), 0 \leq u < t\right) \\
&= P\left(B(s+t) - B(t) \leq y - x\right) \\
&= P\left(B(s+t) - B(t) \leq y - x | B(t) = x\right) \\
&= P\left(B(s+t) \leq y | B(t) = x\right).
\end{aligned}$$

(c) A homogeneous Poisson process also satisfies condition (i),(ii),(iii). Stochastic process satisfying conditions (i),(ii),(iii) are referred to Levy processes.

Example 8.2. Suppose $\{X_n\}_{n \in \mathbb{Z}^+}$ is a sequence of iid random variables, where $P(X_1 = -1) = P(X_1 = 1) = 1/2$. Define

$$X_n(t) = \frac{1}{\sqrt{n}} \sum_{k=1}^{\lfloor nt \rfloor} X_k.$$

For each fix t, by the CLT,

$$X_n(t) \xrightarrow{d} \sqrt{t}Z, \text{ as } n \to \infty.$$

where

$$Z \sim N(0,1).$$

One can show that

$$\{Y_n(t); t \geq 0\} \xrightarrow{d} \{B(t); t \geq 0\} \text{ as } n \to \infty.$$

8.2 Calculations associated with Brownian Motion

Theorem 8.3. *Let $0 \leq t_1 < t_2$,*

$$\begin{aligned} P\left(B(t_1) \leq x, B(t_2) \leq y\right) &= P\left(B(t_1) \leq x, B(t_2) - B(t_1) + B(t_1) \leq y\right) \\ &= \int_{-\infty}^{x} P\left(B(t_2) - B(t_1) \leq y - z\right) f_{B(t_1)}(z)dz \\ &= \int_{-\infty}^{x} \int_{-\infty}^{y-z} f_{B(t_2-t_1)}(v)dv f_{B(t_1)}(z)dz \\ &= \int_{-\infty}^{x} \int_{-\infty}^{y} f_{B(t_2-t_1)}(v-z) f_{B(t_1)}(z)dvdz. \end{aligned}$$

So the joint density function of $(B(t_1), B(t_2))$ with $0 \leq t_1 < t_2$ is

$$f_{B(t_1),B(t_2)}(z_1, z_2) = f_{B(t_1)}(z_1) f_{B(t_2-t_1)}(z_2 - z_1), \ z_1, z_2 \in \mathbb{R}.$$

More generally, for a sequence $0 \leq t_1 < \cdots < t_n$,

$$f_{B(t_1),\ldots,B(t_n)}(z_1, \ldots, z_n) = f_{B(t_1)}(z_1) f_{B(t_2-t_1)}(z_2 - z_1) \cdots f_{B(t_n-t_{n-1})}(z_n - z_{n-1}), \ z_1, \ldots, z_n \in \mathbb{R}.$$

Theorem 8.4. *(a) Let $t > 1$, then*

$$\begin{aligned} B(t)|B(1) = 0 &= B(t) - B(1) + B(1)|B(1) = 0 \\ &= B(t) - B(1)|B(1) = 0 \\ &\stackrel{d}{=} B(t) - B(1) \\ &\stackrel{d}{=} B(t-1) \sim N(0, t-1). \end{aligned}$$

(b) Let $0 < t < 1$, then

$$\begin{aligned} f_{B(t)|B(1)=0}(z) &= \frac{f_{B(t),B(1)}(z, 0)}{f_{B(1)}(0)} \\ &= \frac{f_{B(t)}(z) f_{B(1-t)}(0 - z)}{f_{B(1)}(0)} \\ &= \frac{\frac{1}{\sqrt{2\pi t}} e^{-\frac{z^2}{2t}} \frac{1}{\sqrt{2\pi(1-t)}} e^{-\frac{z^2}{2(1-t)}}}{\frac{1}{\sqrt{2\pi}}} \\ &= \frac{1}{\sqrt{2\pi t(1-t)}} e^{-\frac{z^2}{2t(1-t)}}. \end{aligned}$$

So

$$B(t)|B(1) \sim N\left(0, t(1-t)\right).$$

8.2.1 Hitting times of Brownian Motion

Definition 8.5. Let

$$\tau_a = \inf\{t > 0 : B(t) = a\}$$

be the time that B reaches level a.

Lemma 8.6.

$$P\left(B(t) > a | \tau_a \leq t\right) = 1/2.$$

Proof.

$$\begin{aligned}
P\left(B(t) > a | \tau_a \leq t\right) &= P\left(B(t) - B(\tau_a) + B(\tau_a) > a | \tau_a \leq t\right) \\
&= P\left(B(t) - B(\tau_a) > 0 | \tau_a \leq t\right) \\
&= P\left(B(t - \tau_a) > 0 | \tau_a \leq t\right) \\
&= P\left(B(t - \tau_a) > 0\right) \\
&= 1/2.
\end{aligned}$$

□

Theorem 8.7. *Assume $a > 0$, Then*

$$\begin{aligned}
P\left(B(t) > a\right) &= P\left(B(t) > a, \tau_a \leq t\right) + P\left(B(t) > a, \tau_a > t\right) \\
&= P\left(B(t) > a, \tau_a \leq t\right) \\
&= P\left(B(t) > a | \tau_a \leq t\right) P\left(\tau_a \leq t\right) \\
&= 1/2 P\left(\tau_a \leq t\right).
\end{aligned}$$

So

$$P(\tau_a \leq t) = 2P\left(B(t) \geq a\right) = 2\int_a^\infty \frac{1}{\sqrt{2\pi t}} e^{-\frac{x^2}{2t}} dx = 2\int_{\frac{a}{\sqrt{t}}}^\infty \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} dy.$$

It is easy to show

$$P(\tau_a < \infty) = \lim_{t\to\infty} P\left(\tau_a \leq t\right) = 1.$$

$$\begin{aligned}
E[\tau_a] &= \int_0^\infty P(\tau_a > t) dt = \int_0^\infty \left[1 - P(\tau_a \leq t)\right] dt \\
&= \int_0^\infty \left[1 - 2\int_{\frac{a}{\sqrt{t}}}^\infty \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} dy\right] dt = 2\int_0^\infty \left[\frac{1}{2} - \int_{\frac{a}{\sqrt{t}}}^\infty \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} dy\right] dt \\
&= 2\int_0^\infty \int_0^{\frac{a}{\sqrt{t}}} \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} dy dt = 2\int_0^\infty \int_0^{\frac{a^2}{y^2}} \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} dt dy \\
&= 2\int_0^\infty \frac{1}{\sqrt{\sqrt{2\pi}}} \frac{a^2}{y^2} e^{-\frac{y^2}{2}} dy \geq 2\int_0^1 \frac{1}{\sqrt{2\pi}} \frac{a^2}{y^2} e^{-\frac{y^2}{2}} dy \\
&\geq \frac{2a^2}{\sqrt{2\pi}} e^{-\frac{1}{2}} \int_0^1 \frac{1}{y^2} dy = \infty.
\end{aligned}$$

Remark. (a) The standard BM is null recurrent.

(b)

$$\tau_a \stackrel{d}{=} \tau_{-a}.$$

(c)

$$f_{\tau_a}(t) = \frac{a}{\sqrt{2\pi}} t^{-3/2} e^{-a^2/2t}.$$

Consider for $x \neq 0$,

$$X(t) = B(t) + x.$$

Then

$$\begin{aligned}
\tau_y^* &= \inf\{t > 0; X(t) = y\} \\
&= \inf\{t > 0 : B(t) + x = y\} \\
&= \inf\{t > 0 : B(t) = y - x\} \\
&= \tau_{y-x}.
\end{aligned}$$

Definition 8.8. The *running maximum* of B is defined as

$$M(t) = \max_{0 \le s \le t} B(s).$$

In fact, we have

$$P\left(M(t) < x\right) = P(\tau_x > t).$$

8.2.2 Zeros of a Brownian motion

Theorem 8.9. *Let A denote the event that the standard BM $B(t)$ has at least one zero in the time interval (t_0, t_1). Then*

$$P(A) = \frac{2}{\pi} \arctan\left(\sqrt{\frac{t_1 - t_0}{t_0}}\right).$$

Proof. Assume $x > 0$, then by symmetry and space-invariant property,

$$\begin{aligned}
P\left(A | B(t_0) = x\right) &= P\left(\min_{t_0 < s < t_1} B(s) \le 0 | B(t_0) = x\right) \\
&= P\left(\max_{t_0 < s < t_1} B(s) \ge 0 | B(t_0) = -x\right) \\
&= P\left(\max_{t_0 < s < t_1} B(s) \ge x | B(t_0) = 0\right) \\
&= P\left(\max_{0 < s < t_1 - t_0} B(s) \ge x | B(0) = 0\right) \\
&= P\left(M(t_1 - t_0) \ge x\right) \\
&= P(\tau_x \le t_1 - t_0).
\end{aligned}$$

By symmetry,

$$P\left(A|B(t_0)=x\right)=P\left(A|B(t_0)=-x\right).$$

Then

$$\begin{aligned}
P(A) &= \int_{-\infty}^{\infty} P\left(A|B(t_0)=x\right) f_{B(t_0)}(x) \\
&= 2\int_0^{\infty} P(\tau_x \le t_1 - t_0)\frac{1}{\sqrt{2\pi t_0}} e^{-\frac{x^2}{2t_0}}\,dx \\
&= 2\int_0^{\infty}\int_0^{t_1-t_0} \frac{x}{\sqrt{2\pi}} u^{-\frac{3}{2}} e^{-\frac{x^2}{2u}}\,du \frac{1}{\sqrt{2\pi t_0}} e^{-\frac{x^2}{2t_0}}\,dx \\
&= \frac{1}{\pi\sqrt{t_0}}\int_0^{t_1-t_0}\int_0^{\infty} u^{-\frac{3}{2}} x e^{-\left(\frac{1}{2u}+\frac{1}{2t_0}\right)x^2}\,dx\,du \\
&= \frac{1}{\pi\sqrt{t_0}}\int_0^{t_1-t_0} u^{-\frac{3}{2}}\frac{1}{2}\frac{1}{\frac{1}{2u}+\frac{1}{2t_0}}\,du \\
&= \frac{2}{\pi}\arctan\left(\sqrt{\frac{t_1-t_0}{t_0}}\right).
\end{aligned}$$

□

Theorem 8.10. *For $a > b$ and $a \ge 0$,*

$$\begin{aligned}
P\left(M(t)\ge a, B(t)\le b\right) &= P\left(\tau_a \le t, B(t)\le b\right) \\
&= \int_0^t P\left(B(t)\le b|\tau_a = s\right) dF_{\tau_a}(s) \\
&= \int_0^t P\left(B(t-s)\le b|B(0)=a\right) dF_{\tau_a}(s) \\
&= \int_0^t P\left(B(t-s)\le b-a|B(0)=0\right) dF_{\tau_a}(s) \\
&= \int_0^t P\left(B(t-s)\ge -(b-a)|B(0)=0\right) dF_{\tau_a}(s) \\
&= \int_0^t P\left(B(t-s)\ge a-(b-a)|B(0)=a\right) dF_{\tau_a}(s) \\
&= \int_0^t P\left(B(t)\ge 2a-b|\tau_a=s\right) dF_{\tau_a}(s) \\
&= P\left(B(t)\ge 2a-b, \tau_a\le t\right) \\
&= P\left(B(t)\ge 2a-b, M(t)\ge a\right) \\
&= P\left(B(t)\ge 2a-b\right).
\end{aligned}$$

Then the joint pdf of $(M(t), B(t))$ at $a > b, a \ge 0$ is

$$f_{M(t),B(t)}(a,b) = -\frac{\partial^2}{\partial a\partial b}P\left(M(t)\ge a, B(t)\le b\right) = \frac{2(2a-b)}{\sqrt{2\pi t^3}}e^{-\frac{(2a-b)^2}{2t}}.$$

Definition 8.11. Let

$$B_{\mu,\sigma}(t) = \mu t + \sigma B(t),$$

then then process $\{B_{\mu,\sigma}(t); t \geq 0\}$ is referred to as a Brownian motion with drift.

Remark. Because of the presence of our drift term, we see that a Brownian motion with drift does not posses the same type of symmetry that we used earlier, when calculatiing quantities for standard Brownian motion. Hence, in order to compute many probability values that give interesting insight into the behavior of a Brownian motion with drift, we will have to make use of something known as martingale theory.

Theorem 8.12. *Almost surely, the sample path is always continuous.*
Almost surely, Brownian sample paths are nowhere differentiable.

Lemma 8.13. Let $\{B(t); t \geq 0\}$ be a standard BM. Then the processes obtained from the following transformations are also standard BMs.

(a) Scaling:

$$X(t) = \frac{1}{\sqrt{c}} B(ct).$$

(b) Time inversion:

$$Y(t) = \begin{cases} tB(1/t), & t > 0 \\ 0, & t = 0 \end{cases}.$$

(c) Time -reversal

$$Z(t) = B(T) - B(T-t),\ 0 \leq t \leq T.$$

(d) Symmetry:

$$U(t) = -B(t).$$

Lemma 8.14.

$$\lim_{t\to\infty} \frac{B(t)}{t} = 0.$$

Definition 8.15. A stochastic process $\{X(t); t \geq 0\}$ is called a *Gaussian process* if for any $k \geq 1$, and $0 \leq t_1 < \cdots < t_k < \infty$, the random vector $(X(t_1), \ldots, X(t_k))$ has a joint normal distribution. If the distribution of $(X(t+t_1), \ldots, X(t+t_k))$ does not depend on t, we say that the process is *stationary.*

Theorem 8.16. *A standard BM is a zero mean Gaussian process with*

$$\rho(s,t) = E\left[X(s)X(t)\right] = s \wedge t.$$

Conversely, any zero-mean Gaussian process with a.s. continuous path and covariance function $\rho(s,t) = s \wedge t$ *is a standard BM.*

8.3 Martingale Theory

Definition 8.17 (Discrete Martingales)**.** Suppose that $\{X_n\}_{n\geq 0}$ and $\{Y_n\}_{n\geq 0}$ are two discrete time stochastic processes. We say that $\{X_n\}_{n\geq 0}$ is a *Martingale* w.r.t. $\{Y_n\}_{n\geq 0}$ if

(a)
$$E\left[|X_n|\right] < \infty, \ \forall\ n \in \mathbb{N},$$

(b)
$$E\left[X_{n+1}|Y_0, \ldots, Y_n\right] = X_n, \ \forall\ n \in \mathbb{N}, \text{ w.p.1.}$$

Definition 8.18 (Continuous-time Martingales)**.** Suppose that $\{X(t)\}$ and $\{Y(t)\}$ are two continuous time stochastic processes. We say that $\{X(t)\}$ is a *Martingale* w.r.t. $\{Y(t)\}$ if

(a)
$$E\left[|X(t)|\right] < \infty, \ \forall\ t \geq 0,$$

(b) For each $t, h \geq 0$,
$$E\left[X(t+h)|Y(s), 0 \leq s \leq t\right] = X(t), \text{ w.p.1.}$$

Definition 8.19 (Stopping time)**.** A random variable τ is a *stopping time* w.r.t. a process $\{Y(t)\}$ if for each $t \geq 0$, the event $\{\tau \leq t\}$ depends only on the collection $\{Y(s); 0 \leq s \leq t\}$.

Theorem 8.20 (Optimal sampling theorem)**.** *Let $\{X(t); t \geq 0\}$ be a martingale w.r.t. $\{Y(t); t \geq 0\}$ and let τ be a stopping time w.r.t. $\{Y(t); t \geq 0\}$. If*

(a)
$$P(\tau < \infty) = 1,$$

(b)
$$\lim_{a\to\infty} \sup_{t\geq 0} E\left[|X(t\wedge\tau)|\mathbb{1}_{\{|X(t\wedge\tau)|\geq a\}}\right] = 0,$$

then
$$E\left[X(\tau)\right] = E\left[X(0)\right].$$

Example 8.21. Suppose that $\{B(t)\}$ is a standard BM, and let $a, b \in \mathbb{R}$, where $a < 0 < b$. Define

$$\tau_{a,b} = \inf\{t > 0 : B(t) = a \text{ or } B(t) = b\}.$$

First note $\tau_{a,b}$ is a stopping time w.r.t. $\{Y(t)\}$. Check

(a) since $\{B(t)\}$ is recurrent,
$$P(\tau_{a,b} < \infty) = 1.$$

(b) Since
$$|B(t\wedge\tau_{a,b})| \leq \max\{|a|, b\},$$

we have
$$\lim_{c\to\infty} \sup_{t\geq 0} E\left[|B(t\wedge\tau_{a,b})|\mathbb{1}_{\{|B(t\wedge\tau_{a,b})|\geq c\}}\right] = 0.$$

Then

$$0 = E\left[B(0)\right] = E\left[B(\tau_{a,b})\right] = aP\left(B(\tau_{a,b}) = a\right) + b\left[1 - P\left(B(\tau_{a,b} = a)\right)\right].$$

Thus,

$$P\left(B(\tau_{a,b}) = a\right) = \frac{b}{b-a}.$$

Remark. This problem can be considered to be a "Brownian" version of the classical Gambler's ruin problem.

Theorem 8.22.

$$E[\tau_{a,b}] = |a|b$$

Proof. Let

$$Y(t) = B(t)^2 - t, \ \forall\, t \geq 0.$$

Then $\{Y(t); t \geq 0\}$ is a martingale w.r.t. $\{B(t); t \geq 0\}$ since

$$\begin{aligned}
E\left[Y(t+h)|B(s), 0 \leq s \leq t\right] &= E\left[B(t+h)^2 - (t+h)\middle| B(s), 0 \leq s \leq t\right] \\
&= E\left[\left(B(t+h) - B(t) + B(t)\right)^2 - (t+h)\middle| B(s), 0 \leq s \leq t\right] \\
&= E\left[\left(B(t+h) - B(t)\right)^2 + B(t)^2\right. \\
&\quad \left. +\ 2B(t)\left(B(t+h) - B(t)\right) - (t+h)\right| B(s), 0 \leq s \leq t] \\
&= h + B(t)^2 + 2B(t) \cdot 0 - (t+h) \\
&= B(t)^2 - t.
\end{aligned}$$

Suppose the two conditions of the optimal sampling theorem are satisfied, then

$$0 = E\left[B(0)^2 - 0\right] = E\left[B(\tau_{a,b})^2 - \tau_{a,b}\right].$$

Consider the stopping time $\tau_{a,b} \wedge c$, where $c > 0$ is some fixed number. Since

$$P\left(\tau_{a,b} < \infty\right) = 1,$$

$$P\left(\tau_{a,b} \wedge c < \infty\right) = 1.$$

Since

$$B\left(t \wedge \tau_{a,b} \wedge c\right)^c - t \wedge \tau_{a,b} \wedge c \leq \max\left(a^2, b^2\right) + c < \infty,$$

the second condition is satisfied. Hence, by the optimal samping theorem,

$$E\left[\tau_{a,b} \wedge c\right] = E\left[B(\tau_{a,b})^2 \wedge c\right].$$

Letting $c \to \infty$, by MCT,

$$\begin{aligned}
E\left[\tau_{a,b}\right] &= E\left[B(\tau_{a,b})^2\right] \\
&= a^2 \frac{b}{b + |a|} + b^2 \frac{|a|}{|a| + b} \\
&= |a|b.
\end{aligned}$$

□

8.4 Challenging problems

Example 8.23. Suppose $\{B(t); t \geq 0\}$ is a standard Brownian motion, and consider another stochastic process $\{Y(t); t \geq 0\}$ defined as

$$Y(t) = e^{B(t)},\ t \geq 0.$$

(a) Compute, for each $t > 0$, $E[Y(t)]$.

(b) Compute, for $0 < y < \infty$ and for $t > 0$,

$$\lim_{h \downarrow 0} \frac{E\left[(Y(t+h) - Y(t))^2 \,|Y(t) = y\right]}{h}.$$

(a) Let $M(s)$ be the MGF of of $B(t)$, since $B(t) \sim N(0, t)$,

$$M(s) = E\left[e^{sB(t)}\right] = e^{\frac{ts^2}{2}}.$$

Then

$$E\left[Y(t)\right] = E\left[e^{B(t)}\right] = M(1) = e^{\frac{t}{2}}.$$

(b) Since

$$\begin{aligned}
E\left[(Y(t+h) - Y(t))^2 \middle| Y(t) = y\right] &= E\left[\left(e^{B(t+h)} - e^{B(t)}\right)^2 \middle| e^{B(t)} = y\right] \\
&= E\left[e^{2B(t)}\left(e^{B(t+h)-B(t)} - 1\right)^2 \middle| e^{B(t)} = y\right] \\
&= y^2 E\left[\left(e^{B(t+h)-B(t)} - 1\right)^2 \middle| e^{B(t)} = y\right] \\
&= y^2 E\left[\left(e^{B(t+h)-B(t)} - 1\right)^2\right] \\
&= y^2 E\left[\left(e^{B(h)} - 1\right)^2\right] \\
&= y^2 E\left[e^{2B(h)}\right] - 2E\left[e^{B(h)}\right] + 1 \\
&= y^2\left[e^{\frac{h2^2}{2}} - 2e^{\frac{h}{2}} + 1\right] \\
&= y^2\left[e^{2h} - 2e^{\frac{h}{2}} + 1\right].
\end{aligned}$$

Thus, since $0 < y < \infty$,

$$\begin{aligned}
\lim_{h \downarrow 0} \frac{E\left[(Y(t+h) - Y(t))^2 \,|Y(t) = y\right]}{h} &= \lim_{h \downarrow 0} \frac{y^2\left[e^{2h} - 2e^{\frac{h}{2}} + 1\right]}{h} \\
&= \lim_{h \downarrow 0} y^2\left[2e^{2h} - e^{\frac{h}{2}}\right] \\
&= y^2.
\end{aligned}$$

Example 8.24. Suppose $\{B(t); t \geq 0\}$ is a standard Brownian motion, and define the process $\{X(t); t \geq 0\}$ as

$$X(t) = x + B(t),$$

where $x > 0$ is a fixed positive number. Next define

$$\tau_0 := \inf\{t \geq 0 : X(t) = 0\}$$

and using τ_0, we define the stochastic process $\{Y(t); t \geq 0\}$ as

$$Y(t) = \begin{cases} X(t), & t \leq \tau_0 \\ 0 & t > \tau_0 \end{cases}.$$

Intuitively, the Y process behaves the X process until it reaches level 0: when it reaches this level, it stays there forever.

Compute, for $t > 0$, the probability density function of $Y(t)$. **Hint:** Observe that for each $t > 0, y > 0$,

$$P(Y(t) > y) = P(X(t) > y, \tau_0 > t)$$

and try to simply the probability on the right-hand-side of this equality. Next, use this observation to find the pdf of $Y(t)$. Define

$$\tau_x = \inf\{t \geq 0 : B(t) = x\}.$$

If $y > 2x$,

$$\begin{aligned} P(Y(t) > y) &= P(X(t) > y, \tau_0 > t) \\ &= P(B(t) > y - x, \tau_x > t) \\ &= 0. \end{aligned}$$

If $y < 2x$,

$$\begin{aligned} P(Y(t) \leq y) &= P(X(t) \leq y, \tau_0 > t) \\ &= P(B(t) \leq y - x, \tau_x > t) \\ &= P(B(t) \leq y - x) - P(B(t) \leq y - x, \tau_x \leq t) \\ &= \int_0^t P(B(t) \leq y - x | \tau_x = s)\, dF_{\tau_x}(s) \\ &= \int_0^t P(B(t-s) \leq y - x | B(0) = x)\, dF_{\tau_x}(s) \\ &= \int_0^t P(B(t-s) \leq y - 2x | B(0) = 0)\, dF_{\tau_x}(s) \\ &= \int_0^t P(B(t-s) \leq -(y - 2x) | B(0) = 0)\, dF_{\tau_x}(s) \\ &= \int_0^t P(B(t-s) \geq x - (y - 2x) | B(0) = x)\, dF_{\tau_x}(s) \\ &= \int_0^t P(B(t) \geq 3x - y | \tau_x = s)\, dF_{\tau_x}(s) \\ &= P(B(t) \geq 3x - y, \tau_x \leq t) \\ &= P(B(t) \geq 3x - y). \end{aligned}$$

Example 8.25. Let $\{B(t); t \geq 0\}$ represent standard Brownian motion. Assume that all paths of $\{B(t); t \geq 0\}$ are continuous on $[0, \infty)$.

(a) Determine, given a fixed $t > 0$ and a partition $0 = t_0 < t_1 \cdots t_n = t$ of $[0, t]$, the distribution of the random variable

$$\sum_{k=1}^{n} B(t_{k-1})(t_k - t_{k-1}).$$

(b) Given a fixed $t > 0$, determine the distribution of

$$\int_0^t B(s)ds.$$

Hint: This integral is well defined for each possible path. Furthermore, because the path of $\{B(t); t \geq 0\}$ is continuous on $[0, t]$, we have

$$\int_0^t B(s) = \lim_{n\to\infty} B\left(\frac{(k-1)t}{2^n}\right)\left(\frac{kt}{2^n} - \frac{(k-1)t}{2^n}\right).$$

This also implies that the limiting cdf of the random variable on the right-hand-side coincides with the cdf of the left-han-side.

(a)

$$\begin{aligned}
\sum_{k=1}^{n} B(t_{k-1})(t_k - t_{k-1}) &= \sum_{k=1}^{n} B(t_{k-1})t_k - \sum_{k=1}^{n} B(t_{k-1})t_{k-1} \\
&= \sum_{k=1}^{n} B(t_{k-1})t_k - \sum_{k=0}^{n-1} B(t_k)t_k \\
&= \sum_{k=1}^{n} B(t_{k-1})t_k - \sum_{k=1}^{n-1} B(t_k)t_k \\
&= \sum_{k=1}^{n} \left(B(t_{k-1}) - B(t_k)\right) t_k + B(t_{n-1})t_n \\
&= \sum_{k=1}^{n} \left(B(t_{k-1}) - B(t_k)\right) t_k - \sum_{k=1}^{n-1} \left(B(t_{k-1}) - B(t_k)\right) t \\
&= \left(B(t_{n-1}) - B(t)\right) t + \sum_{k=1}^{n-1} \left(B(t_{k-1}) - B(t_k)\right)(t_k - t) \\
&\sim N\left(0, t^2(t - t_{n-1}) + \sum_{k=1}^{n} (t_k - t)^2 (t_k - t_{k-1})\right).
\end{aligned}$$

Example 8.26. Let $\{B(t); t \geq 0\}$ be a standard Brownian motion.

(a) Let

$$Y(t) = \exp\left(B(t) - t/2\right).$$

Show that the process $\{Y(t); t \geq 0\}$ is a martingale.

(b) Let $a < 0 < b$. Find the first two moments of $\tau_{a,b}$.

(c) Give an example of random time S, with $P(0 \le S < \infty) = 1$ such that the process $\{B(S+t) - B(S); t \ge 0\}$ is not a standard Brownian motion. Hint: S need not be a stopping time.

(a) *Proof.*

$$\begin{aligned} E\left[Y(t+h)|Y(s), 0 \le s \le t\right] &= E\left[Y(t+h) - Y(t) + Y(t)|Y(s), 0 \le s \le t\right] \\ &= Y(t) + E\left[e^{B(t+h)-\frac{(t+h)}{2}} - e^{B(t)-\frac{t}{2}} \middle| Y(s), 0 \le s \le t\right] \\ &= Y(t) + Y(t)E\left[e^{B(t+h)-B(t)-\frac{h}{2}} - 1\right] \\ &= Y(t). \end{aligned}$$

□

(b) Similarly, we can show $Z(t) = B(t)^4 - 3t^2$ is a martingale.

(c)

$$S = \arg\max_{0 \le s \le t} B(s).$$

Then the process cannot be a standard BM since for $t \le T - S$, we clearly have $B(S+t) - B(s)$

Example 8.27. Let $\{B_1(t); t \ge 0\}$ and $\{B_2(t); t \ge 0\}$ be two independent standard Brownian motions. Assume that for $i = 1, 2$,

$$X_i(t) = \mu_i t + \sigma_i B_i(t), \ \forall\, t > 0.$$

Let $x > 0$ and c_1 and c_2 are two fixed constants. Then

$$Z(t) = \frac{c_1 X_1(t) - c_2 X_2(t)}{\sqrt{c_1^2\sigma^2 + c_2^2\sigma_2^2}} = \frac{c_1\mu_1 - c_2\mu_2}{\sqrt{c_1^2\sigma_1^2 + c_2^2\sigma_2^2}} t + \frac{1}{\sqrt{c_1^2\sigma_1^2 + c_2^2\sigma_2^2}} B(t) = ut + \sigma B(t).$$

Thus

$$P\left(\sup_{0 \le s \le t} (c_1 X_1(s) - c_2 X_2(s)) > x\right) = P\left(\sup_{0 \le s \le t} Z(s) > \frac{x}{\sqrt{c_1^2\sigma_1^2 + c_2^2\sigma_2^2}}\right) = P(\tau \le t),$$

where the stopping time

$$\tau = \inf\left\{t \ge 0 : Z(t) \ge \frac{x}{\sqrt{c_1^2\sigma_2^2 + c_2^2\sigma_2^2}}\right\}.$$